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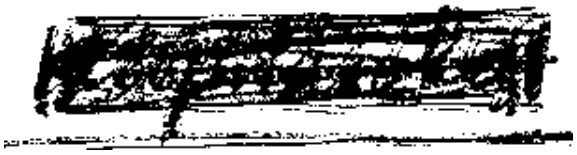
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# Chapter 7

GELFAND-P000501

[source number: 1]

3/IX-73. Sem. of



**Undeciphered passage.** The entry between “Sem.” and “Gelfand” is heavily crossed out. Its reading is unresolved.

Original source details: GELFAND-P000501

Gelfand.

Plan:

1. Derived category (Kirillov, Bernstein).

2. Ahlfors — Nevanlinna theory for many variables (and according to Grauert). (Henkin, Shabat).

3. $K$-theory (simplicial version) according to Quillen (Fuchs).



**Uncertain reading.** The faint words are provisionally read as “simplicial version”.

Original source details: GELFAND-P000501

4. Lichtenbaum’s conjecture on the connection between the $\zeta$-fn. at integers and higher $K$-theories (Manin).

5. Weil: the ideas of «Герстенцуса».



**Uncertain reading.** The Cyrillic spelling is provisionally read as «Герстенцуса».

Original source details: GELFAND-P000501

Interlude:



**Uncertain reading.** The number at the left is heavily retraced, possibly 6 or 8.

Original source details: GELFAND-P000501

2 problems of Serre:

1) $k[t]$, $k$ — finite; $SL_n(k[t])$, $n \geq 3$; $SL_n(k[t, t^{-1}])$, $n \geq 2$

These groups have finitely many generators. Are they finitely presented? (Do there exist generators with finitely many rels. among them.).

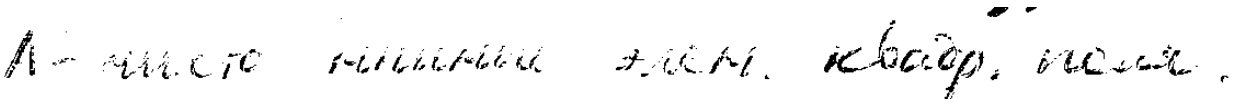
2) Coher. group: $\forall$ subgrp. with finitely many gens. is finitely presented. Free and nilpot. are coherent.



Original source details: GELFAND-P000501

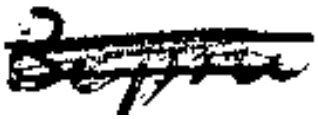
**Uncertain reading.** The faint abbreviation reads “nilpot.”.

Fund. groups of 3-dimensional manifolds are coherent (G. P. Scott, J. Lond. Math. Soc., (2)6 (1973), 437–440). Article. $SL_2(\Lambda)$ is coher., where $\Lambda$ — purely imaginary elem. of a quadr. field.



Original source details: GELFAND-P000501

**Uncertain reading.** The faint entry is provisionally read as “purely imaginary elem. of a quadr. field”.

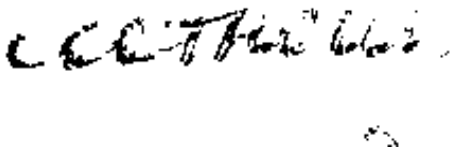


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**Uncertain reading.** The crossed-out word is provisionally read as “also”.

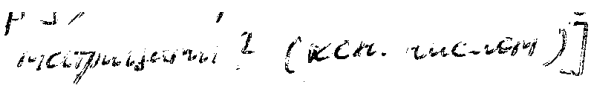
Are $SL_3(\mathbb{Z})$ or $SL_2(\mathbb{Z}[\frac{1}{p}])$ coherent? $GL_n(\mathbb{Q})$?

[are there many rels. among distinct irreducibles? (a fin. number)]



Original source details: GELFAND-P000501

**Uncertain reading.** The faint abbreviation is provisionally read as “rels.”.



Original source details: GELFAND-P000501

**Uncertain reading.** The word is provisionally read as “irreducibles”.

[Which fund. groups occur?]

6. Smale: optimization of several functions.



Original source details: GELFAND-P000501

**Uncertain reading.** The digit is heavily retraced. Its outlines and the adjacent numbering support 6.

7. Brown, Douglas, Fillmore  Proper $K$-theory of several variables (Kirillov).

8. Mishchenko: representations in his sense.

9. Lusztig: higher sign. of Novikov  J. Diff. Geom. 197(1)



**Uncertain reading.** The faint name is provisionally read as “Lusztig”.

Original source details: GELFAND-P000501

10. Atiyah, Bott, Patodi on $L$-functions associated with eqs. (spectral asymmetry, etc.) (Shubin)

11. Donovan, Freilich a (V. Ponomarev) — a quadruple of subspaces, etc.



**Uncertain reading.** The small mark is provisionally read as «a».

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Modules, etc.

12. Yu. I. Manin.

13. Vershik, Gelfand, Graev — 6 variations on $SL_2$.

Yu. I. Manin.  $p$-adic measures

Setting: $\mathbb{Q}_p$;  $K$ — a fin. ext. $X$ — a compact $p$-adic analytic set over $K$. Example: integers in $K$, their direct prod., etc.

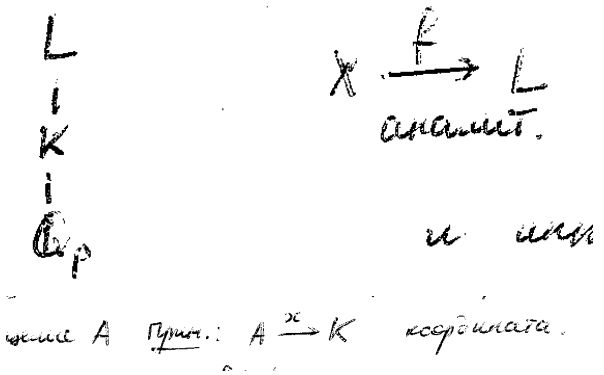


**Diagram and typed description.** The field tower at the left has no arrows. Beside it are an analytic map and an example of a coordinate.

$$\begin{array}{c} L \\ | \\ K \\ | \\ \mathbb{Q}_p \end{array} \qquad X \xrightarrow[\text{analyt.}]{f} L \qquad \text{integers } A \qquad \text{Ex.: } A \xrightarrow{x} K \quad \text{coordinate.}$$

Original source details: GELFAND-P000501

Problem: define $\int f\, d\mu$, i.e. determine a reasonable and inter. class of measures of $p$-adic type on $X$.



**Uncertain reading.** The abbreviation is provisionally read as “inter.”.

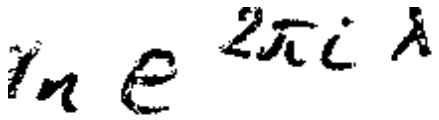
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GELFAND-P000502

Motivation: a) there are $\zeta$-fns. $L(s)$ $\quad \mathbb{C} \longrightarrow \mathbb{C}$

$$\sum n^{-s}; \qquad \sum \chi(n) n^{-s}; \qquad \sum a_n n^{-s},$$

where $\sum a_n e^{2\pi i \lambda}$ is an autom. form.

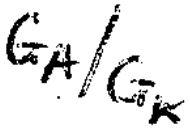


**Uncertain reading.** The last sign in the exponent is provisionally set as $\lambda$.

Original source details: GELFAND-P000502

For them, a class of int. formulas with adelic integration

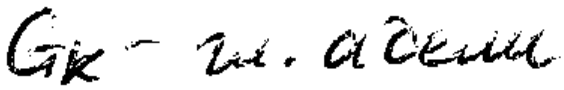
$$L(\chi) = \int_{G_A/G_K} \chi \, d\mu$$



**Uncertain reading.** The denominator is provisionally set as $G_K$. There is an additional loop at the base of $G$.

Original source details: GELFAND-P000502

$G_K$ — princ. adeles

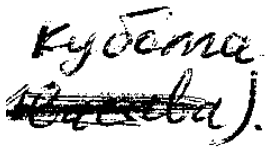


**Uncertain reading.** The faint note is provisionally read as "$G_K$ — princ. adeles".

Original source details: GELFAND-P000502

In fact, the argument is a character $L(\chi)$

b) there are $p$-adic analogues of $L(\chi)$ and $\zeta(s)$ (Leopoldt — ~~Iwasawa~~ Kubota).



**Uncertain reading.** "Iwasawa" is crossed out. "Kubota" is inserted above.

Original source details: GELFAND-P000502

$$\zeta(2m) = B_{2m}\frac{\pi^{2m}}{\dots}$$

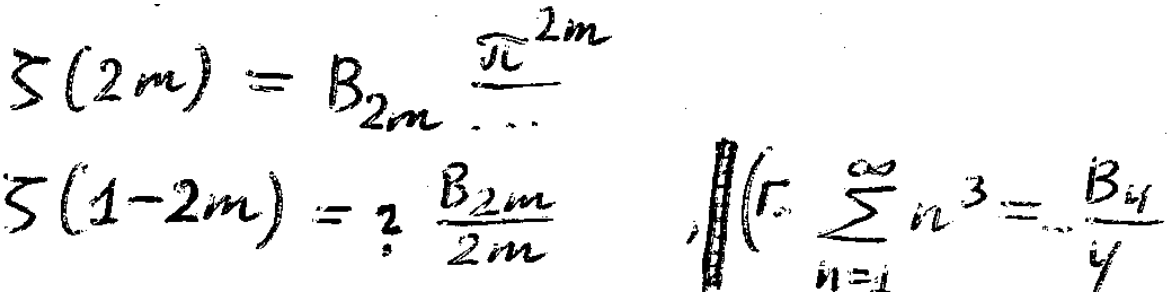
$$\zeta(1-2m) = ?\,\frac{B_{2m}}{2m},$$

$$\| \quad \left(\text{G.}\sum_{n=1}^{\infty} n^3 = \frac{B_4}{4}\right.$$



**Uncertain reading.** The letter after the parenthesis is provisionally read as «Г.». A hatched stroke is at the left.

Original source details: GELFAND-P000502



Original source details: GELFAND-P000502

**Uncertain reading.** There are faint marks below the line between the equals sign and $\frac{B_4}{4}$. Their reading is unresolved.

even and odd differ:

$$-y'' + p(x)y = \lambda y, \qquad \sum \lambda_n^k$$

integers and half-integers differ)

Still unknown why such a $\sum n^3$ is expr. through $B$ .)

The point of the next plan — pres. of Kronecker's formulas.



**Uncertain reading.** The abbreviation before "formulas" is provisionally read as «пред.».

Original source details: GELFAND-P000502

Choose $p$ from the Euler prod.

$$1-2m \longrightarrow \frac{B_{2m}}{2m}(1-p^{2m-1}) \qquad \text{Fix } p!$$

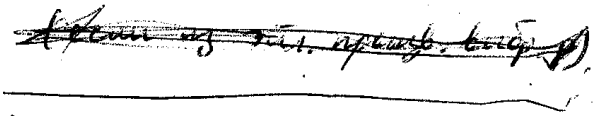
$p$-adic cont. and even analytic !!! ~~(from the Euler prod. we removed the chosen $p$)~~.

---



**Uncertain reading.** The ending of the last word is provisionally read as «аналитичне».

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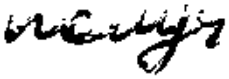


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**Uncertain reading.** The crossed-out entry is provisionally reconstructed. A long stroke runs below the line in the manuscript.

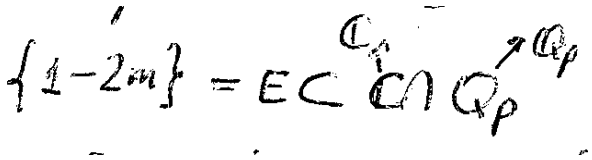
$$\begin{array}{c}\zeta(s)(1-p^{-s})\\ \mathbb{C}\longrightarrow\mathbb{C}\end{array} \quad \text{between.} \quad \begin{array}{c}\zeta^{(p)}(s_p)\\ \mathbb{Z}_p\longrightarrow\mathbb{Q}_p\end{array}$$

**Uncertain reading.** The faint word between the two displays is provisionally read as “between”.



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there is an intersection of $\mathbb{C}$ and $\mathbb{Z}_p$ (integers), the values there are the same.



Original source details: GELFAND-P000502

**Diagram and typed description.** The lower left letter is provisionally set as $\mathbb{C}$. A short arrow leads upward from it. The right arrow goes up and to the right.

$$\{1-2m\} = E \subset \begin{array}{ccc}\mathbb{C} & & \mathbb{Q}_p\\ \uparrow & \nearrow & \\ \mathbb{C}\cap\mathbb{Q}_p & & \end{array}$$

and they are analytic

In fact, the correct argument is apparently adelic.

$$\mathbb{Q}_p \longrightarrow \mathbb{Q}_\ell \quad (p\neq\ell) \quad \text{apparently simpler (but unknown)}$$

Difficult $\mathbb{Q}_p \longrightarrow \mathbb{Q}_p$,

Was begun by Mazur.

Measures on compact $p$-adic spaces.

$K$ — fin. ext. of $\mathbb{Q}_p$; $A$ — integers in $K$; $A \supset \overline{m}$ (max. ideal)

(roots of eqs. with integral coeffs.)

$$\text{Card}(A/\overline{m}) \underset{\text{def}}{=} q.$$

Compact anal. set over $K$.

$$A^n = \underbrace{A\times\ldots\times A}_{n} \quad \text{— } n\text{-dimensional ball}$$

Anal. set: space and a sheaf of fns., loc. structure — a ball.

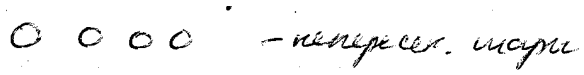
Theorem (Serre): $X$ — a comp. an. set over $K$ of dim. $n$. Then

a) $X \simeq rA^n$ (analytic isom.), $r\in\mathbb{N}$



Original source details: GELFAND-P000502

**Uncertain reading.** The number of balls is provisionally set as $r$, possibly $\tau$.



Original source details: GELFAND-P000502

**Diagram and typed description.** Four ovals with a label at the right.

◯ ◯ ◯ ◯ — disjoint balls

b) ~~if~~

$$rA^n \simeq r'A^n \quad \Leftrightarrow \quad r \equiv r' \mod(q-1)$$



Original source details: GELFAND-P000502

**Uncertain reading.** The crossed-out word is provisionally read as “if”. The equivalence sign is encircled.

GELFAND-P000503

[source number: 2]

$$A = \{a \in K \mid a \text{ is integral}\}. \qquad A = \bigcup_{a_i \bmod m} \{a \equiv a_i \bmod m\}.$$

How to compute $r$?

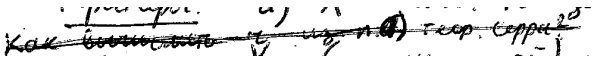


Original source details: GELFAND-P000503

**Uncertain reading.** The letter is provisionally set as $r$.

Examples: a) $X$ — an alg. set over $A$ (a smooth scheme over Spec $A$).

~~How to compute $r$ from pt. a) of Serre's theor.?~~



Original source details: GELFAND-P000503

**Uncertain reading.** The crossed-out question is read provisionally. The strokes over the part number are particularly heavy.

Reduction $X_0$ (closed fibre) — an alg. set over $A/\bar{m}$
over Spec $A$.

$$\text{Card}\, X_0(A/\bar{m}) = r(X).$$

Example of a scheme: $X = P^1$ or a circle. 2 aff. lines glued by $x = y^{-1}$
Reduction: proj. line over a fin. field of $q$ elems. — there are $q+1$ elems.

$$q + 1 = 2 \bmod(q-1).$$

So 2 balls:



Original source details: GELFAND-P000503

**Diagram and typed description.** The top of the left contour is blacked out. The arrows of the right contour point towards its upper gap. At the right are “and” and a small round sign.


and

$$P^1(K) = \mathbb{Q}_p \cup \infty, \qquad pA \cup A^{-1}.$$

b) $X$ — $p$-adic comp. Lie grp.; it is profinite, $\forall$ factor has an order.

Order: $2^2\, 3^\infty\, 5^1\, 7^0\, 11^\infty \ldots$ — a supernatural number (since they divide one another).



**Manuscript excerpt.** The stroke left of the exponent belongs to the digit 5. The exponent reads 1.

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It turns out ord $X = p^\infty \cdot N$, where $(p, N) = 1$, $N$ — an ord. integer.

$$\underline{\underline{N = r(X)}}$$

$SL_2$ — $(p-1)$ balls $(= p^2 - 1 \bmod p)$



Original source details: GELFAND-P000503

**Manuscript excerpt.** The manuscript reads $p^2 - 1$ mod $p$.



**Diagram and typed description.** The mark between $L$ and $K$ is blacked out. Its shape and direction are unresolved.

$L$

•

$K$

Original source details: GELFAND-P000503

Definition: $L$-measure on $X$ — an $L$-valued finitely addit. fn. on open-and-closed subsets of $X$.

($X$ — a comp. anal. set).



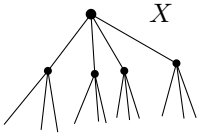
**Uncertain reading.** The start of "anal." is heavily overwritten.

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Examples: a) $X = \mathbb{Z}_p$




**Diagram and typed description.** Four branches descend from the vertex $X$. Some lower strokes merge.



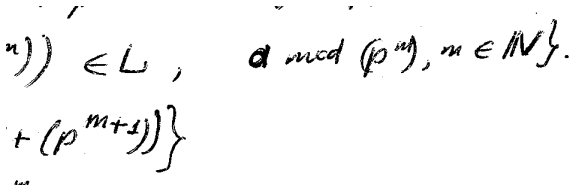

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The vertices are open-and-closed subsets.

Construct these measures successively. We obtain a fund. syst. of neighbourhoods.

$$\text{Measure on } X \leftrightarrow \{\mu(a+(p^m)) \in L; \quad a \bmod (p^m),\ m \in \mathbb{N}\}.$$

$$\mu(a+(p^m)) = \sum_{b \equiv a \bmod p^m} \mu(b+(p^{m+1}))\}$$



**Manuscript excerpt.** Each of the two written lines ends in a closing brace.

Original source details: GELFAND-P000503

And this applies to any compact subset.

$$\text{b) } \mu(a+(p^m)) = \underbrace{\text{const} \cdot p^{-m}}_{\in L} \quad \text{— invar. measures.}$$

There is on $L$ a metric ($p$-adic absolute value).

$$|\text{const} \cdot p^{-m}|_L \longrightarrow \infty \quad \text{as } m \longrightarrow \infty.$$

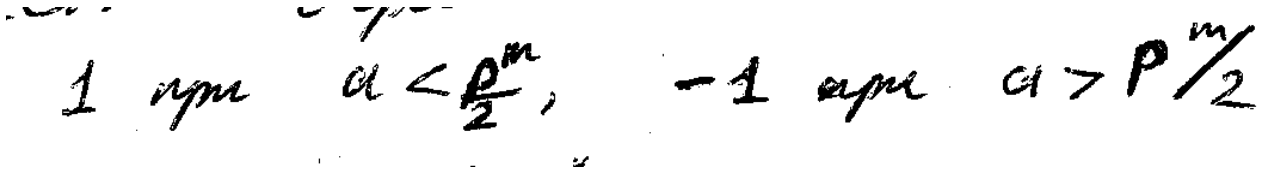
The smaller the sets, the larger the measure!!

It cannot tend to zero! It would follow from the non-Archim. property that the measure of the whole equals $0$.

Invar. measures are bad.

The slowest-growing is bounded.

$$\text{c) } \mu(a+(p^m)) = 1 \text{ if } a < \frac{p^m}{2}, \qquad -1 \text{ if } a > p^m/2$$



Original source details: GELFAND-P000503

**Manuscript excerpt.** The left value is $1$ and the right value is $-1$. The first threshold uses a stacked fraction and the second a slash.

|| nontriv. bounded measure

$0 < a < p^m, \quad (a,p) = 1$ Measure on $\mathbb{Z}_p^*$.



**Undeciphered passage.** A short entry between $0 < a < p^m$ and $(a,p) = 1$ is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000503

GELFAND-P000504

Exercise: check additivity. Interference occurs.

Integr. against it — $\zeta$-fns. of Leopoldt — Kubota.

Mazur devised this measure.

Integrals. $X$, $\mu$ — a measure with vals. in $L$.



**Diagram and typed description.** In the right margin an arrow points down from $L$ to $K$.

$$\begin{array}{c} L \\ \downarrow \\ K. \end{array}$$

Original source details: GELFAND-P000504

$X \xrightarrow{f} L$, analyt. fn.

$$\int_X f\, d\mu$$

$$R = \{(U_i), \quad \underset{\text{open-and-closed}}{U_i} \text{ — a partition of } X, \quad a_i \in U_i\}$$

$$S(f, R) = \sum f(a_i)\mu(U_i).$$

$$\int_X f\, d\mu = \lim_R S(f; R)$$

if the limit exists under refinement

Example. $X = \mathbb{Z}_p$, $\mu$ — invar. $(\mu(a + (p^m)) = p^{-m}.)$,

$L = \mathbb{Q}_p$. $f = 1$ is integrable

$f = x$ — simple nonconst.

$$\int f\, d\mu \quad \text{does not exist!}$$

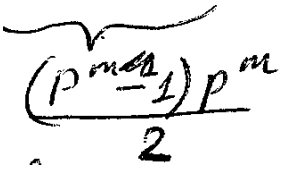
$$R = \{0 \le a \le p^m - 1, \quad a \in U_a \equiv \{b \mid b \equiv a \bmod p^m\}.$$



**Uncertain reading.** An entry resembling $-1$ is crossed out after the exponent $m$. The next $-1$ is part of the bound itself.

Original source details: GELFAND-P000504

$$S(f,R) = \frac{1}{p^m} \underbrace{\sum_{a=0}^{p^m-1} a}_{\frac{(p^m-1)p^m}{2}} \longrightarrow -\frac{1}{2}$$



Original source details: GELFAND-P000504

**Uncertain reading.** Below the brace, an entry resembling $-1$ is crossed out after the exponent $m$ of the first $p$.

But this is not the only limit point.

$$R' = \{p^m \in U_0 \quad \text{the rest as in } R\}$$

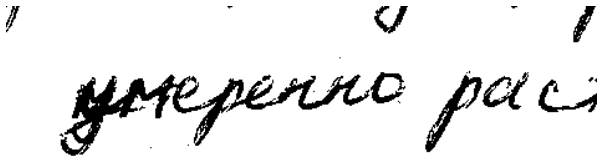
The limit will be different: variation of the fn. on the interval $\approx$ measure of the interv.

hence a cond. on the measure: under refinement the ratio of the measure to the invar. measure must $\longrightarrow 0$ (this is in the case $n = 1$).

Theorem: let $X$ be one-dimensional over $\mathbb{Q}_p$

1) $\mu$ — a measure on $X$ with vals. in $L$, moderately growing, i.e.

$$U \longrightarrow \underset{\text{point}}{\bullet} \Longrightarrow \left|\frac{\mu(U)}{\mu_{\text{inv}}(U)}\right|_L \longrightarrow 0$$



Original source details: GELFAND-P000504

**Uncertain reading.** The beginning of "moderately" is heavily retraced.

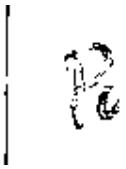
($\mu_{\text{inv}}$ — the usual inv. meas. on $\mathbb{Z}_p$).

2) $f : X \longrightarrow L$ satisfies the Lipschitz cond. with exponent $1$

Then $\int f\, d\mu$ exists.

|| Next time: concrete measures and mod. forms.

GELFAND-P000505



Original source details: GELFAND-P000505

**Uncertain reading.** The entry beyond the right border is cut off by the edge of the scan. Its reading is unresolved.

10.9.73. Gelfand Sem.

Manin (continued).

$X$ — 1-dimensional $p$-adic space;

$\mu$ — $K$-measure on $X$ ($K$ — a fin. ext. of $\mathbb{Q}_p$)

$f : X \longrightarrow K$

$$R: \quad \text{Riemann data: } \{X = \bigcup_{i=1}^{n} U_i\}, \quad a_i \in U_i\}.$$



**Uncertain reading.** A crossed-out closing brace is visible after $U_i$.

Original source details: GELFAND-P000505

$$S(f, R) = \sum \mu(U_i) f(a_i).$$

Theorem:

$\exists \int_X f\, d\mu$, if

a) $f$ is Lipschitz with exp. 1

b) $(U \mapsto \text{point}) \Longrightarrow |\mu(U)/\mu_{\text{inv}}(U)| \longrightarrow 0$.

Example. $X = \mathbb{Z}_p$;

$$\mu(a + (p^m)) = \begin{cases} -1 & \text{if } 0 \le a < \dfrac{p^m}{2} \\ +1 & \text{if } \dfrac{p^m}{2} < a < p^m. \end{cases}$$

Tree of open-and-closed subsets:

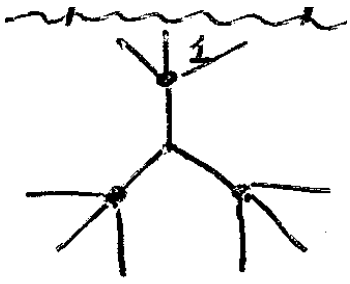
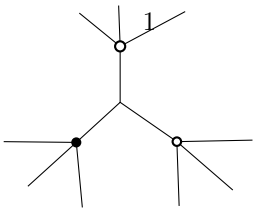
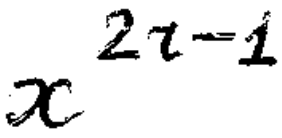



**Diagram and typed description.** Three strokes extend upward from the top circle. There are two three-ray forks below. A 1 is written at the upper branch.




Original source details: GELFAND-P000505

$f(x) = x^{2r-1}$ (for $f(x) = x^{2r}$ one gets 0).



**Uncertain reading.** The letter resembles $\tau$ and is provisionally set as $r$.

Original source details: GELFAND-P000505

$$\int_{\mathbb{Z}_p} x^{2r-1}\,d\mu = (-1)^r(1-2^{-r})\frac{B_{r-1}}{2r}$$

$$\mathbb{Z}_p = \bigcup_{0\le a<p^{m-1}} \{a + p^m\mathbb{Z}_p\}.$$



**Manuscript excerpt.** In the upper bound, $-1$ is written at the level of the exponent $m$.

Original source details: GELFAND-P000505

$R$



**Uncertain reading.** The sign in the left margin is heavily retraced. Provisionally read as $R$.

Original source details: GELFAND-P000505

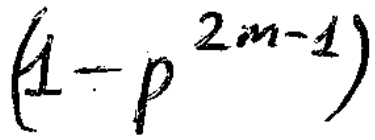
$$S(f,R) = -\sum_{0\le a<p^m/2} a^{2r-1} + \sum_{p^m>a>p^m/2} a^{2r-1}$$

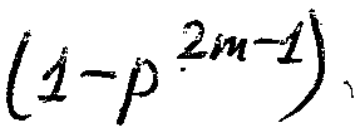
There are Voronoi formulas interpreted as evaluations of integrals of this kind.

$$\int_{\mathbb{Z}_p^*} = \int_{\mathbb{Z}_p} - \int_{p\mathbb{Z}_p}$$

$$\int_{\mathbb{Z}_p^*} x^{2r-1}\,d\mu = (1-p^{2m-1})\int_{\mathbb{Z}^p}$$

$$= (-1)^r(1-p^{2m-1})\cdot(1-2^{-r})\frac{B_{r-1}}{2r} =$$





**Manuscript excerpt.** The exponent is written as $2m-1$ in both factors.

Original source details: GELFAND-P000505



**Manuscript excerpt.** The $p$ is written to the upper right of $\mathbb{Z}$.

Original source details: GELFAND-P000505

(computed by the Mellin transf. of characters)

$$= (\ldots)\zeta_{(p)}(1-2r)$$

GELFAND-P000506

[source number: 5]

Consider $X = \mathrm{Hom}(\mathbb{Z}_p^*; K^*)$.



**Uncertain reading.** The mark above $K$ merges with the parenthesis. Provisionally read as a star.

Original source details: GELFAND-P000506

$$\mathbb{Z}_p^* = (\mathbb{Z}/p\mathbb{Z})^* \times (1+p)^{\mathbb{Z}_p}.$$

$\chi \in X \Longrightarrow \chi = \chi_0 \cdot \chi_1$, where

$$\chi_0 = 1 \quad \text{on } (1+p)^{\mathbb{Z}_p}.$$
$$\chi_1 = 1 \quad \text{on } (\mathbb{Z}/p\mathbb{Z})^*.$$

$$\forall u \in \mathbb{Z}_p^* : \quad u = \varepsilon \cdot (1+p)^a, \qquad \text{where } \varepsilon^{p-1} = 1, \quad a \in \mathbb{Z}_p$$

Therefore

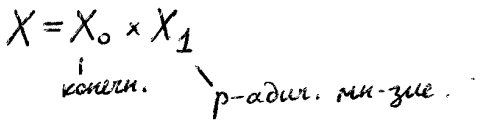



**Diagram and typed description.**

$X = X_0 \times X_1$

finite — $p$-adic set.

Original source details: GELFAND-P000506

Thus: $X$ — $p$-adic set

$$\chi_1(1+p) = 1+t, \qquad t \in \overline{m}. \qquad \text{(notation: } \chi_{(t)})$$

Hence $X_1 \cong \overline{m}$ (but with a different law of composition)



**Manuscript excerpt.** The first letter of the Russian word for “but” is heavily retraced.

Original source details: GELFAND-P000506

Theorem. If $\mu$ grows moderately (more slowly than an invari. one), then

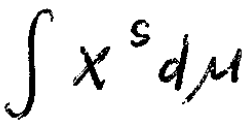
$$\int_{\mathbb{Z}_p^*} \chi \, d\mu \quad \text{is analytic in } \chi.$$

In more detail:

$$\int \chi_{(t)} \, d\mu = F(t) \in K[[t]]$$

converges for $|t| < 1$.

Next: $\int x^s \, d\mu$ is analytic in $s$ for $s \equiv 0 \ (\mathrm{mod}(p-1))$ and for $s = k \, \mathrm{mod}(p-1)$



**Manuscript excerpt.** The integrand reads $x^s$.

Original source details: GELFAND-P000506

— · —

Modular forms. $\Gamma = SL(2,\mathbb{Z})/(\pm 1)$ acts on

$$H = \{z \in \mathbb{C} \mid \operatorname{Im} z > 0\}, \qquad w \equiv 0\ (2), \qquad w \geq 0$$

$P_w$: Parab. forms of weight $w+2$ on $H$:

a) $\varphi(z)(dz)^{\frac{w+2}{2}}$ $\Gamma$-invariant

b) $\varphi(z) = \sum_{n=1}^{\infty} a_n e^{2\pi i n z}$

1st nontriv. fn. at $w+2 = 12$ with gaps up to $26$ (1-dimensional), then a fin.-dim. space throughout [Crossed out:



**Undeciphered passage.** The reading under the heavy strikeout is unresolved.

Original source details: GELFAND-P000506

]

$$w+2 = 12 \quad \Longrightarrow \quad e^{2\pi i z} \prod_{n=1}^{\infty} (1 - e^{2\pi i n z})^{24} = \sum_{n=1}^{\infty} \tau(n) e^{2\pi i n z}$$

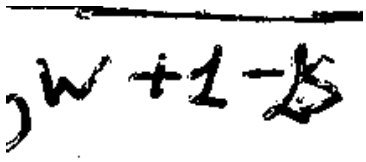
— · —

In $P_w$ there exists a basis of forms with algebraic integer coefficients

GELFAND-P000507

and such that if $\sum_{1}^{\infty} a_n e^{2\pi i n z}$ — an elem. of this basis, then

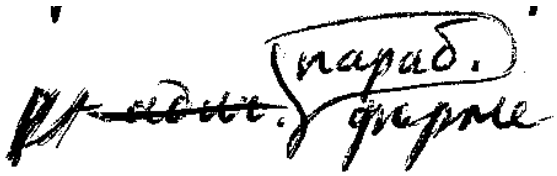
$$\sum a_n n^{-s} = \prod_p \frac{1}{1 - a_p p^{-s} + p^{w+1-[?]}}.$$



**Uncertain reading.** The end of the exponent in the first product is overwritten. Readings: $2s$ or a sign resembling $k$.

Original source details: GELFAND-P000507

(For each [Crossed out: $p$-adic.; above the line, circled: parab.] form we will construct a par. measure below)



**Manuscript excerpt.** The circled insertion "parab." points to the crossed-out word.

Original source details: GELFAND-P000507

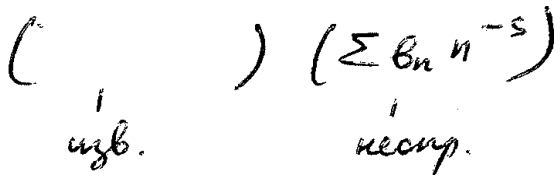
Hecke op.: let us see what the values of $a$, are if

$$\sum a_n n^{-s} = \frac{1}{(1-\lambda_p p^{-s}+p^{w+1-2s})}\left(\sum \dots\right)$$

i.e.

$$\sum a_n n^{-s} = (\quad)\left(\sum b_n n^{-s}\right)$$

known undet.




**Manuscript excerpt.** The empty parentheses are labelled “known” and the series “undet.”.

Original source details: GELFAND-P000507

$$1-\lambda p^{-s}+p^{w+1-2s} = (1-\alpha_1 p^{-s})(1-\alpha_2 p^{-s}).$$

We get a lin. condition on the coeffs. $a_n$.
All together — an infin. matrix.



**Uncertain reading.** The mark after the period is heavily overwritten. Its reading is unresolved.

Original source details: GELFAND-P000507

This cond. $\Longleftrightarrow$ The coeff. vector is an eigenvector of the Hecke operator.
Let us write its action on the fn. instead of on the Fourier coeffs.

$$(\Phi \mid T_p)(z) \underset{\text{def}}{=} p^{w+1}\Phi(pz) + p^{-1}\sum_{k=0}^{p-1}\Phi\left(\frac{z+k}{p}\right).$$

Eigenfns. $\Longleftrightarrow$ extract the $p$-factor from the corr. Dirichlet series.

$$(P_w \mid T_p) = P_w.$$

Measures: on $\mathbb{Z}_p^*$, assoc. with $\varphi$, $\varphi \mid T_p = \lambda_p \varphi \quad \forall p$.
[Crossed out:



**Uncertain reading.** Possibly $\mathbb{Q}_p$. The reading under the strikeout is unresolved.

Original source details: GELFAND-P000507

] with values in $\{\mathbb{Q}_p(\lambda_1, \ldots, \lambda_p, \ldots)\}_p$

Half-measures with values in [Crossed out:

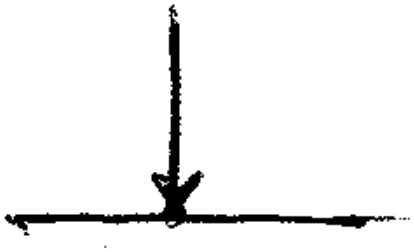


Original source details: GELFAND-P000507

**Uncertain reading.** Possibly $\mathbb{Q}$. The reading under the strikeout is unresolved.

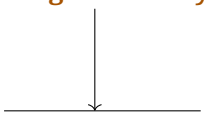
] $\mathbb{C}$.

$x \in \mathbb{Q}$. $\quad P(x) = \int_x^{i\infty} \Phi(z)\, dz$



Original source details: GELFAND-P000507

**Diagram and typed description.**



Convergence: obvious at the upper end ($e^{2\pi i n z} \to 0$), and unclear at the lower end. From the funct. eq.: it converges at the lower end too, but the dependence on $x$ is discontinuous (from subtle arithmetic). At irr. points it generally diverges. $P(x+1) = P(x)$.

$\lambda_p P(x) = \int_x^{i\infty} \Phi(z) \mid T_p\, dz$ we substitute and get a funct. eq.

$$\lambda_p P(x) = p^w P(px) + \sum_{k=0}^{p-1} P\left(\frac{x+k}{p}\right)$$



Original source details: GELFAND-P000507

**Uncertain reading.** A separate sign resembling $p$ or $P$ stands over the argument in the right-hand summand.

$$\frac{a}{p^m} \sim a + p^m \mathbb{Z}_p.$$

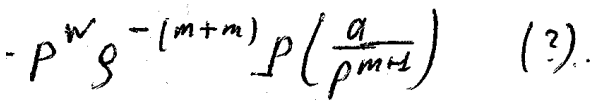
The measure $P\left(\frac{a}{p^m}\right)$ gives almost fin. additivity.

GELFAND-P000508

[source number: 6]

Correction:

$$\mu_\Phi(a + p^m\mathbb{Z}_p) = \varrho^{-m} P\left(\frac{a}{p^m}\right) - p^w \varrho^{-(m+m)} P\left(\frac{a}{p^{m+1}}\right) \qquad (?).$$

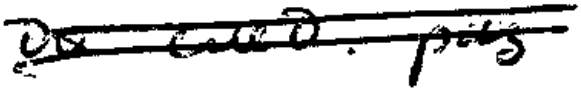


**Manuscript excerpt.** The second exponent on $\varrho$ reads $-(m+m)$. The last denominator reads $p^{m+1}$. The $(?)$ is in the manuscript.

Original source details: GELFAND-P000508

We get a measure on $\mathbb{Z}_p^*$.

~~Until next time~~



**Manuscript excerpt.** Two strokes cross out the phrase. The last word reads “раз”.

Original source details: GELFAND-P000508

— · —

GELFAND-P000509

17.9.73. Gelfand Sem.



**Uncertain reading.** The second digit of the day is overwritten. A 7 is visible over the earlier strokes.

Original source details: GELFAND-P000509

Yu. I. Manin (continued)

$$P(x) = \int_x^{i\infty} \Phi(z)\, dz$$

Consider the eq. $\varrho^2 - \lambda_p \varrho + p^{w+1} = 0$. and $\varrho$ is any root of it

Proposition. Put

$$\mu_\Phi\big(a + (p^{m+1})\big) = \varrho^{-m} P\left(\frac{a}{p^{m+1}}\right) - p^w \varrho^{-(m+1)} P\left(\frac{a}{p^m}\right).$$

Then this is a half-measure (on $\mathbb{Z}_p^*$ taking values in $\mathbb{C}$ and fin.-additive).

Consider the periods of the form:

$$\tau_k(\varphi) = \int_0^{i\infty} \varphi(z) z^k\, dz, \qquad 0 \le k \le w.$$

$$L_\varphi(s) = \sum_{n=1}^{\infty} \lambda_n n^{-s} = \frac{(2\pi)^s}{\Gamma(s)} \int_0^\infty \varphi(iy) y^{s-1}\, dy$$

$$\tau_k(\varphi) = \frac{k!\, i^{k+1}}{(2\pi)^{k+1}} L_\varphi(k+1)$$

$[1, w+1]$ — crit. strip (funct. eq. relative to it)

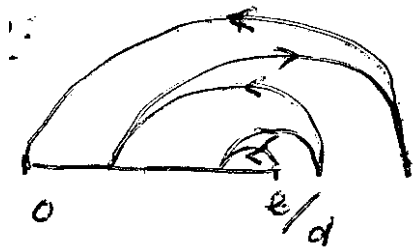
Proposition. $\forall x \in \mathbb{Q},\ P(x) \in \sum_{k=0}^{w} \mathbb{Z} \cdot \tau_k(\Phi)$

Re and Im have a common transcend. factor. Dividing by it gives algebraic

$$\int_{b/d}^{i\infty} \Phi(z)\, dz = \int_0^{i\infty} - \int_0^{b/d}$$

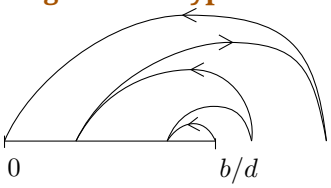
(in neighbourhoods of points of the axis, integrate along geodesics).

Contour:




Original source details: GELFAND-P000509

**Diagram and typed description.**



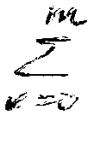

The inner arcs partly overlap. The direction of the fourth arc from the top is unresolved.

from the cont. fraction exp.

$$\frac{b}{d} = \frac{b_m}{d_m}, \quad \frac{b_{m-1}}{d_{m-1}}, \quad \dots, \quad \frac{b_0}{d_0} = \frac{0}{1}$$

$$g_k = \begin{pmatrix} b_k & (-1)^{k-1} b_{k-1} \\ d_k & (-1)^{k-1} d_{k-1} \end{pmatrix} \in SL(2, \mathbb{Z}). \qquad \text{(choose signs).}$$

$$\begin{aligned} \int_0^{b/d} \varphi(z)\, dz &= \sum_{k=0}^{m} \int_{b_{k-1}/d_{k-1}}^{b_k/d_k} \Phi(z)\, dz \\ &= \sum_{k=1}^{m} \int_{g_k(0)}^{g_k(i\infty)} \varphi(z)\, dz \\ &= \sum_{k=1}^{m} \int_0^{i\infty} \varphi(z) \big(d_k z + (-1)^{k-1} d_{k-1}\big)^w\, dz. \end{aligned}$$



**Manuscript excerpt.** The first sum reads $k = 0$. The next two read $k = 1$.

Original source details: GELFAND-P000509

This is an integral lin. comb. of $\tau_k(\varphi)$.

Summary:

$$P\left(\frac{b}{d}\right) = \sum F(d_k, d_{k-1})$$

$F$ — polynomial (form) of deg. $w$

Hence the arithmetical char.

$$P\left(\frac{b}{d}\right) = \sum_{k=0}^{w} A_k \left(\frac{b}{d}\right) \tau_k(\varphi)$$

— separating out the irrationalities

In this expr. addit. is unclear...



**Manuscript excerpt.** The abbreviation before "addit." reads "выраж." in Russian.

Original source details: GELFAND-P000509

$$\begin{aligned} k \text{ — even} \quad &\Rightarrow \quad \tau_k(\varphi) \text{ purely imaginary} \\ k \text{ — odd} \quad &\Rightarrow \quad \tau_k(\varphi) \text{ real} \end{aligned}$$

GELFAND-P000510



**Uncertain reading.** The mark in the upper right has two strokes joined by a third. It may read "II" or Cyrillic "И".

Original source details: GELFAND-P000510

Proposition.

$$\exists\, \omega^+, \omega^- \in \mathbb{C}, \qquad \left(\frac{\tau_0}{\omega^+}, \frac{\tau_2}{\omega^+}, \ldots, \frac{\tau_w}{\omega^+}\right), \quad \left(\frac{\tau_1}{\omega^-}, \ldots, \frac{\tau_{w-1}}{\omega^-}\right) \quad -$$

— algebraic (from the field containing the coeffs. of $\Phi$)

Proof.

Linear eqs. for $\tau_k(\Phi)$.

a) Eichler — Shimura; depend only on $w$:

$$SL(2,\mathbb{Z})/(\pm 1). \qquad \text{Consider } s = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad t = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{(generators).}$$

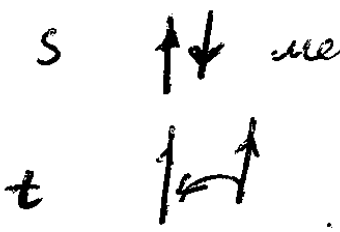
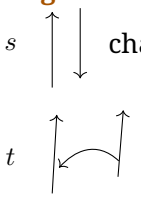



**Diagram and typed description.**




Original source details: GELFAND-P000510

$$\left(\int_0^{i\infty} + \int_{s(0)}^{s(i\infty)}\right) \Phi(z)\, dz = 0 = \left(\int_0^{i\infty} + \int_{t(0)}^{\ldots} + \ldots\right)$$

$$\tau_k(\Phi) + (-1)^k \tau_{w-k}(\varphi) = 0 \qquad \text{(also follows from the funct. eq.)}$$

$$\int_0^{i\infty} \varphi(z) \left[z^k + z^{w-k}(z-1)^k + (-1)^k (z-1)^{w-k}\right] dz = 0.$$

This gives the sought lin. eq. for $\tau_k$.

$S_{w+2} = R$-space of forms $\varphi$ with real coeffs.

$$S_{w+2} \longrightarrow R$$

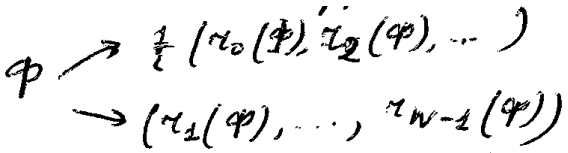
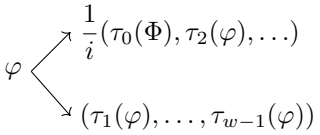
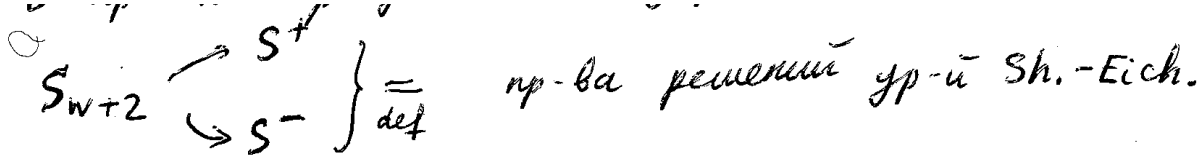


**Diagram and typed description.**




Original source details: GELFAND-P000510

In the period spaces take the subspaces given by the eqs.




Original source details: GELFAND-P000510

**Diagram and typed description.** $S_{w+2} \begin{matrix} \nearrow S^+ \\ \searrow S^- \end{matrix} \Big\} \underset{\text{def}}{=}$

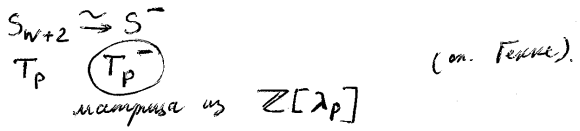
spaces of solutions of the eqs. of Sh.-Eich.

Theorem.

a) $S_{w+2} \xrightarrow{\sim} S^-$

b) $S_{w+2} \hookrightarrow S^+$ embedding codim 1.

Theorem.



Original source details: GELFAND-P000510

**Diagram and typed description.**

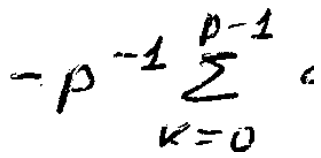
$S_{w+2} \xrightarrow{\sim} S^-$

$T_p$ $T_p^-$ (Hecke op.).

matrix over $\mathbb{Z}[\lambda_p]$

Here

$$(\Phi \mid T_p)(z) = p^{w+1}\varphi(pz) - p^{-1}\sum_{k=0}^{p-1}\varphi\left(\frac{z+k}{p}\right)$$



Original source details: GELFAND-P000510

**Manuscript excerpt.** A minus sign is written before $p^{-1}\sum$.

Eigenfns. of the Hecke op. $\Leftrightarrow$ [Crossed out:



Original source details: GELFAND-P000510

**Undeciphered passage.** The reading under the heavy strikeout is unresolved.

] factor into an Euler prod.

$T_p^-$ — expression in terms of periods (the lin. form is uniquely det.)

Application.

$$w = 10, \qquad S_{12} = R \cdot e^{2\pi iz}\prod_{n=1}^{\infty}(1-e^{2\pi inz})^{24}$$

$$= R\left(\sum_{n=1}^{\infty}\tau(n)e^{2\pi inz}\right)$$

(Ramanujan fn.)

Ramanujan identity

$$\tau(n) \equiv \sum_{d|n} d^{11} \pmod{691}$$

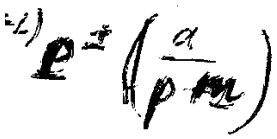
GELFAND-P000511

Measures.

$$P^+(x) = \frac{i}{\omega^+} \operatorname{Im} \int_x^{i\infty} \varphi(z)\, dz,$$

$$P^-(x) = \frac{1}{\omega^-} \operatorname{Re} \int_x^{i\infty} \Phi(z)\, dz$$

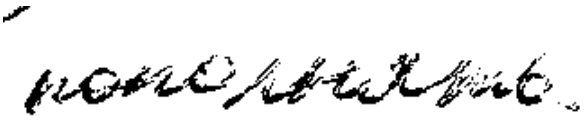
$$\mu_\varphi^\pm\left(a + (p^{m+1})\right) \underset{\text{def}}{=} \varrho^{-m} P^\pm\left(\frac{a}{p^{m+1}}\right) - p^w \varrho^{-(m+1)} P^\pm\left(\frac{a}{p^m}\right)$$



**Uncertain reading.** The $P$ and the exponent on $p$ in the last term are overwritten. Provisionally read as $P^\pm(a/p^m)$.

Original source details: GELFAND-P000511

measures with alg. values, and one can complete.



**Uncertain reading.** The last Russian verb is provisionally read “пополнять”. Another possible reading is “пополнить”.

Original source details: GELFAND-P000511

Growth. $P^\pm$ have bounded denominators. They may arise from $\varrho$.

$$\varrho^2 - \lambda_p \varrho + p^{w+1} = 0$$

$$\begin{cases} \varrho_1 \cdot \varrho_2 = p^{w+1} \\ \varrho_1 + \varrho_2 = \lambda_p. \end{cases}$$

Typ. case: $\lambda_p$ copr. to $p \Rightarrow$ one root gives a bounded measure, the other a growing one!
Unknown relation [crossed out:



**Undeciphered passage.** The reading between “relation” and the next symbol is unresolved.

Original source details: GELFAND-P000511

] $\lambda_p$ and $p$.

Integrals (Mellin transf. of $p$-adic char.)

$$\dots \int_{\mathbb{Z}_p^*} \chi \, d\mu_\Phi^\pm$$

Manin can do it only for a char. of finite ord. $\chi : \mathbb{Z}_p^* \to K^*$, $\chi^m = 1$ and primitive.

Theorem

$$\dots \int_{\mathbb{Z}_p^*} \chi \, d\mu_\varphi^\pm = \dots L_\varphi(1, \chi),$$

where

$$L_\Phi(s, \chi) = \sum_{n=1}^{\infty} \chi(n) a_n n^{-s}$$

$$\dots \dot{L}_\varphi(1, \chi) = \dots \int_{R_+^*} \varphi_\chi(z) \, dz, \qquad \text{where } \Phi_\chi = \sum \chi(n) a_n e^{2\pi i n z}.$$



**Uncertain reading.** A separate dot is visible above $L$ in the manuscript. Its meaning is unresolved.

Original source details: GELFAND-P000511

On Lichtenbaum's conjectures: on the connection of $K$-groups of alg. fields with $\zeta$-fns. at int. points.

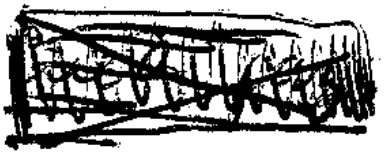
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \qquad \sum_{n=1}^{\infty} \frac{1}{n^3} = 2\pi^2 |K_4\mathbb{Z}| \cdot e_5(\alpha_5)$$

For the first time there is a conjecture

$\alpha_5$ — a generator of $K_5\mathbb{Z}$. (of the free part)

$e_5$ — a homom. into the real numbers (a certain one)

It is unproved that $\sum_1^{\infty} \frac{1}{n^3}$ is transcend. Euler's const. also unproved.



**Undeciphered passage.** The text inside the rectangle is heavily crossed out. Its reading is unresolved.

Original source details: GELFAND-P000511

GELFAND-P000512

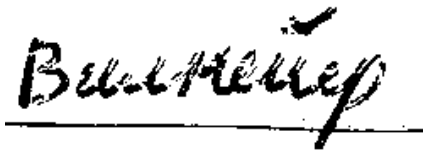


**Uncertain reading.** Clipped strokes are visible at the upper right. The reading is unresolved.

Original source details: GELFAND-P000512

24.9.73. Gelfand Sem.

Pincus Douglas, Coburn, Vilneyer and Singer Helton and Roger Howe. (Howe)



Original source details: GELFAND-P000512

**Uncertain reading.** The surname after Coburn is provisionally transcribed “Вилнейер” and romanized “Vilneyer”. The middle letters run together.

(interesting work in funct. analysis).

Theorem (Helton, How). close to the work of Krein, Gohberg, Simonenko, etc.

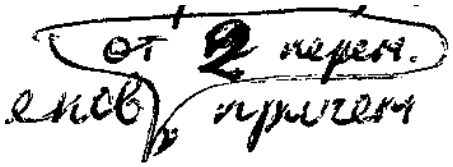
Let $B$ — a bilin. form on 2 polynomials [inserted above the line: of 2 var.], with

$$B(P_1(Q), P_2(Q)) = 0,$$

if $P_1(z), P_2(z)$ — polyn. in one var. $z$, and $Q$ — a polyn. in any number of var. Then

$$B(R, S) = L(\{R, S\}), \qquad \{R, S\} \text{ — Poisson bracket}$$

$L$ — lin. form.



Original source details: GELFAND-P000512

**Manuscript excerpt.** The insertion “of 2 var.” is circled and points to the place after “polynomials”.

— · —

Let $U, V$ — self-adj. and commuting mod nuclear operators. Then

$$\mathrm{Tr}\{[R(U,V), S(U,V)]\} = \iint \{R, S\}\, d\sigma$$

$R, S$ — polynomials. The order of $U$ and $V$ in them is immaterial.

— · —

Book by Douglas Banach algebras technique.
$\Rightarrow$ in the $p$-adic case suitably chosen normed rings are needed in place of diff. operators.

— · —

Kirillov.
L. Brown, R. Douglas, P. Fillmore.
Extensions of Banach algebras (of a nontraditional type).
$\mathcal{K}$ — alg. of comp. operators on a sep. Hilbert space.
Def. 1.

$$0 \to \mathcal{K} \to \mathcal{A} \to C(X) \to 0, \qquad X \text{ — compact}$$

Exts. form a group $\mathrm{Ext}(X) = \mathrm{Ext}_1(X)$.
Another definition gives the same group:
Def. 2.

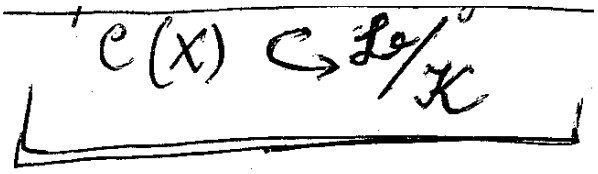


**Diagram and typed description.**

$$C(X) \hookrightarrow \mathcal{L}/\mathcal{K}$$

Original source details: GELFAND-P000512

1) consider all embeddings.
2) identify embeddings differing by a unit. op. (unit. eq. $UAU^{-1}$).
3) Embeddings of the form $C(X) \to \mathcal{L} \to \mathcal{L}/\mathcal{K}$ are regarded as triv.
Equiv. classes then form a group.

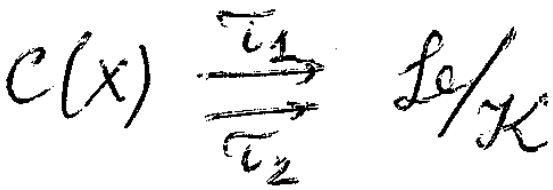


**Diagram and typed description.**

$$C(X) \underset{\tau_2}{\overset{\tau_1}{\rightrightarrows}} \mathcal{L}/\mathcal{K}$$

Original source details: GELFAND-P000512

$\tau_1 \oplus \tau_2$ already in $H_1 \oplus H_2 \cong H$

Functor

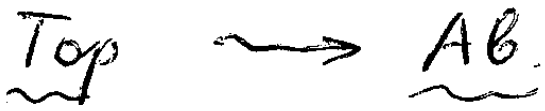
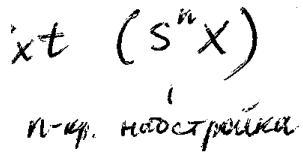


**Diagram and typed description.**
Top ⟶ Ab.

Original source details: GELFAND-P000512

Main result: the functor $\mathrm{Ell}(X)$ reduces to $\mathrm{Ext}(X)$ (answer to a question of Atiyah).

$$\mathrm{Ext}_{1-n}(X) = \mathrm{Ext}\left( \begin{array}{c} S^n X \\ | \\ n\text{-fold suspension} \end{array} \right)$$



**Manuscript excerpt.** A short stroke under $S^n X$ points to the label.

Original source details: GELFAND-P000512

GELFAND-P000513

[source number: 14]

Th. 1. Exact seq.

$$\mathrm{Ext}_1(A) \to \mathrm{Ext}_1(X) \to \mathrm{Ext}_1(X/A) \to \mathrm{Ext}_0(A) \to \dots$$

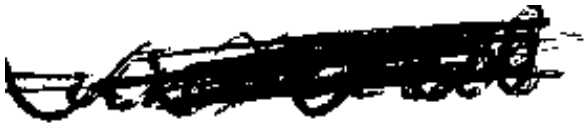
$A \subset X$

Th. 2. There is a pairing

$$K^1 \times \mathrm{Ext}_1 \to \mathrm{Ext}_0$$

$$K^1(X \times Y) \times \mathrm{Ext}_1(X) \to K^0(Y)$$

~~The functor $\mathrm{Ext}_1$ is computed for all~~



**Undeciphered passage.** The end of the crossed-out line is illegible.

Original source details: GELFAND-P000513



**Uncertain reading.** The crossed-out verb is provisionally read as “computed”.

Original source details: GELFAND-P000513

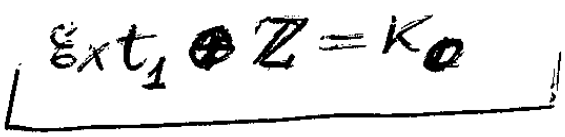


**Diagram and typed description.**
$\mathrm{Ext}_1 \oplus Z = K_0$

Original source details: GELFAND-P000513

for simpl. complexes.



Original source details: GELFAND-P000513

**Uncertain reading.** The sign before $Z$ is heavily retraced. Provisionally set as $\oplus$. It could be $\otimes$.

Example of an extension.

$$0 \to \mathcal{K} \to C^*(\text{„}ax + b\text{”}) \to C(\infty) \to 0$$

$$\text{Ext}(\infty) = Z + Z.$$

What is this pair of numbers for a specific op.?

— · —

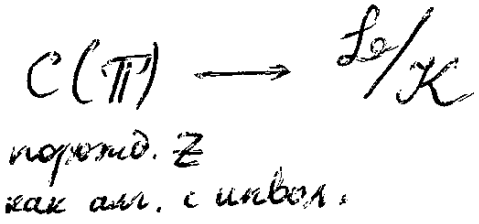
~~$X = S^1$~~



Original source details: GELFAND-P000513

**Manuscript excerpt.** $X = S^1$ is crossed out on the left.

$$X = \mathbb{T} = S^1$$



Original source details: GELFAND-P000513

**Diagram and typed description.**

$C(\mathbb{T}) \longrightarrow \mathcal{L}/\mathcal{K}$

gen. by $z$
as an alg. with invol.

Let $T$ — the image of $z$.

$$TT^* = 1 \bmod \mathcal{K}$$

up to unit. equiv. and mod $1 + \mathcal{K}$.

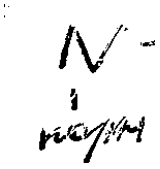
Answer: index (classific.).

— · —

Th.

$$\left\{ \begin{matrix} N \\ | \\ \text{norm.} \end{matrix} \quad + \mathcal{K} \right\}$$

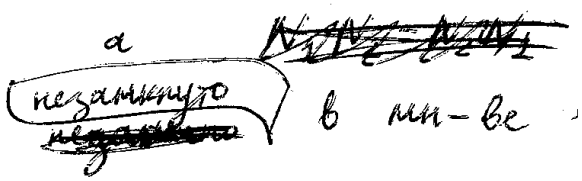
is norm-closed among all op.,



**Manuscript excerpt.** A short stroke connects $N$ to the label “norm.” below.

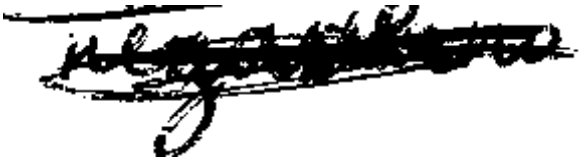
Original source details: GELFAND-P000513

but ~~$N_1N_2 = N_2N_1$~~ pairs $\{N_1 + K_1,\ N_2 + K_2\}$ with cond. $N_1N_2 = N_2N_1$ [inserted and circled at the left: not closed] in the set of all pairs of this kind.



**Manuscript excerpt.** The equality is crossed out before “pairs”. The insertion at the left is circled.

Original source details: GELFAND-P000513



**Uncertain reading.** The crossed-out word below the circle is provisionally read as “not closed”.

Original source details: GELFAND-P000513

Next time: Fuchs's talk pres. on Mishchenko.

Fuchs. Quillen Higher algebraic K-theory I

(Proc. of symp. on $K$-theory)

Introduction. $M$ — a category (abelian or spec. additive). We define $K_i(M)$.

Constructions

$QM$ — cat. of quotients $\Rightarrow$ classif. space $BQM \Rightarrow$

$$\Rightarrow \quad K_i(M) = \pi_{i+1}(BQM).$$

Essential part.

A simpl. object in a category $S$ — a contravar. functor Ord $\to S$, where Ord — the can. category of nonnegative integers.

$$\mathrm{Ord}^{\circ} \longrightarrow S$$



**Uncertain reading.** There is a small oval above Ord. Provisionally set as $\circ$. It could also be $o$ or $0$.

Original source details: GELFAND-P000514

(Functor always means covar. functor).

Ord objects — $0, 1, 2, 3, \ldots$

Morphism [crossed out:



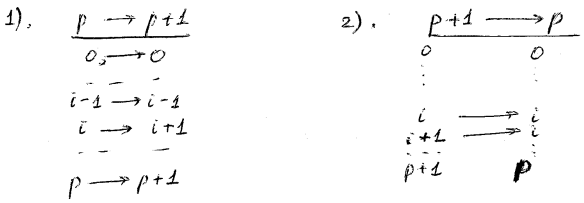
**Undeciphered passage.** The reading under the strikeout is unresolved.

Original source details: GELFAND-P000514

] $p \to q$ is called a monotonically nondecr. map

$$(0, 1, \ldots, p) \longrightarrow (0, 1, \ldots, q)$$

All are repr. as compositions of morphisms of 2 kinds:



Original source details: GELFAND-P000514

**Diagram and typed description.**

$$\begin{array}{cccc} & \underline{p \longrightarrow p+1} & & \underline{p+1 \longrightarrow p} \\ & 0, \longrightarrow 0 & & 0 \quad\quad 0 \\ & \vdots \quad\quad \vdots & & \vdots \quad\quad \vdots \\ 1). & i-1 \longrightarrow i & & i \longrightarrow i \\ & i \longrightarrow i+1 & & i+1 \longrightarrow i \\ & \vdots \quad\quad \vdots & & \vdots \quad\quad \vdots \\ & p \longrightarrow p+1 & & p+1 \quad\quad p \end{array}$$

To specify a simpl. obj. it suffices to give a seq. of objects

$$S_0, S_1, S_2, \ldots$$

and morphisms

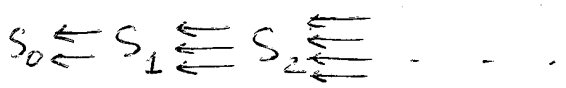


**Diagram and typed description.**

$S_0 \leftleftarrows S_1 \;\substack{\longleftarrow\\ \longleftarrow\\ \longleftarrow}\; S_2 \;\substack{\longleftarrow\\ \longleftarrow\\ \longleftarrow\\ \longleftarrow}\; \cdots$

Original source details: GELFAND-P000514

(correspond to embeddings of type 1))

and



**Diagram and typed description.**

$S_0 \longrightarrow S_1 \;\substack{\longrightarrow\\ \longrightarrow}\; S_2 \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; S_3 \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \cdots$

Original source details: GELFAND-P000514

(corr. to maps of type 2))

(The relations must be satisfied).

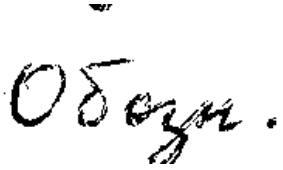
they are easy to write down

$S$ — either the cat. of sets or the cat. of top. spaces.

Groups, smooth manifolds, etc. are also possible.

Important example: sing. complex of a top. space. $S$ — cat. of sets.

$X$ — top. space. Denote by $S_q$ — all sing. simplices of $X$

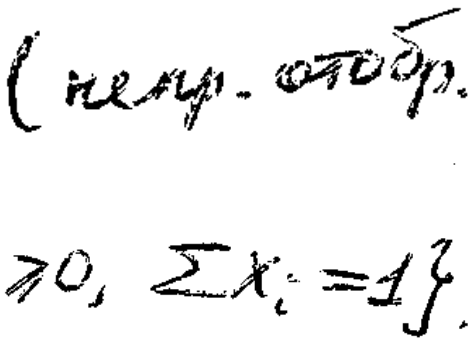


Original source details: GELFAND-P000514

**Manuscript excerpt.** The Russian abbreviation reads “Обозн.”.

(cont. maps $\Delta^q \to X$, where

$$\Delta^q = \{(x_1, \ldots, x_{q+1}) \in R^{q+1}, \quad x_i \geq 0, \quad \sum x_i = 1\}.$$

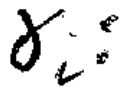


Original source details: GELFAND-P000514

**Manuscript excerpt.** An opening parenthesis precedes “cont. maps”. There is no visible matching parenthesis before the next formula.

$$\exists\ \delta_i : \Delta^{q-1} \longrightarrow \Delta^q$$

canon. map onto the $i$-th face (vertices ordered)



Original source details: GELFAND-P000514

**Uncertain reading.** The letter with index $i$ is provisionally set as $\delta_i$. It could be $\gamma_i$.

Sing. simplex — composition

$$\Delta^{q-1} \longrightarrow \Delta^q \longrightarrow X$$



Original source details: GELFAND-P000514

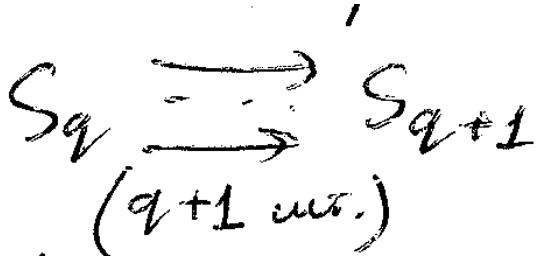
**Uncertain reading.** The initial Russian abbreviation is provisionally read as “Син.”. Its last letter is uncertain.

We get $q+1$ maps



Original source details: GELFAND-P000514

**Diagram and typed description.**
$S_q \overset{\longrightarrow}{\underset{\longrightarrow}{\cdots}} S_{q-1}$ (faces).



Original source details: GELFAND-P000514

**Diagram and typed description.**
$S_q \overset{\longrightarrow}{\underset{\longrightarrow}{\cdots}} S_{q+1}$
($q+1$ in all)

from $\Delta^{q+1} \longrightarrow \Delta^q$ affine projections ($i$-th identifies the $i$-th vertex with the $(i+1)$-st).

— · —

GELFAND-P000515

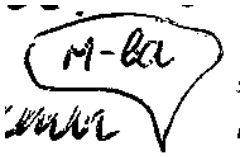
[source number: 15]

Realization.

Let $S$ — the cat. of sets or top. spaces.

A top. space is associated with a simpl. obj. in the cat. of sets or top. spaces.

Elements of [inserted above the line, circled: the set] $S_i$ are called simplices

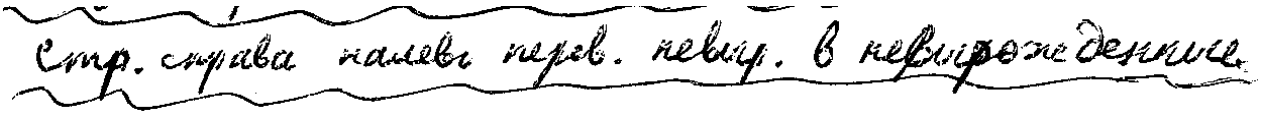


Original source details: GELFAND-P000515

**Manuscript excerpt.** The circled abbreviation for “set” points to the place after “Elements”.

The images of arrows from left to right are called degenerate simplices.

The others are called nondegenerate. Arrows from right to left take nondeg. to nondegenerate simplices.



Original source details: GELFAND-P000515

**Manuscript excerpt.** The manuscript states that the right-to-left arrows take nondeg. to nondegenerate simplices.

Consider $\Delta^q \times S^0_q$

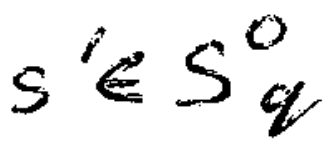
nondeg. simpl. from $S_q$. ($S^0_q$ — a discrete top. space).

In $\coprod_q \Delta^q \times S^0_q$ (disjoint union) an identification.

$$\Delta^{q-1} \times s, \qquad s \in S^0_{q-1}$$

is identified with part of $\Delta^q \times s'$, $s' \in S^0_q$, when $s$ — the image of $s'$ under the map ($i$-th)

$$S_{q-1} \xleftarrow{i} S_q \qquad (i\text{-th face})$$



**Manuscript excerpt.** A prime is visible in the membership condition: $s' \in S^0_q$.

Original source details: GELFAND-P000515

We get a top. space. This is a cell space (the interiors are left unglued).

But: boundary points can be glued together (so it is not a simplicial space).

The same ex. if $S$ — top. spaces. A top. struct. no longer arises, since in $\Delta^q \times S^0_q$ there is the direct product top.



Original source details: GELFAND-P000515

**Manuscript excerpt.** The manuscript reads “A top. struct. no longer arises”.

Ex.: for a good top. space the realization of the sing. [crossed out:



**Undeciphered passage.** The reading under the strikeout is unresolved.

Original source details: GELFAND-P000515

] complex is homotopy equiv. to $X$.

To reconstr. a polyhedron, take for $S_i$ the aff. maps of simplices into it



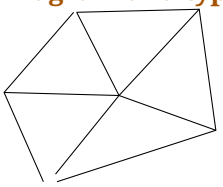
**Manuscript excerpt.** The Russian abbreviation before “polyhedron” reads “восстан.”.

Original source details: GELFAND-P000515

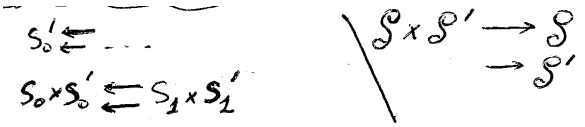


Original source details: GELFAND-P000515

**Diagram and typed description.**



Realization commutes with multiplication.

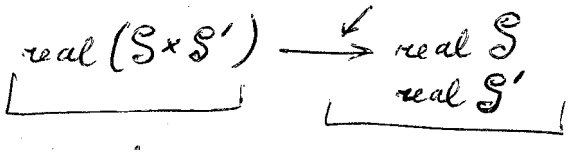


Original source details: GELFAND-P000515

**Diagram and typed description.**

$$\begin{array}{ccc} S_0' \longleftarrow \dots & & \mathcal{S}\times\mathcal{S}' \longrightarrow \mathcal{S} \\ & & \qquad\quad \longrightarrow \mathcal{S}' \\ S_0\times S_0' \longleftarrow S_1\times S_1' & & \end{array}$$

hence

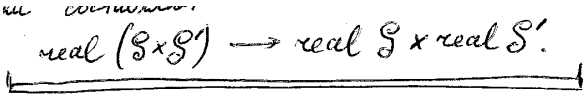


Original source details: GELFAND-P000515

**Diagram and typed description.**

$$\begin{array}{ccc} \operatorname{real}(\mathcal{S}\times\mathcal{S}') & \longrightarrow & \operatorname{real}\mathcal{S} \\ & & \operatorname{real}\mathcal{S}' \end{array}$$

and together they form



Original source details: GELFAND-P000515

**Diagram and typed description.**

$$\operatorname{real}(\mathcal{S}\times\mathcal{S}') \longrightarrow \operatorname{real}\mathcal{S}\times\operatorname{real}\mathcal{S}'.$$

Prop. It turns out to be — a homeomorphism.

Example.



Original source details: GELFAND-P000515

**Diagram and typed description.** ⟼

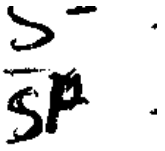
Ordered segment.

$S^0$ — 2 elements

$S^1$ — 3 elem. (counting degenerate ones)

$S^p = (p+2)$ elem. $i$-th:

$$[0, i-1] \longrightarrow 0, \qquad [i, p] \longrightarrow 1$$



Original source details: GELFAND-P000515

**Uncertain reading.** The raised sign to the right of $S$ is heavily retraced. Provisionally set as $p$. A short stroke appears above $S$ on the left.

$S_p \times S_p$



**Uncertain reading.** Faint detached strokes follow the product. Their reading is unresolved.

Original source details: GELFAND-P000515

(next time.)

GELFAND-P000516

§2. Classif. space in a category.

$C$ — a category (small). Associated to it is its nerve $NC$ (simpl. object of the cat. of sets).

$$NC_q = \{[?] \quad X_0 \longrightarrow \cdots \longrightarrow X_q\} \qquad \text{set of such diagrams}$$



**Uncertain reading.** An overwritten mark is visible after the opening brace before $X_0$. The reading is unresolved.

Original source details: GELFAND-P000516

$$NC_q \longrightarrow NC_{q-1}$$

0th

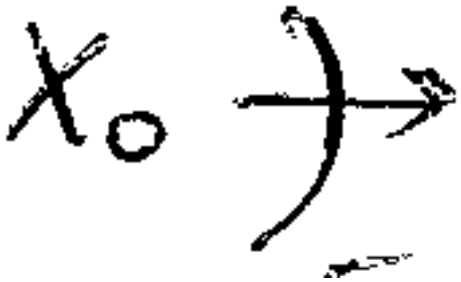


**Diagram and typed description.** $X_0$ (discard it).

Original source details: GELFAND-P000516

$q$th — discard $X_q$

$i$th — compose the $i$th and $(i+1)$st arrows

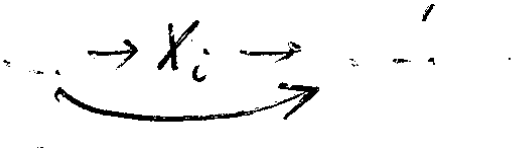


**Diagram and typed description.**

$$\cdots \longrightarrow X_i \longrightarrow \cdots$$

Original source details: GELFAND-P000516

$$NC_q \longrightarrow NC_{q+1}$$

one of the $X$ is doubled and the id. morphism is inserted.

~~Nerve~~ Real $NC = BC$ — classif. space of the category.



**Uncertain reading.** The crossed-out word is provisionally read as “Нерв.” (“Nerve”).

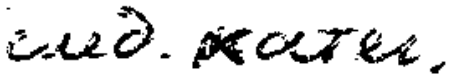
Original source details: GELFAND-P000516

Example: the classif. space of the group $G$ is obtained from the following cat.:



Original source details: GELFAND-P000516

**Undeciphered passage.** The reading under the strikeout after the colon is unresolved.



Original source details: GELFAND-P000516

**Manuscript excerpt.** The Russian abbreviations after “from” read “след. катег.”.

1 object, with morphisms the elements of $G$.

$$NC_q = \underbrace{G \times \cdots \times G}_{q}$$

— · —

$G = Z_2$. Simplices:

Nondeg. simpl. — do not contain 1.

One nondeg. simpl. of each dimension.



Original source details: GELFAND-P000516

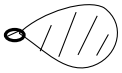


**Diagram and typed description.** Three long strokes and one short stroke in an oval. Dark overlapping marks at the left edge.

Ex. Wind the 2-dimensional one twice. One gets $RP_\infty$.



Original source details: GELFAND-P000516

**Manuscript excerpt.** The manuscript reads “2-мерную намот. дважды”.



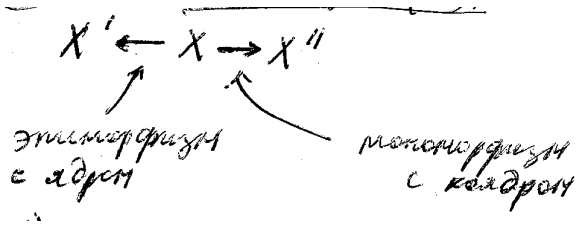
Original source details: GELFAND-P000516

**Manuscript excerpt.** The phrase ends with “дважды” (“twice”).

— · —

Let $M$ — a category that is either abelian or a full subcat. of an abelian cat., closed under extensions. Suppose the isom. classes of objects form a set. Then it is equivalent to a small category. Category $QM$. $\mathrm{Ob}(QM) = \mathrm{Ob}(M)$.

$$\mathrm{Hom}_{QM}(X', X'') \Rightarrow \text{diagrams}$$




Original source details: GELFAND-P000516

**Diagram and typed description.**

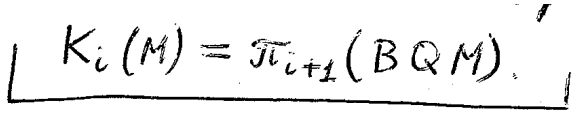
$X' \longleftarrow X \longrightarrow X''$

epimorphism with kernel

monomorphism with cokernel

This is a category.

Definition.




Original source details: GELFAND-P000516

**Diagram and typed description.**

$K_i(M) = \pi_{i+1}(BQM).$

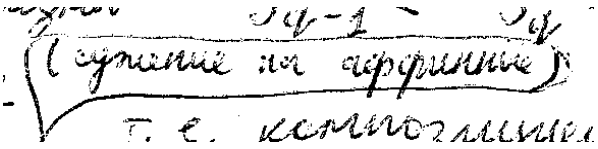
GELFAND-P000517

Continuation of Fuchs's talk (following Marchenko).

Example of realization

$S_q$ consists of $q+2$ elements. Interpret it as the set of aff. maps from $\Delta^q$ into the segment $\Delta^1$, compatible with the ordering of the vertices of $\Delta^q$ and $\Delta^1$.

The morphisms $S_{q-1} \leftarrow S_q \rightarrow S_{q+1}$ are def. as for the singular simplices of the space $I = \Delta^1$ [inserted above the line: (restriction to affine maps)], i.e. by composition with an embedding and restrictions.




Original source details: GELFAND-P000517

**Manuscript excerpt.** The circled insertion "restriction to affine maps" points to the place after the formula.

Nondegenerate simplices — 2 0-dimensional $* \to I$ and 1 1-dimensional one.

$S_q \times S_q$ when is an element of $S_q \times S_q$ nondeg.?

Elem. — a pair $(\varphi, \psi)$, it is nondeg. when there are no 2 vertices in $\Delta^q$, mapped by $\varphi$ and $\psi$ [crossed out:



Original source details: GELFAND-P000517

**Undeciphered passage.** The reading of the crossed-out sign is unresolved.

] into one [crossed out:



**Undeciphered passage.** The reading of the crossed-out word is unresolved.

Original source details: GELFAND-P000517

] vertex.

$$S_0 \times S_0 \quad \text{— 4 elems.} \qquad * \to (0,0),(0,1),(1,0),(1,1)$$

$$S_1 \times S_1 \qquad \begin{pmatrix}1\\0\end{pmatrix} \to \begin{matrix}\varphi & \psi\\ 0 & 0\\ 0 & 1\end{matrix} \qquad \begin{pmatrix}0 & 1\\ 0 & 1\end{pmatrix} \text{degenerate}\Big)$$

$$\begin{pmatrix}0 & 0\\ 1 & 1\end{pmatrix}\begin{pmatrix}0 & 0\\ 1 & 0\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 1\end{pmatrix}\begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}$$



**Uncertain reading.** A stroke resembling a closing parenthesis is crossed out after the second column, before “degenerate”.

Original source details: GELFAND-P000517

$$S_2 \times S_2 \qquad \Delta \to \begin{matrix}0 & 0 & 0 & 0\\ 1 & 0 & 0 & 1\\ 1 & 1 & 1 & 1\end{matrix}$$



Original source details: GELFAND-P000517

**Manuscript excerpt.** The manuscript has four columns with no dividers.

Realization of $S \times S$.

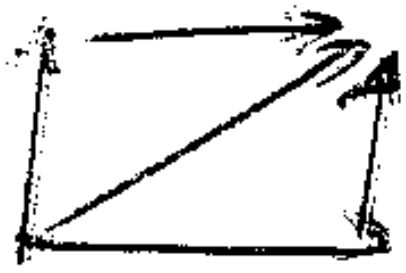
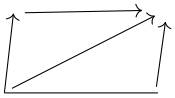


**Diagram and typed description.** 

Strokes overlap at the right end of the lower side. Its direction is unresolved.

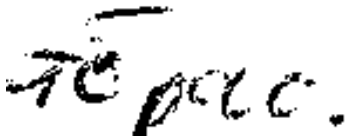
Original source details: GELFAND-P000517

Classif. space of a category.

$C$ — a small cat., $NC$ — the nerve of the category

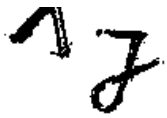
$(NC)_q$ $(= S_q)$ — all diagrams in $C$ of the form $X_0 \to \ldots \to X_q$

the maps — taking compositions, discarding endpoints and inserting id $X_j$.



Original source details: GELFAND-P000517

**Uncertain reading.** The Russian abbreviation before “endpoints” is provisionally read “отбрс.”. It could be “отброс.”.



Original source details: GELFAND-P000517

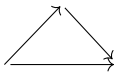
**Uncertain reading.** The subscript on $X$ is provisionally set as $j$. It could be a capital $J$.

$$BC = \text{Real}\, NC.$$

The vertices of $BC$ — objects of $C$, 1-dimensional simplices are nonidentity morphisms. $\Delta$ are stretched over commutative simplices



Original source details: GELFAND-P000517

**Diagram and typed description.** 

and so on.

Barycentr. subdiv. of a polyhedron of high deg. is the nerve of some category.



Original source details: GELFAND-P000517

**Uncertain reading.** The Russian abbreviation is provisionally read “подразд.”. The last letter could be “б”.

Properties:

$$B(C \times C') \cong BC \times BC'$$

$$f : C \rightsquigarrow C' \quad \Rightarrow \quad Bf : BC \to BC'$$

If there is also $g : C \to C'$ and a nat. map from $f$ to $g$, then $Bg : BC \to BC'$ is connected to $Bf$ by a canonical homotopy.

Let $\{0, 1\}$ — a cat. of 2 objects with one morphism $0 \to 1$

A morphism of the functors $f$ and $g$ — a functor from $C \times \{0, 1\}$ to $C'$.

GELFAND-P000518

[source number: 21]

$F : C \times \{0,1\} \to C'$. At the endpoints $f, g$; morph. $0, 1$ — nat. map.
It determines a map, a homotopy connecting $f$ with $g$

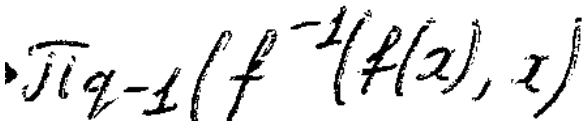
$$BC \times I = B(C \times I) \to BC'.$$

If $C$ has a terminal or an initial object, then $BC$ is contractible —
— it is a cone.
If $f : C \rightsquigarrow C'$ has a left or right adjoint, then $Bf$ — a homotopy equivalence.

$$BC^{\circ} = BC, \qquad B(C \times C') = BC \times BC'.$$

Analogue of the homotopy sequence of a fibration.
Let $f : X \to Y$ — a Serre fibration. Then

$$\ldots \to \pi_q(X, x) \to \pi_q(Y, f(x)) \to \pi_{q-1}(f^{-1}(f(x), x)$$
$$\to \pi_{q-1}(X, x) \to \ldots$$

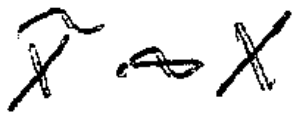


**Manuscript excerpt.** The parentheses follow the manuscript. One closing parenthesis is visible before the comma.

Original source details: GELFAND-P000518

If $f$ does not satisfy the homotopy lifting axiom, then after replacement by an equivalent map it will satisfy it and we get the sequence. Here is the construction:
Let $\widetilde{X} = \{(x, s),\ x \in X,\ s : I \to Y,\ s(0) = f(x)\}$.
$\widetilde{X} \underset{\sim}{\supset} X$. The map $\widetilde{f} : \widetilde{X} \to Y$ is constructed as follows:

$\widetilde{f}(x, s) = s(1), \qquad \widetilde{f}|_X = f$ but now the homotopy lifting axiom holds.
Because $\widetilde{X} \simeq X$ we have an exact sequence.



**Uncertain reading.** The sign between $\widetilde{X}$ and $X$ is provisionally set as $\simeq$. A wavy stroke crosses a straight one in the manuscript.

Original source details: GELFAND-P000518

$$\ldots \to \pi_q(X) \to \pi_q(Y) \to \pi_{q-1}(\text{homot. fibre}) \to \pi_{q-1}(X) \to \ldots$$

Now let $g : C \rightsquigarrow C'$, then we have

$Bg : BC \rightsquigarrow BC'$. We would like to construct a "homotopy fibre" of $Bg$.

This has not been done. There is a surrogate.

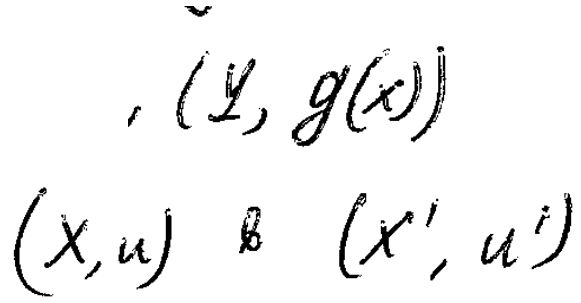
The constr. also depends on some object $Y \in \mathrm{Ob}\, C'$.

Def. the category $Y\backslash g$. $\mathrm{Ob}(Y\backslash g)$ — the pairs $(X, u)$, where

$$X \in \mathrm{Ob}\, C, \qquad u \in \mathrm{Mor}_{C'}(Y, g(x)).$$

A morphism $(X, u)$ to $(X', u')$ — a morphism $\varphi : X \to X'$, compat. with $u$ and $u'$. There is a nat. map

$$B(Y\backslash g) \to \text{homot. fibre } Bg : BC \to BC'$$



**Manuscript excerpt.** The argument of $g(x)$ shows a lowercase $x$. The letters in the pairs in the description of the morphism read as uppercase $X$ and $X'$.

Original source details: GELFAND-P000518

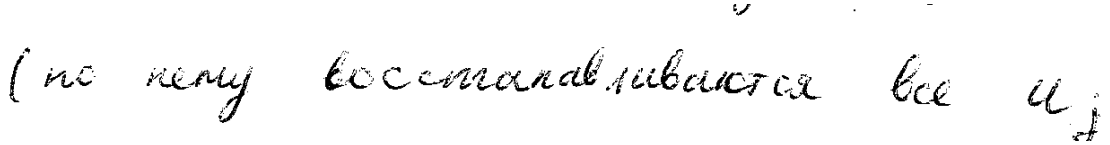
Theorem. Let the functor $g$ be such that

$\forall v \in \mathrm{Mor}_{C'}(Y, Y')$ the resulting map is $Y'\backslash g \xrightarrow{v\backslash g} Y\backslash g$ and $B(v\backslash g) : B(Y'\backslash g) \to B(Y\backslash g)$ is a homot. equivalence (which in general is false). Then $B(Y\backslash g)$ — homot. fibre $BC \overset{Bg}{\rightsquigarrow} BC'$.

GELFAND-P000519

Construct the map $B(Y\backslash g)$ — homot. fibre of $Bg$.

If there is $(X_0, u_0) \leftarrow \cdots \leftarrow (X_\tau, u_\tau)$ (a simplex of $B(Y\backslash g)$) then there is also $X_0 \leftarrow \cdots \leftarrow X_\tau$ and one only needs to know $Y \to g(X_\tau)$ (from it all the following are recovered: $u_j$



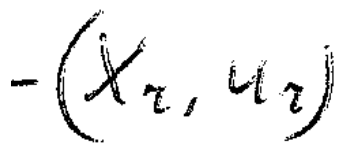
Original source details: GELFAND-P000519

**Manuscript excerpt.** The manuscript has no closing parenthesis after this line.

$X_0 \leftarrow \cdots \leftarrow X_\tau$ — a simplex of $BC$ of dimension $\tau$. Together with $u_\tau$ it defines a simplex

$$g(X_0) \leftarrow \cdots \leftarrow g(X_\tau) \overset{u}{\leftarrow} Y$$

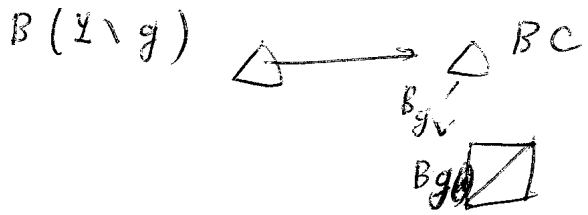
in $BC'$.



**Uncertain reading.** The index is provisionally set as $\tau$. The same letter is used for the dimensions.

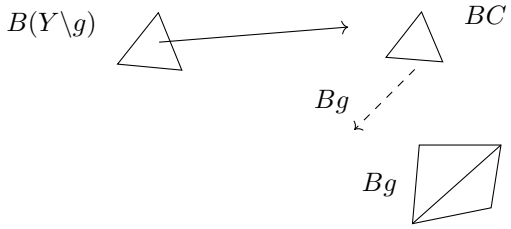
Original source details: GELFAND-P000519

Thus, from a simplex in $B(Y\backslash g)$ we have defined 2 simplices, in $BC$ and in $BC'$ of dimensions $\tau$ and $\tau+1$ resp.. A simplex from $B(Y\backslash g)$ is mapped as follows: the point of the space $BC$ corresponding to that point (in barycentric coord.) and the straight path in the simplex of $BC'$, connecting $Bg$ of this point to $Y$:



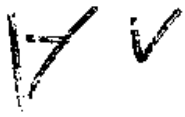
Original source details: GELFAND-P000519

**Diagram and typed description.**




Overlapping strokes follow $Bg$ on the left side of the quadrilateral. Their reading is unresolved.

This is a homot. equiv. $\forall v$ iff the conditions of the theorem hold.



Original source details: GELFAND-P000519

**Manuscript excerpt.** The beginning of the second line to the right of the sketch reads $\forall v$.

Remark by Fuchs: probably $Y\backslash g$ should be def. as the inverse limit over the spectrum of all $Y\backslash g$ with different $Y$.

Mishchenko. (Representations according to Mishchenko)

$\Lambda$ — a ring with involution, $H$ — a Hilbert space and 2 reprs. $T_1$ and $T_2$ of the ring $\Lambda$ into the algebra of bounded op. in $H$, $F$ — a Fredholm operator in $H$ and let

$$\forall \lambda \in \Lambda \qquad FT_1(\lambda) - T_2(\lambda)F \in \text{Comp}(H)$$

This system $\{T_1, T_2, F\}$ is called a Fredholm repr. of $\Lambda$.

Atiyah considered $\Lambda = C(M)$, $M$ — a manifold (symposium in Tokyo).



Original source details: GELFAND-P000519

**Manuscript excerpt.** The parenthetical in the manuscript reads “symposium in Tokyo”.

Constr. with these representations

1) Atiyah (1970) $\Lambda = C(M)$ $\quad \forall$ repr., assign index $T \in K_0(M)$

2) $\Lambda = \mathbb{Z}[\pi] \quad \Rightarrow \quad \forall$ repr., construct $\xi_T \in K^0(B\pi)$.

3) $\Lambda$ — arbitrary, $\quad A = (a_{ij})$ — a quadr. form on $\Lambda^n \quad (a_{ij} \in \Lambda)$

$$A^* = A, \quad \exists A^{-1} \quad \Rightarrow \quad \text{Construct sign}_T A \in \mathbb{Z}.$$



Original source details: GELFAND-P000519

**Undeciphered passage.** Crossed-out number before “1)”. Possibly “4)”.

1) $\Lambda = C(M)$, $T$ — a Fredh. repr. of $\Lambda$ in $H$, $\quad T = (T_1, T_2, F)$.

$H_i$ — modules over $\Lambda$. ind $F$ — a number. Let $\eta$ — a bundle, $\Gamma(\eta)$ — a proj. module over $C(M)$. Consider

GELFAND-P000520

[source number: 22]

$$H_1 \otimes_{C(M)} \Gamma(\eta) \xrightarrow{F \otimes 1_\eta} H_2 \otimes_{C(M)} \Gamma(\eta)$$

$$\text{ind}(F \otimes 1_\eta) : K^0(M) \to \mathbb{Z}, \quad \text{i.e. an element of } K_0(M)$$

This is somewhat crude because $\exists$ elements of finite order in $K^0(M)$.

More precisely: $T$ determines $\text{index}_T : K^0(M \times Y) \to K^0(Y)$; $\quad \forall Y$. This is already an element of $K_0(M)$.

If $\sigma(A) \in K^0(T^*M)$, $\quad A : H \to H$, $\quad$ since $[f, A]$ — compact, there is



**Undeciphered passage.** The short crossed-out group of signs before “index” cannot be read.

Original source details: GELFAND-P000520

index$(A) \in K_0(M)$.



**Uncertain reading.** The stroke after the commutator has been retraced. It is provisionally set as a dash.

Original source details: GELFAND-P000520

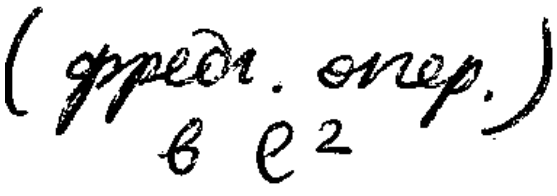
Thus these elems. are Poincaré dual to one another $K^0(T^*M) \underset{\mathcal{D}}{\sim} K_0(M)$. Then

index $A = \mathcal{D}(\sigma(A))$

(NB one can constr. the direct image of $A$)

2) $\pi$ — an arbitrary discr. group, $\Lambda = \mathbb{Z}[\pi]$ — its group ring, $B_\pi$ —
— the classif. space. $T$ — a Fredh. repr.: $T = (T_1, F, T_2)$. We need to constr. $\xi_T \in K(B_\pi)$.

$K(B_\pi)$ is understood as $[B_\pi, BU]$, but $BU \simeq \mathcal{F}$ (Fredh. op. [inserted below: in $\ell^2$])



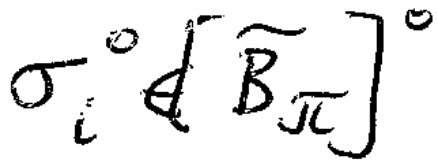
**Manuscript excerpt.** “In $\ell^2$” is written below “Fredh. op.”.

Original source details: GELFAND-P000520

$$B_\pi \ni x \to \Phi_x \in \mathcal{F}.$$

Let. $\varphi_x : H_1 \to H_2$ where $F$ from $T$ acted, $\widetilde{B}_\pi$ — the universal cover of $B_\pi$, on which there acts $\pi$. Take $\widetilde{B}_\pi \times H_i$ with $\pi$ acting on $\widetilde{B}_\pi$ in the standard way, and on $H_i$ via $T_i$. Let $\widetilde{B}_\pi$ — a cell complex (even simplicial).

Constr. $\varphi_x : \widetilde{B}_\pi \times H_1 \to \widetilde{B}_\pi \times H_2$. Let $[\widetilde{B}_\pi]^k$ — the $k$-dim. skeleton of $\widetilde{B}_\pi$. Choose one $\sigma_i^0 \in [\widetilde{B}_\pi]^0$ in any orbit of the action of $\pi$ and set $\Phi_{\sigma_i^0} \overset{\text{def}}{=} F$.



Original source details: GELFAND-P000520

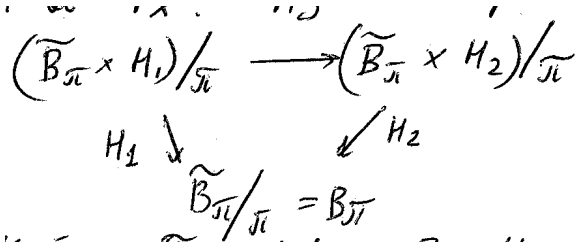
**Manuscript excerpt.** The $\in$ sign touches the left bracket of the skeleton.

We have:

a) equivariance

b) $\forall x, y \in [\widetilde{B}_\pi]^0 \quad \Rightarrow \quad \varphi_x - \varphi_y \in \text{Comp}$

Extend this to $[\widetilde{B}_\pi]^1$; by the simplicial action of $\pi$ we can choose one simplex in $\forall$ orbit and extend over the simplex as $p\varphi_1 + q\varphi_2$, $p + q = 1$ with equivariance satisfied. By ind. we obtain the whole family $\varphi_x$. From equivariance we have

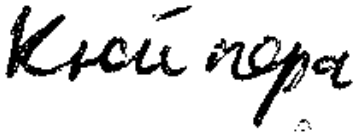


Original source details: GELFAND-P000520

**Diagram and typed description.**

$$\begin{array}{ccc} (\widetilde{B}_\pi \times H_1)/\pi & \longrightarrow & (\widetilde{B}_\pi \times H_2)/\pi \\ & {\scriptstyle H_1} \searrow \quad \swarrow {\scriptstyle H_2} & \\ & \widetilde{B}_\pi/\pi = B_\pi & \end{array}$$

But by Kuiper $(\widetilde{B}_\pi \times H_i)/\pi \approx B_\pi \times H_i$.



Original source details: GELFAND-P000520

**Manuscript excerpt.** The manuscript gives the Russian genitive “Кюйпера” for Kuiper.

After identification we have $B_\pi \times H_1 \to B_\pi \times H_2$, which defines $\varphi_x : H_1 \to H_2$.

GELFAND-P000521

15/$\overline{X}$-73 Gelfand's seminar.

Rudakov. On representations of infinite-dimensional Lie algebras.

Consider $W_n$ — the Lie alg. of form. vector fields ([inserted below: cont.] deriv. of the ring $k[[x_1, \ldots, x_n]]$).



Original source details: GELFAND-P000521

**Manuscript excerpt.** The circled “cont.” is inserted from below before “deriv.”.

$$\left(\sum_{i=1}^{n} a_i(x_1, \ldots, x_n)\frac{\partial}{\partial x_i}\right)$$

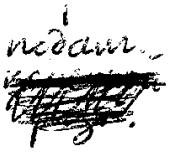
$W_n = \mathcal{L} \supset$ [crossed out] $M \supset M'$.



**Uncertain reading.** Lower loops are visible under the crossing out before $M$. This could also be $M$.

Original source details: GELFAND-P000521

[under $M$: subalg.]



**Undeciphered passage.** The crossed-out words under "subalg." cannot be read.

Original source details: GELFAND-P000521

[under $M'$:] an ideal of finite codim. in $M$.

Suppose we are given a repr. of

$$M/M' \longrightarrow \text{repr. of } M \underset{\text{induc.}}{\longrightarrow} \text{representation of } \mathcal{L}.$$

Thm.: all irred. reprs. of $W_n$ are obtained this way, with finitely many exceptions.

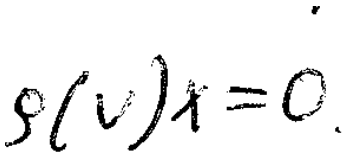
Spaces countable-dimensional discrete (without top.). $E$

[under «countable-dimensional»: (vect. spaces)]

Repr. $\varrho : \mathcal{L} \longrightarrow \operatorname{Aut} E$, with

$$x \in E,\ \exists V \subset \mathcal{L} \qquad \varrho(v)x = 0. \qquad \text{(loc. const. functions).}$$

[under $V$: nbhd. of $0$]



**Manuscript excerpt.** The argument of $\varrho$ is a lowercase $v$. The condition on the left uses a capital $V$.

Original source details: GELFAND-P000521

Filtrations of nbhds. of zero

$$\mathcal{L} = \mathcal{L}_{-1} \supset \mathcal{L}_0 \supset \ldots \supset \mathcal{L}_k \supset \ldots$$

$$\mathcal{L}_k = \left\{ \sum a_i \frac{\partial}{\partial x_i} \text{ [crossed out]}, \text{ deg. } a_i \geqslant k+1 \right\}.$$



**Undeciphered passage.** The crossed-out notation between the derivative and the comma cannot be read.

Original source details: GELFAND-P000521

Assertion: consider cont. reprs. of $\mathcal{L}_0/\mathcal{L}_k$ and irred. reprs. (and inducing representations).
Exceptions: in the dual spaces to spaces of closed forms
Suppose there is a repr. of the alg. $\mathcal{L}$.

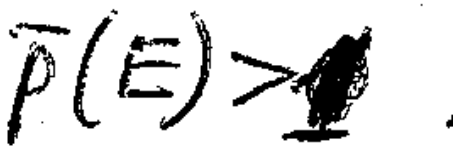
$$x \in E; \quad p(x) = \{\min p : \mathcal{L}_p x = 0\}$$

($\exists$ there a bounded cont. repr.?).
Consider

$$\bar{p}(E) = \min_{x \in E,\ x \neq 0} p(x).$$

$$E_0 = \{x \in E,\ p(x) = \bar{p}(E)\} \text{ — } \mathcal{L}_0\text{-submodule}.$$

If $\bar{p}(E) > 1$, then the repr. is induced: $E = \mathrm{Ind}(E_0)$



**Uncertain reading.** The digit is retraced. The bottom stroke resembles the base of a 1. The overlapping strokes make this reading provisional.

Original source details: GELFAND-P000521

— · —

Exceptions:

$\Omega^p$ — form. forms of deg. $p$, $\quad d\Omega^p$ — exact forms,

$(d\Omega^p)^*$ — the dual space (with the natural topology)

— · —

There are corresponding results for Hamiltonian ones as well.

— · —

GELFAND-P000522

[source number: 23]

Kirillov. A similar result about the group
Unproved, but well known
$M$ — a manifold, Diff $M$ — the group of diffeomorphisms.



**Uncertain reading.** The Russian abbreviation is “мн-зие”, probably short for “manifold”.

Original source details: GELFAND-P000522

describe those reprs. that have finite funct. dimension.

$$\mathrm{Diff}\, M \supset H_{x_1,\ldots,x_n}$$

[under $H_{x_1,\ldots,x_n}$:] keeps fixed $x_1, \ldots, x_n$.
Consider cont. reprs. of $H/H_0$ and induce.
[on the right:] $H_0$ preserves infin. nbhds. of order $n$.
Fuchs. (continuation).
$f : C \longrightarrow C'$ a functor
[under $C$, $C'$:] small cats.

$$Y \in \mathrm{Ob}\, C', \qquad Y\backslash f$$

$$\mathrm{Ob}(Y\backslash f) = \{(X, u),\ X \in \mathrm{Ob}\, C,\ u : Y \longrightarrow f(X)\}.$$

$$v : Y \longrightarrow Y' \quad \rightsquigarrow \quad v\backslash f : Y'\backslash f \longrightarrow Y\backslash f$$

$\exists$ a nat. map

$$B(Y\backslash f) \longrightarrow \text{homot. fibre of the map } Bf : BC \longrightarrow BC'$$

over the point $Y \in BC'$.

$$\longleftarrow \cdot \longrightarrow$$

new cat. of pairs $(X, u)$, where $X \in \mathrm{Ob}\, C$, $u : f(X) \longrightarrow Y$

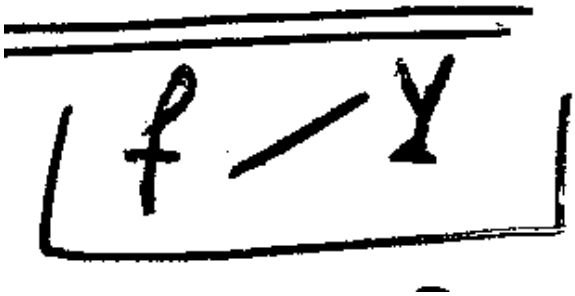
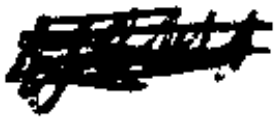



**Diagram and typed description.** $f/Y$

Original source details: GELFAND-P000522

another model of the homot. fibre.
The thm. also holds (dual)
Theorem A. This condition holds if all cats. $Y\backslash f$ are contractible (i.e.



**Undeciphered passage.** The word after "i.e." is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000522

have contractible classif. spaces). Then the conclusion also holds (in particular, the exact homot. seq.) $\Rightarrow Bf : BC \longrightarrow BC'$ induces isos. of homot. groups.

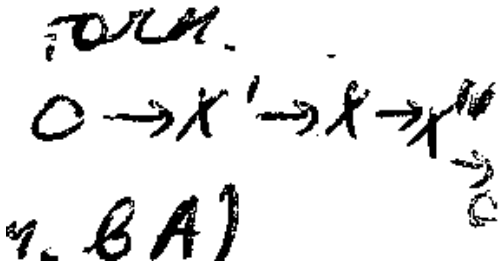
— · —

$\underline{\underline{K_i(M)}}$

$M$ — a full subcat. of the ab. cat. $A$, closed under extensions, i.e. from

$$0 \longrightarrow X' \longrightarrow X \longrightarrow X'' \longrightarrow 0, \quad X', X'' \in \mathrm{Ob}\, M \quad \Rightarrow \quad X \in \mathrm{Ob}\, M.$$

If a morphism $i : X' \longrightarrow X$ (from $M$) is included in a short exact seq., it is called an admissible monomorphism.



Original source details: GELFAND-P000522

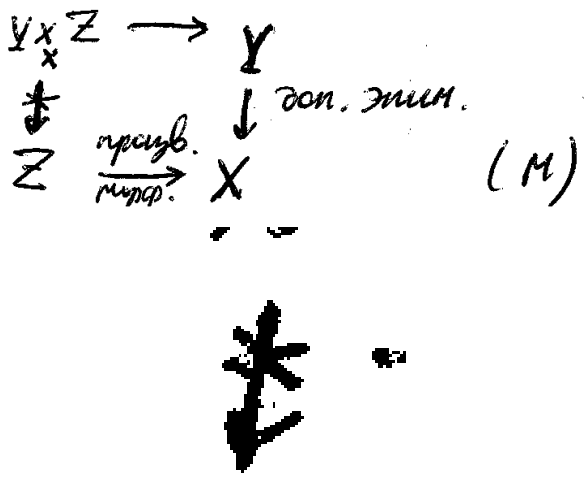
**Diagram and typed description.**

$$0 \to X' \to X \to X''$$
$$\downarrow$$
$$0$$

The final sign in the right-hand insertion is provisionally read as zero.

(exactness is understood in $A$)

Dual def. if a morph. $p : X \longrightarrow X''$ is included in a short exact seq., it is called an adm. epimorphism.



Original source details: GELFAND-P000522

**Diagram and typed description.**

$$\begin{array}{ccc} Y \times_X Z & \longrightarrow & Y \\ \downarrow & & \downarrow \text{adm. epim.} \\ Z & \xrightarrow[\text{morph.}]{\text{arbitr.}} & X \end{array} \qquad (M)$$

Then $Y \times_X Z \in \mathrm{Ob}\, M$ and $\downarrow$ — adm. epim..

GELFAND-P000523

From $M$ we construct the cat. $QM$:

$$\mathrm{Ob}\, M = \mathrm{Ob}\, QM.$$

$$\mathrm{Mor}_{QM}(X_1, X_2) = \left\{ X_1 \xleftarrow{\text{a.e.}} X \xrightarrow{\text{a.m.}} X_2 \right\}.$$

Def. equivalence in the obvious way.

([crossed out] $X_1$ — an admiss. subquotient of $X_2$)



Original source details: GELFAND-P000523

**Undeciphered passage.** The crossed-out entry before $X_1$ cannot be read.

Def. of composition:

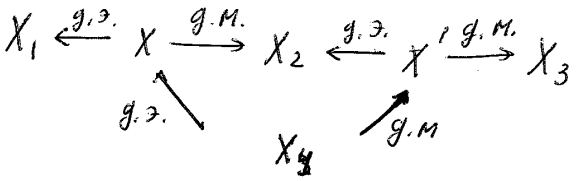


**Diagram and typed description.**

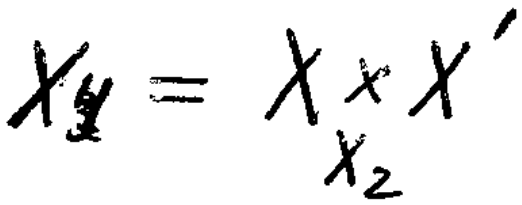
$$X_1 \xleftarrow{\text{a.e.}} X \xrightarrow{\text{a.m.}} X_2 \xleftarrow{\text{a.e.}} X' \xrightarrow{\text{a.m.}} X_3$$

$$\text{a.e.} \nwarrow \qquad \nearrow \text{a.m.}$$

$$X_4$$

$X_4 = X \underset{X_2}{\times} X'$.

The subscript $4$ in the lower vertex and the product formula overwrites an earlier sign. The earlier sign is unreadable. A short stroke is visible at the lower end of the left diagonal arrow.



Original source details: GELFAND-P000523

and one must take

$$X_1 \xleftarrow{\text{a.e.}} X_4 \xrightarrow{\text{a.m.}} X_3.$$

By definition

$$K_i(M) = \pi_{i+1}(BQM, 0),$$

where $0$ — any of the zero objects of the category $M$.

Isom. classes of objects of $M$ must form a set.



Original source details: GELFAND-P000523

**Undeciphered passage.** The crossed-out entry before “Special case” cannot be read.

Special case: (the main one).

$A$ — an arbitrary ring, $PA$ — the cat. of finitely generated projective left $A$-modules.

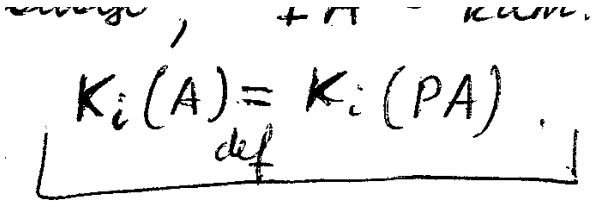


**Diagram and typed description.**

$K_i(A) = K_i(PA)$.
def

Original source details: GELFAND-P000523

~~Properties of a certain $K$-functor.~~



**Uncertain reading.** The crossed-out abbreviation between “Properties” and K is provisionally read as “некотор.” (“a certain”).

Original source details: GELFAND-P000523

$$\varinjlim \mathrm{GL}(n, A) = \mathrm{GL}(A).$$

$$B\,\mathrm{GL}(A)$$

We construct $B_+\,\mathrm{GL}(A)$ with the properties

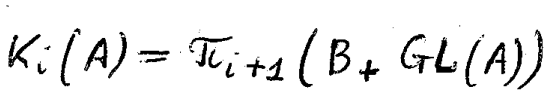
1) it is an $H$-space,

2) $B\,\mathrm{GL}(A) \longrightarrow B_+\,\mathrm{GL}(A)$

3) the homot. fibre is acyclic

4) surj. map on $\pi_1$.

$K_i(A) = \pi_{i+1}(B_+\,\mathrm{GL}(A))$ — the old def. (Quillen’s)



**Manuscript excerpt.** The subscript $i+1$ in this formula follows the manuscript.

Original source details: GELFAND-P000523



**Uncertain reading.** A crossed-out entry before “Apparently” may be the beginning of “По-видимому”.

Original source details: GELFAND-P000523

Apparently $BQPA$ is $B_+ \mathrm{GL}(A)$. This is false, but there is a map.

$$B\,\mathrm{GL}(A) \longrightarrow BQPA.$$

Let $S_n$ — a free $A$-mod. with $n$ gens..

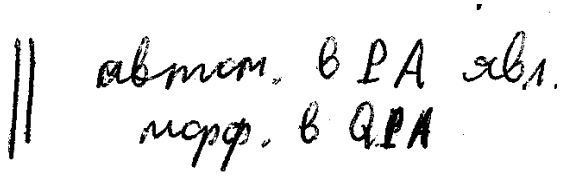
$$S_n \in \mathrm{Ob}(PA) = \mathrm{Ob}(QPA).$$

Elems. of the group $\mathrm{GL}(n, A)$ — the automs. $S_n \longrightarrow S_n$. This lets us def. the functor

$$\mathrm{GL}(n, A) \rightsquigarrow PA.$$

[under $\mathrm{GL}(n, A)$:] corr. cat. (a single obj.)

$$\text{obj.} \longrightarrow S_n$$




**Manuscript excerpt.** ‖ automs. in $PA$ are morphs. in $QPA$

Original source details: GELFAND-P000523

Functor $\mathrm{GL}(n, A) \longrightarrow QPA$.

GELFAND-P000524

[source number: 24]

Hence we obtain a map of classif. spaces

$$B\,\mathrm{GL}(n, A) \longrightarrow BQPA$$

It remains to pass to the limit. (They are incompatible).

$$\begin{array}{c} QPA \longrightarrow QPA \\ X \longrightarrow X \oplus A \end{array}$$ is a homotopy equivalence.

Diagr.

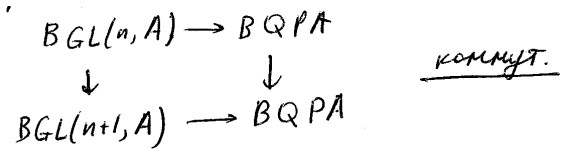



**Diagram and typed description.**

$$\begin{array}{ccc} B\,\mathrm{GL}(n, A) & \longrightarrow & BQPA \\ \downarrow & & \downarrow \\ B\,\mathrm{GL}(n+1, A) & \longrightarrow & BQPA \end{array}$$ commutes.

Original source details: GELFAND-P000524

and hence the limit and the map

$$B\,\mathrm{GL}(A) \longrightarrow BQPA$$

GELFAND-P000525

22.10.73 Gelfand's seminar.



**Undeciphered passage.** The crossed-out entry before "Gelfand" cannot be read.

Original source details: GELFAND-P000525

Gelfand. Publicizing work on nonlin. scattering theory. (Straus)




Original source details: GELFAND-P000525

**Manuscript excerpt.** The surname in parentheses is written “Straus”.

$$U(t)U(s) = U(t+s),$$

Relation for $U(t)f$ as $t \longrightarrow \pm\infty$

Typical features: $E(U(t)f) = E(f)$; $E$ — some functional

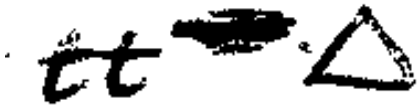
$\exists$ a simpler group $U_0(t)$, such that

$$E\big(U(t)f - U_0(t)f_\pm\big) \underset{t\to\pm\infty}{\longrightarrow} 0 \qquad (\exists f_\pm)$$

$S$ — the relation between $f_+$ and $f_-$.

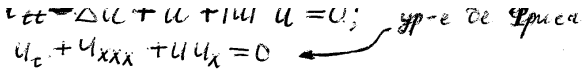
Examples: wave eq.

$$u_{tt} - \Delta u + u + |u|^2 u = 0;$$



Original source details: GELFAND-P000525

**Uncertain reading.** The sign before the Laplacian is retraced. The horizontal stroke supports a provisional minus.



Original source details: GELFAND-P000525

**Diagram and typed description.**

de Vries eq.

$$u_\tau + u_{xxx} + uu_x = 0$$

The first subscript is provisionally set as $\tau$ from its shape. It differs from the $tt$ on the line above.

— • —

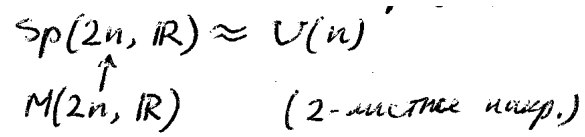
Kirillov.

1. Kostant’s work.

$M$ — a symplectic manifold (a nondeg. closed 2-form is given $\omega$).

$TM$ — a bundle over $M$ with struct. group $\mathrm{Sp}(2n,\mathbb{R})$. It is not simply connected.

Topol.



Original source details: GELFAND-P000525

**Diagram and typed description.**

$\mathrm{Sp}(2n,\mathbb{R}) \approx U(n)$

$\uparrow$

$M(2n,\mathbb{R})$ (2-sheeted cover)

Sometimes the struct. group lifts to



Original source details: GELFAND-P000525

**Undeciphered passage.** The crossed-out sign before $M(2n, \mathbb{R})$ cannot be read.

$M(2n, \mathbb{R})$.

Then on $M$ one constructs a spinor bundle (if $C_1$ is divisible by 2).

$M$ has a special



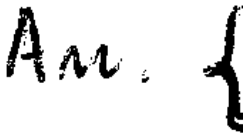
Original source details: GELFAND-P000525

**Undeciphered passage.** The crossed-out word before "repr. $T$" cannot be read.

repr. $T$, called the spinor repr. (or the Weyl repr.)

Alg. $\{p_1, \ldots, p_n, q_1, \ldots, q_n\}$ and the rel.

$$p_i q_j - q_j p_i = \delta_{ij}$$



Original source details: GELFAND-P000525

**Manuscript excerpt.** The abbreviation at the start of the line reads "Ал.".

(if $+$, then the spin repr. of the orthog. group).

$\mathrm{Sp}(2n, \mathbb{R})$ linearly changes $p$ and $q$.

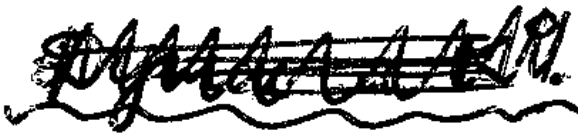
Realiz. $p_i \longrightarrow \frac{\partial}{\partial x_i}$, $q_i \longrightarrow x_i$ in $L^2(\mathbb{R}^n)$.

Automs. of the alg. $\Longrightarrow$ automs. of the alg. of bounded operators. But they are all inner: $A \longrightarrow UAU^{-1}$ ($U$ up to a factor) $\Rightarrow$ a projective representation. The factors reduce to $\pm 1$. The repr. becomes single-valued on the 2-sheeted cover.

GELFAND-P000526

[source number: 28]

2. [crossed out] $p$-adic measures.



Original source details: GELFAND-P000526

**Undeciphered passage.** The crossed-out part of the heading cannot be read.

$p$-adic measures; $\mathbb{Z}_p$ — integers;



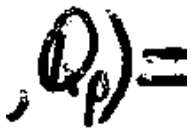
**Undeciphered passage.** The crossed-out entry after “integers” cannot be read.

Original source details: GELFAND-P000526

Consider $C(\mathbb{Z}_p, K)$, $K$ — a finite ext. of the $p$-adic field
(cont. funcs. with values in $K$)

$$\mathrm{Hom}\big(C(\mathbb{Z}_p, K), \mathbb{Q}_p\big) = C(\mathbb{Z}_p, K)' = M(\mathbb{Z}_p, K)$$

(measures with $p$-adic values)



**Manuscript excerpt.** The second argument of Hom reads $\mathbb{Q}_p$.

Original source details: GELFAND-P000526

There is a convolution on measures

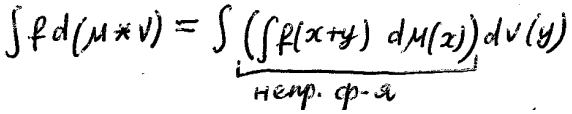
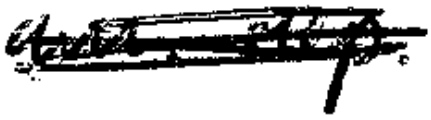



**Diagram and typed description.**

$$\int f\, d(\mu * \nu) = \int \underbrace{\left( \int f(x+y)\, d\mu(x) \right)}_{\text{cont. func.}} d\nu(y)$$

Original source details: GELFAND-P000526

Theorem 1.



**Undeciphered passage.** The crossed-out entry after “Theorem 1” cannot be read.

Original source details: GELFAND-P000526

$K$-algebra of measures $M(\mathbb{Z}_p, K) = K[[T]]$.

$$\left\| \sum a_k T^k \right\| = \sup_k |a_k| \qquad \text{(series of finite norm)}$$

Form. series are obtained from unbounded measures.

Theorem 2. Consider $M(\mathbb{Z}_p^*, K) = (p-1)K[[T]]$.

Unbounded measures are finitely add. on elem. sets (dual to loc. const. funcs.).

Proof of Thm. 1.

$$\mu \longrightarrow \sum a_k T^k, \qquad \text{where } a_k = \int_{\mathbb{Z}_p} \frac{x(x-1)\cdots(x-k+1)}{k!}\, d\mu(x)$$

$$\sum a_k T^k = \int_{\mathbb{Z}_p} (1+T)^x \, d\mu(x).$$

it follows that convolution of measures $\longrightarrow$ product of series.



Original source details: GELFAND-P000526



**Diagram and typed description.**
A tilted shaded quadrilateral between the formulas. Its purpose is unclear.

$$1 + T = e^{\tau} \qquad (\tau \text{ — form. var.}) \qquad \int_{\mathbb{Z}_p} e^{\tau x}\, d\mu(x) = \widetilde{\mu}(\tau)$$

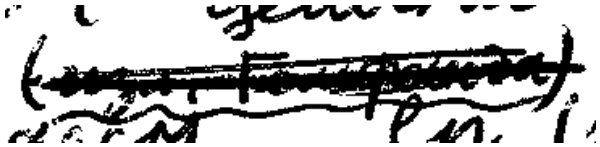
(rewrite the analysis!)

$\tau$ can be treated as a number.

$$\tau = \ln(1+T) = T - \frac{1}{2}T^2 + \frac{1}{3}T^3 - \dots$$

It is unclear what range of funcs. will result (after substitution one should get a series with integer coeffs.)

Example ([crossed out]) (Cartier) as presented by Gelfand

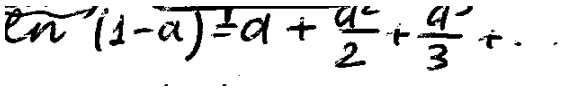


Original source details: GELFAND-P000526

**Undeciphered passage.** The entry in the first parentheses after “Example” is crossed out and cannot be read.

Consider

$$\ln(1-a) = a + \frac{a^2}{2} + \frac{a^3}{3} + \dots \qquad \text{and } a = 2 \in \mathbb{Z}_2.$$



Original source details: GELFAND-P000526

**Manuscript excerpt.** The manuscript has $a$ after the equals sign, followed by plus signs.

The series converges [inserted below: and equals] $\ln(-1)$.



Original source details: GELFAND-P000526

**Manuscript excerpt.** The circled insertion “and equals” points to the line above.

$$\ln[(-1)^2] = 2\ln(-1) = 0 \quad \Rightarrow \quad \ln(-1) = 0.$$

hence the series converges to 0

$$2 + \frac{2^2}{2} + \frac{2^3}{3} + \frac{2^4}{4} + \frac{2^5}{5} + \ldots + \frac{2^n}{n} + \ldots = 0$$

$$S_n = 2 + \frac{2^2}{2} + \frac{2^3}{3} + \ldots + \frac{2^n}{n} \qquad \text{for large } n \text{ is divisible by } 2^k$$

2, 4, $\frac{20}{3}$, $\frac{32}{3}$ and so on.

— · —

Theorem.

$$\|\mu\| = \sup_k |a_k| \qquad (\approx \text{ Plancherel})$$

$$\|\mu\| \underset{\text{def}}{=} \sup_{|f|\le 1} \left| \int f \, d\mu \right|$$

From the formula $\|\mu\| \ge \sup_k |a_k|$

We need to check

$$|f| \le 1 \quad \Rightarrow \quad f(x) = \sum c_k P_k(x), \qquad |c_k| \le 1,$$

$$P_k(x) = \frac{x(x-1)\cdots(x-k+1)}{k!}$$

(Proof: approx. by polynomials and apply the thm.)

$$c_k = \sum_{i=0}^{k} (-1)^{i+k} C_k^i f(i) = \Delta^k f(0) \qquad \text{(Taylor formula)}$$

$$X^{\textcircled{k}} = x(x-1)\cdots(x-k+1),$$

$$\Delta X^{\textcircled{k}} = kx^{\textcircled{k-1}}, \qquad \text{where } \Delta f(x) = f(x+1) - f(x).$$

$i = 0$

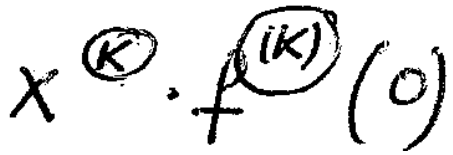
$X^{\textcircled{k}} = x(x-1) \ldots (x-k+1)$,

$\Delta X^{\textcircled{k}} = kX^{\textcircled{k-1}}$, где $\Delta f(x) = f(x+1) - f(x)$.

Original source details: GELFAND-P000526

**Uncertain reading.** The first two letters with a circled index are provisionally set as $X$. The letter after the factor $k$ is set as $x$.

$$f(x) = \sum_{k=0}^{\infty} \frac{1}{k!} X^{\textcircled{k}} \cdot f^{\textcircled{k}}(0), \qquad \text{where } f^{\textcircled{k}}(x) = (\Delta^k f)(x)$$



**Manuscript excerpt.** The letter with circled $k$ in the first product reads as $X$. Both indices $(k)$ on $f$ are also circled.

Original source details: GELFAND-P000527

Assn.: $\forall$ cont. funcs.

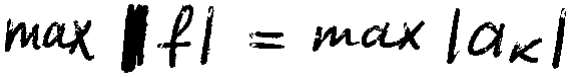


**Undeciphered passage.** The crossed-out word before "series" cannot be read.

Original source details: GELFAND-P000527

the Maclaurin series converges uniformly, and,

$$\max |f| = \max |a_k|.$$



**Uncertain reading.** There is a thick stroke before $f$ in the manuscript. It is provisionally set as one absolute-value bar.

Original source details: GELFAND-P000527

Next time Kirillov (3 more min.) and Mishchenko.

Fuchs. (continuation).

$M$ — an add. full subcat. of an abelian category, closed under extensions

$$QM: \qquad \operatorname{Ob} QM = \operatorname{Ob} M.$$

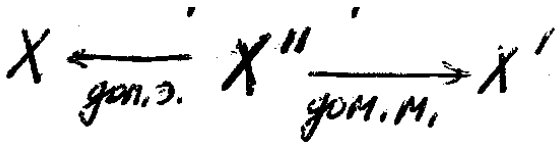


**Diagram and typed description.**

$$X \xleftarrow[\text{adm. e.}]{} X'' \xrightarrow[\text{adm. m.}]{} X'$$

Original source details: GELFAND-P000527

$$\pi_i(BQM)$$

$BQM$ is homol. equiv. to $B\,\mathrm{GL}(\infty, A)$

Def. Grothendieck gr. of the category $M$. Gens. — the objects of $M$, rels.

$$0 \to X' \to X \to X'' \to 0, \qquad \Rightarrow \quad X = X' + X''$$

The gr. is commutative, since $\exists\ X = X' \oplus X''$.

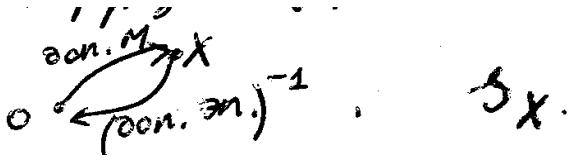
Theorem:

$$\underset{\pi_1(BQM,0)}{K_0(M)} \cong \text{Grothendieck gr. of the cat. } M.$$

Zero-dim. skeleton $= \mathrm{Ob}(M)$

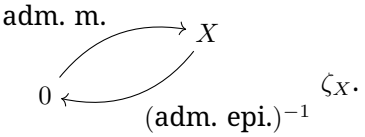
1-dim. skeleton — the morphisms of $QM$

$\forall X$ a loop is def.



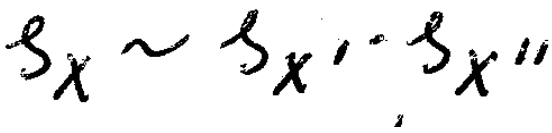
Original source details: GELFAND-P000527

**Diagram and typed description.**



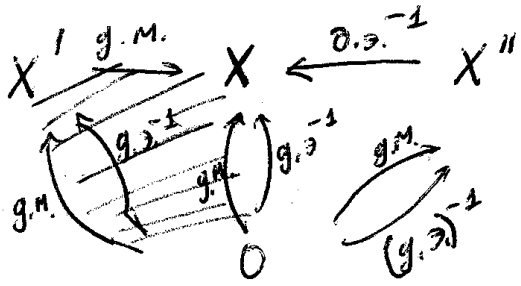

If in $M$ there is an exact seq. (short) $0 \to X' \to X \to X'' \to 0$, then

$$\zeta_X \sim \zeta_{X'} \cdot \zeta_{X''}.$$



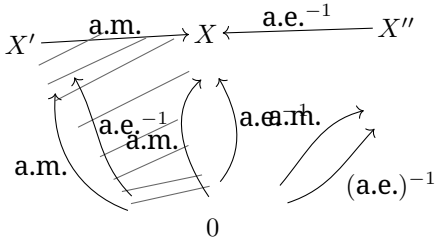
Original source details: GELFAND-P000527

**Uncertain reading.** The letter denoting the loop is provisionally set as $\zeta$.



Original source details: GELFAND-P000527

**Diagram and typed description.**




The lower ends of the arcs are left separate as in the sketch. The thin lines represent the diagonal hatching.

and recall the 2-simplices.

— • —

GELFAND-P000528

29/$\overline{X}$-73. Gelfand's seminar.

A. A. Kirillov.

There is a symplectic manifold $M$

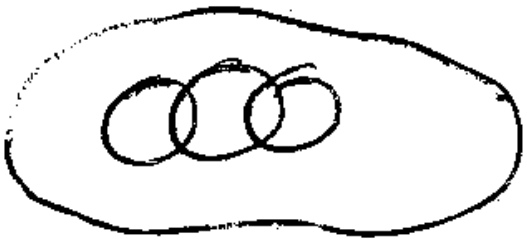
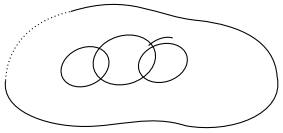


**Diagram and typed description.**



Original source details: GELFAND-P000528

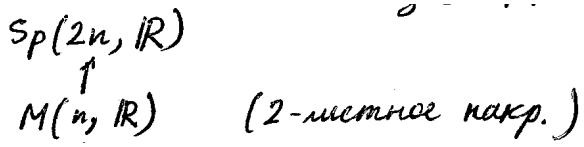



**Diagram and typed description.**

$\mathrm{Sp}(2n,\mathbb{R})$

$\uparrow$

$M(n,\mathbb{R})$ (2-sheeted cover)

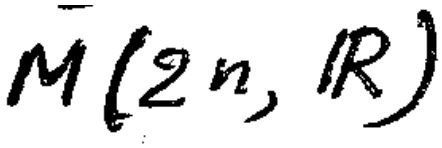
Original source details: GELFAND-P000528

loc. we may take

$$\omega = \sum_{k=1}^{n} dp_k \wedge dq_k$$

Transition funcs. (in $TM$) $g_{ij} \in \mathrm{Sp}(2n,\mathbb{R})$

Suppose they are in the 2-sheeted cover $M(2n,\mathbb{R})$




**Manuscript excerpt.** The line reads $M(2n,\mathbb{R})$. The diagram above and the insertion below read $M(n,\mathbb{R})$.

Original source details: GELFAND-P000528

**The spinor representation** [inserted below: $M(n,\mathbb{R})$] in Hilb. space



**Undeciphered passage.** The crossed-out sign after "space" cannot be read.

Original source details: GELFAND-P000528

induces a Hilbert bundle $HM$ over $M$




**Manuscript excerpt.** The circled insertion $M(n,\mathbb{R})$ points to "representation".

Original source details: GELFAND-P000528

$$SM \subset HM \subset S'M$$



**Uncertain reading.** The crossed-out sign before $S'M$ resembles $H$.

Original source details: GELFAND-P000528

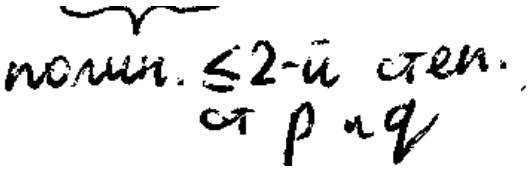
[under $SM$: rapidly decreasing]

Symplectic transf. $\rightsquigarrow$ op. in Hilb. space (can. transf. with $p_i, q_i$)

We obtain a projective representation

$$\underbrace{GM} = TM \oplus \mathbb{R} \times M$$

[under the brace:] polyn. of deg. $\leq 2$ in $p$ and $q$ (Heisenberg Lie alg.)



**Manuscript excerpt.** The sign before 2 reads $\leq$.

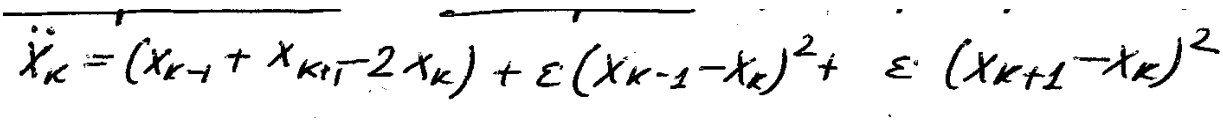
Original source details: GELFAND-P000528

$GM$ acts on $HM$ compat. with the bundle structure.

$RM$, $R_x$ — Lagrangian subspaces in $T_xM$ (max. dim.)

A. Shnirelman. On Fermi: equipartition of energy among degrees of freedom.

$$\ddot{x}_k = (x_{k-1} + x_{k+1} - 2x_k) + \varepsilon(x_{k-1} - x_k)^2 + \varepsilon(x_{k+1} - x_k)^2$$

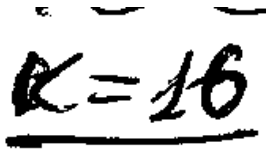


Original source details: GELFAND-P000528

**Manuscript excerpt.** Both quadratic terms have plus signs in the manuscript.

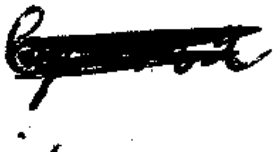
Solved num. on a computer

$k = 16$ after fin. ~~time~~ time intervals they examined the Fourier transf..



**Uncertain reading.** The number 16 is legible. The retraced letter is provisionally set as $k$.

Original source details: GELFAND-P000528



**Uncertain reading.** The crossed-out word is provisionally read as “time”.

Original source details: GELFAND-P000528

Init. cond. — sin (all the energy in the 1st harm.)

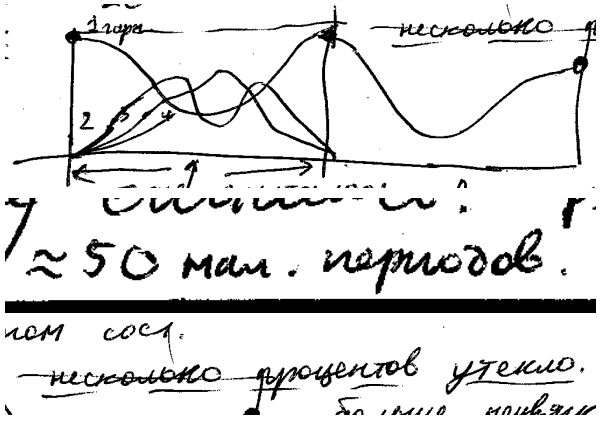
By the estimate $t = 10^{-35}$ in this state.




Original source details: GELFAND-P000528

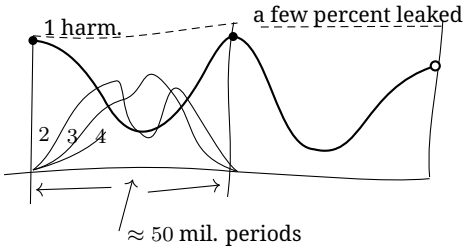
**Manuscript excerpt.** The exponent on the line reads $-35$.

Calculations:




Original source details: GELFAND-P000528

**Diagram and typed description.**




The numerals are placed by the curves as in the manuscript. Their precise assignment and the continuation of the short rising line among the crossings are unresolved.

larger residual

10 steps — per. $1/3$

Another 10 steps — again
the same thing!!

Cubic nonlin. was also computed. The same result was obtained.

GELFAND-P000529

Zakharov: approx. by a diff. system with first integrals. (Possible cause)

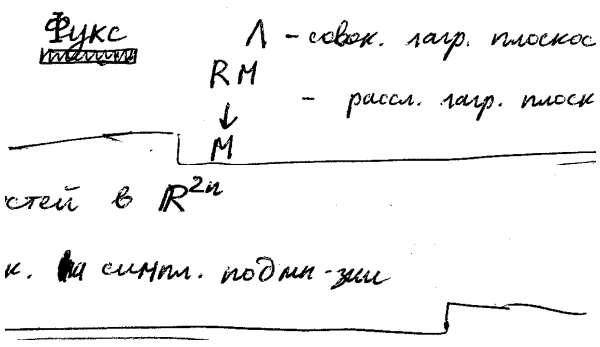
A string with sin — likewise.

There are init. data for which uniform distribution occurs.

$$u_{tt} = u_{xx} \qquad \text{on } S^1 \quad \text{all traj. are closed}$$

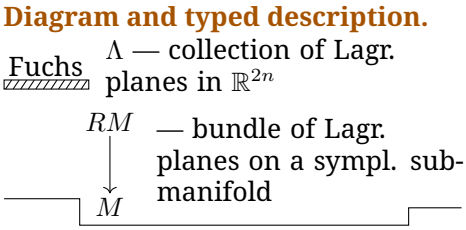
$$u_{tt} = u_{xx} + \varepsilon f'(u)$$

First-order drift of the trajectories: also periodic for $u_{tt} = u_{xx} - \varepsilon u^3$.



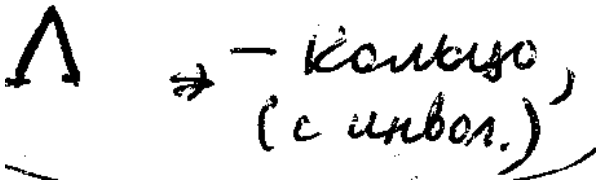

Original source details: GELFAND-P000529

**Diagram and typed description.**




The first stroke in “on” is heavily retraced. The Russian abbreviation reads “подмн-зии”.

Mishchenko.



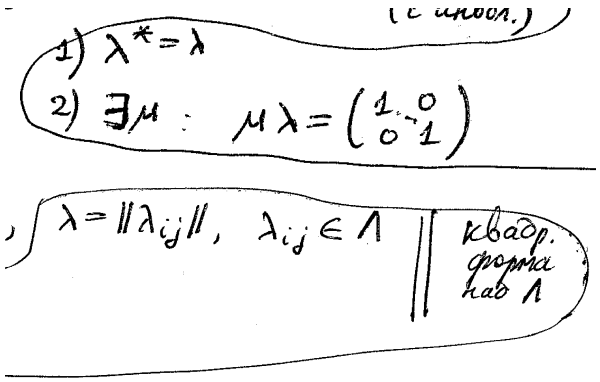
Original source details: GELFAND-P000529

**Diagram and typed description.**

$\Lambda$ — a ring,

$\Rightarrow$ (with invol.)

The sign below and to the right of $\Lambda$ is rendered as $\Rightarrow$.



Original source details: GELFAND-P000529

**Diagram and typed description.**

$\lambda = \|\lambda_{ij}\|, \quad \lambda_{ij} \in \Lambda$

1) $\lambda^* = \lambda$

2) $\exists \mu : \quad \mu\lambda = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$

quadr. form over $\Lambda$

The short diagonal trace between the rows of the identity matrix is rendered as $\ddots$.

A Fredholm rep. is a triple $T = \{T_1, F, T_2\}$;

$T_i$ — a rep. of $\Lambda$ in $\mathcal{L}(H)$ (with involution), where

$F$ — a Fredholm operator

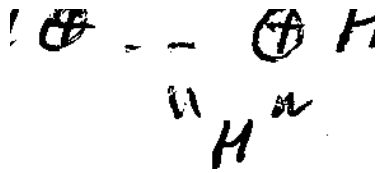
$FT_1(x) - T_2(x)F \quad \in \text{comp} \qquad \forall x \in \Lambda.$



Original source details: GELFAND-P000529

**Undeciphered passage.** After the second $F$, before $\in$, a parenthesized fragment is heavily crossed out.

Construct $\lambda \in \mathcal{L}(H \oplus \cdots \oplus H)$ from the quadr. form.



Original source details: GELFAND-P000529

**Manuscript excerpt.** $\backslash\backslash\, H^n$

Below the direct sum are two slanted strokes and $H^n$. A rotated equality sign represents the strokes.

$$\lambda_{T_1} = (T_1(\lambda_{ij})), \qquad \lambda_{T_2} = (T_2(\lambda_{ij}))$$

[at right:] — invertible and self-adj.

They are bounded, invertible, self-adjoint.

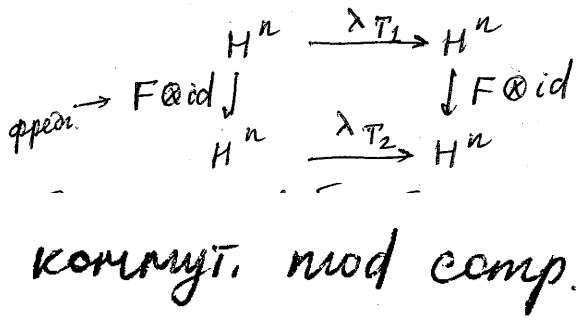






**Diagram and typed description.**

$$\begin{array}{ccc} H^n & \xrightarrow{\lambda_{T_1}} & H^n \\ {\scriptstyle F\otimes id}\downarrow & & \downarrow{\scriptstyle F\otimes id} \\ H^n & \xrightarrow{\lambda_{T_2}} & H^n \end{array}$$

Fredh.

commutes mod comp.

Original source details: GELFAND-P000529

Choose bases so that

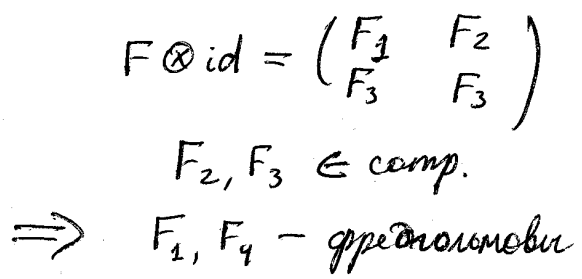
$$\lambda_{T_1} = \begin{pmatrix} A & 0 \\ 0 & -B \end{pmatrix} \qquad \lambda_{T_2} = \begin{pmatrix} A' & 0 \\ 0 & -B' \end{pmatrix}$$

$A, B$ — inv. and self-adj. and $> 0$.

$$F \otimes id = \begin{pmatrix} F_1 & F_2 \\ F_3 & F_3 \end{pmatrix}$$

$F_2, F_3 \in$ comp.

$\Rightarrow$ $F_1, F_4$ — Fredholm




**Manuscript excerpt.** The lower-right entry of the matrix reads $F_3$. The final line reads $F_4$.

Original source details: GELFAND-P000529

GELFAND-P000530

[source number: 32]

Definition. $\operatorname{sign}_T \lambda = \operatorname{index} F_1 - \operatorname{index} F_4$.




Original source details: GELFAND-P000530

**Manuscript excerpt.** The definition gives the difference of the indices of $F_1$ and $F_4$.

It does not depend on the rep. as a direct sum or on the rep. in the class of equiv. quadr. forms.



**Diagram and typed description.**

Original source details: GELFAND-P000530

Let $M$ — a manifold (comp.); $\pi_1(M) = \pi$, $\dim M = 4K$; $\Lambda = \mathbb{Z}[\pi]$.



**Uncertain reading.** The letter after $4$ is provisionally set as $K$. The handwriting leaves the case unresolved.

Original source details: GELFAND-P000530

There is a quadr. form $\sigma(M)$ over $\Lambda$. Formula:

$$\operatorname{sign}_T(\sigma(M)) = \langle \operatorname{ch} \xi_T \cdot L(M), [M] \rangle$$

$\xi_T$ — a bundle on $M$

Hirzebruch proved for $(\operatorname{ch} \xi_T = 1.)$

Finite-dimensional reps. are insufficient.



**Diagram and typed description.**

Original source details: GELFAND-P000530



**Uncertain reading.** A short isolated slanted stroke in the blank right margin. Its reading as the numeral “1” remains unresolved.

Original source details: GELFAND-P000530

GELFAND-P000531

[source number: 37]

12.11.73. Gelfand's sem..

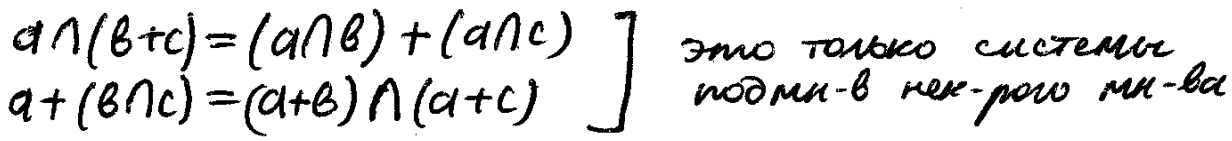
V. Ponomarev.

Suppose the set has 2 operations $\cap$ and $+$ with the natural properties (assoc., commut., $x \cap x = x$, $x + x = x$, $a + (a \cap b) = a$, $a \cap (a + b) = a$) this is a lattice.

Distributive lattice:

$$\left.\begin{aligned} a \cap (b + c) &= (a \cap b) + (a \cap c) \\ a + (b \cap c) &= (a + b) \cap (a + c) \end{aligned}\right]$$

[at right:] these are only systems of subsets of some set.



Original source details: GELFAND-P000531

**Manuscript excerpt.** The right square bracket joins the two identities to the note.

example: the lattice of subspaces with intersection and sum.

order relation in the lattice: $a \leq b \iff a \cap b = a$

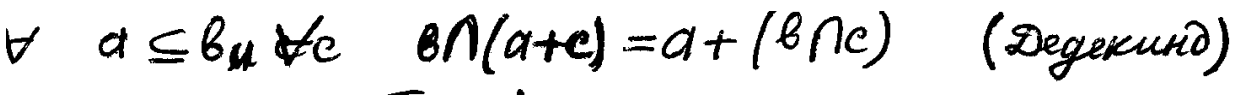


**Uncertain reading.** The heavily retraced sign between $a$ and $b$ is provisionally set as $\leq$.

Original source details: GELFAND-P000531

Modular latt.:

$$\forall\, a \subseteq b \text{ and } \forall c \qquad b \cap (a + c) = a + (b \cap c) \qquad \text{(Dedekind)}$$



Original source details: GELFAND-P000531

**Manuscript excerpt.** The condition in the modular law reads $a \subseteq b$.

($\forall$ distributive is modular, but not conversely).



**Uncertain reading.** The heavily retraced sign before "distributive" resembles $\forall$.

Original source details: GELFAND-P000531

the lattice of subspaces is modular, but not distributive.

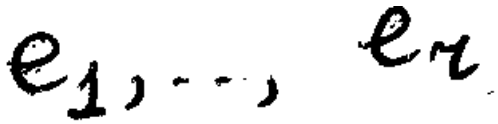
the lattice of submodules of a module is modular too.

Question: are there more identities or is this all? (Conjecture: $\forall$ modular latt. is the latt. of submod. of a module).

For latt. of subspaces there are more!!

---

Let $e_1, \ldots, e_r$ — be letters.



**Manuscript excerpt.** The first subscript has the same shape as $r$ in the second list and in the notation for generators below.

Original source details: GELFAND-P000531

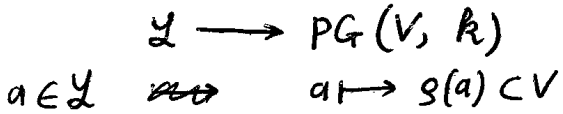
Consider the free modular lattice generated by them (polyn. in $e_1, \ldots, e_r$ with relations) linearly equiv. polynomials $P(e_1, \ldots, e_r) \sim Q(e_1, \ldots, e_r)$, if it holds on substitution of subspaces.

---

$V$ — a fin.-dim. space over the field $k$.

A representation of the latt. is a map into the lattice of subspaces of $V$ (which is denoted $PG(V, k)$)

$$\begin{array}{cc} \mathcal{L} \longrightarrow PG(V,k) & ab \mapsto s(a)s(b) \\ a \in \mathcal{L} \quad \longrightarrow \quad a \mapsto s(a) \subset V & a + b \mapsto s(a) + s(b) \end{array} \Big\|$$



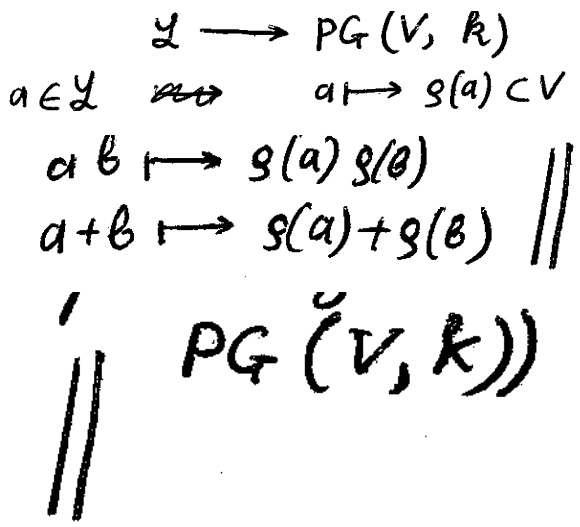
**Manuscript excerpt.** The arrows are written in two rows. A double vertical stroke follows the right column.

Original source details: GELFAND-P000531



**Manuscript excerpt.** The arrow after $a \in \mathcal{L}$ is overwritten or struck through with slanted loops.

Original source details: GELFAND-P000531



**Uncertain reading.** The looped letter is provisionally set as $s$ throughout the notation for representations.

Original source details: GELFAND-P000531

$\mathcal{D}^r$ — the free modular lattice on $r$ generators $e_1, \ldots, e_r$.

A rep. — is $V$ and $s(e_1), \ldots, s(e_r)$ — a collection of subspaces $E_1^0, \ldots, E_r^0$, and conversely such a collection determines a representation.



**Undeciphered passage.** A heavily crossed-out fragment lies between "A rep. —" and "is $V$".

Original source details: GELFAND-P000531

Problem: describe all reps. of $\mathcal{D}^r$.

~~An elem. $a$ is called perfect if~~



Original source details: GELFAND-P000531

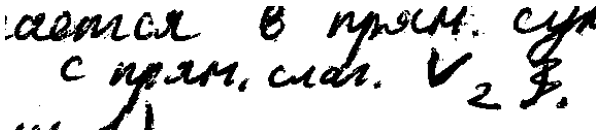
**Manuscript excerpt.** A single stroke crosses out the beginning of the definition.

$s_1, V_1$; $s_2, V_2$

$$s_1 \oplus s_2 \text{ — a rep. in } V_1 \oplus V_2 : \qquad (s_1 \oplus s_2)(a) = s_1(a) \oplus s_2(a)$$

Decomposable representation is defined naturally.

An elem. $a \in \mathcal{L}$ (inserted above: modular latt.) is called perfect, if for $\forall$ rep. $s$ in a fin.-dim. space $V$ over $\forall$ field $k$ the subspace $s(a)$ can be complemented by a subspace $V_2$ and the rep. decomposes as a direct sum [inserted below:] with direct summand $V_2$.



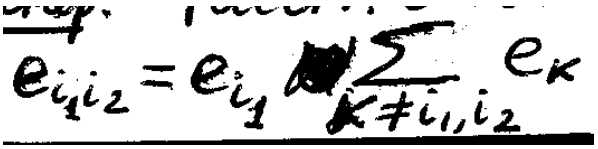
**Uncertain reading.** After $V_2$ a curl with a dot adjoins the descender from the line above.

Original source details: GELFAND-P000531

(Or: in an indecomp. rep. it maps to $0$ or $1$). If $a \sim b$ (lin.), then both are perfect at the same time.

Example. Consider the free mod. latt. on $r$ generators $e_1, \dots, e_r$. Consider the elements

$$e_{i_1 i_2} = e_{i_1} \text{ [overwritten sign] } \sum_{k \neq i_1, i_2} e_k$$

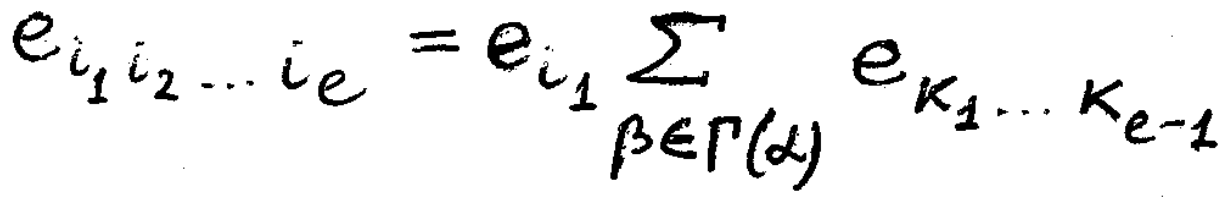


**Uncertain reading.** The sign between $e_{i_1}$ and the sum is heavily overwritten. Its reading remains unresolved.

Original source details: GELFAND-P000531

GELFAND-P000532

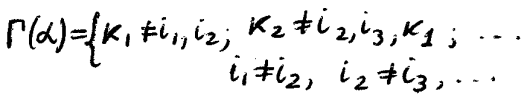
$$e_{i_1 i_2 \dots i_e} = e_{i_1} \sum_{\beta \in \Gamma(\alpha)} e_{k_1 \dots k_{e-1}}$$



Original source details: GELFAND-P000532

**Uncertain reading.** The length letter resembles $e$ and is provisionally set that way. There is no separate sign between $e_{i_1}$ and the sum.

$$\begin{aligned} \alpha &= (i_1, \dots, i_e), \qquad \beta = (k_1, \dots, k_{e-1}) \\ \Gamma(\alpha) &= \{\, k_1 \neq i_1, i_2; \quad k_2 \neq i_2, i_3, k_1; \quad \dots \\ &\qquad i_1 \neq i_2, \quad i_2 \neq i_3, \quad \dots \end{aligned}$$



**Manuscript excerpt.** The line of conditions on $i_1, i_2, i_3$ is indented. There is no closing brace.

Original source details: GELFAND-P000532

. . . . . .

Triv. perfect elements: $m_t = \sum_{i \neq t} e_i$ — perfect elements.

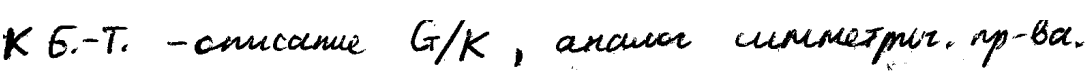
$m_t$ generate a Boolean alg. with $r$ generators.

S. Gelfand.

Bruhat–Tits complex.

$G$ — a $p$-adic group, $K$ — a max. comp. subgroup.

B.–T. cplx. — description of $G/K$, analogue of a symmetric space.



Original source details: GELFAND-P000532

**Manuscript excerpt.** The manuscript abbreviation reads “К Б.-Т.”.

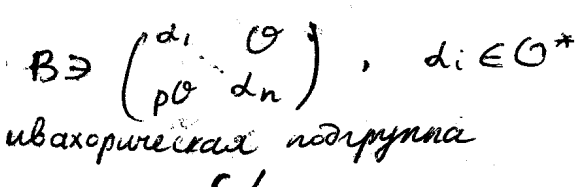
$K$ is open (contains a nbhd. of 1) $\Rightarrow G/K$ is discrete.

example: $G = \mathrm{GL}(2, Q_p)/C$ ($C$ — the centre).

$$K = \mathrm{GL}(2, \mathcal{O}) \qquad \mathcal{O} \text{ — integers}$$

$$\mathrm{GL}(n, Q_p) \supset B, \qquad B \ni \begin{pmatrix} \alpha_1 & & \mathcal{O} \\ & \ddots & \\ p\mathcal{O} & & \alpha_n \end{pmatrix}, \qquad \alpha_i \in \mathcal{O}^*$$

[under the matrix:] Iwahori subgroup.



**Manuscript excerpt.** The diagonal letter matches the shape of $\alpha$ at the start of the frame. At upper right is $\mathcal{O}$ and at lower left is $p\mathcal{O}$.

Original source details: GELFAND-P000532

$x_1, \ldots, x_e$ — points of $G/K$

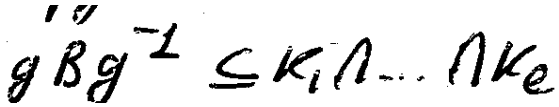
$K_1, \ldots, K_e \leftarrow$ the same ones, subgroups conjugate to $K$



**Manuscript excerpt.** A short leftward arrow precedes “the same ones”.

Original source details: GELFAND-P000532

They span a simplex if $gBg^{-1} \subseteq K_1 \cap \ldots \cap K_e$



**Manuscript excerpt.** The sign between $gBg^{-1}$ and the intersection reads $\subseteq$.

Original source details: GELFAND-P000532

$n = 2 \Rightarrow$ it is contractible.



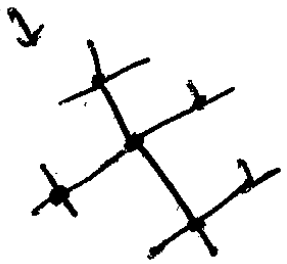
**Uncertain reading.** The heavily overwritten letter before $= 2$ is provisionally set as $n$.

Original source details: GELFAND-P000532



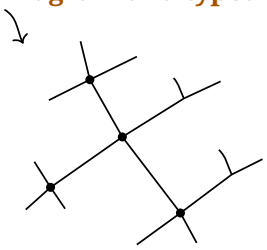
**Undeciphered passage.** A heavily struck-out rectangular fragment sits to the left of the tree sketch. The writing inside is illegible.

Original source details: GELFAND-P000532



Original source details: GELFAND-P000532

**Diagram and typed description.**

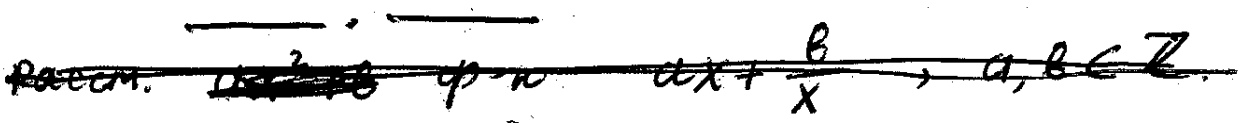


The slanted arrow at left is separate from the upper node. Two hooked strokes appear on the right branches.

Consider $V_3(F_q)$. Points of type $1$ — lines in $V_3$, $2$ — planes in $V_3$.
Incidence $\Rightarrow$ spans a simplex.

D. Kazhdan.

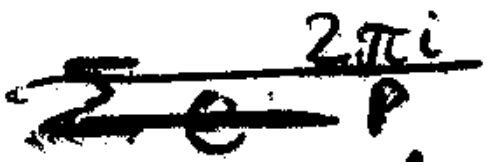
~~Consider the func.~~ $\sim\!\!\sim ax + \frac{b}{x},\ a, b \in \mathbb{Z}. \sim\!\!\sim$



Original source details: GELFAND-P000532

**Undeciphered passage.** In the crossed-out line between “Consider” and “the func.” there is a raised $2$. The writing below it remains undeciphered.

[in the left margin, crossed out:] $\sum e^{\frac{2\pi i}{p}}$.

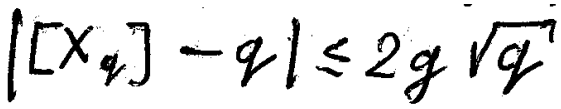


Original source details: GELFAND-P000532

**Manuscript excerpt.** In the crossed-out expression the numerator reads $2\pi i$ and the denominator $p$.

$X$ — a curve over $F_q$. How many points are on it?

$$\left|[X_q] - q\right| \leq 2g\sqrt{q}$$

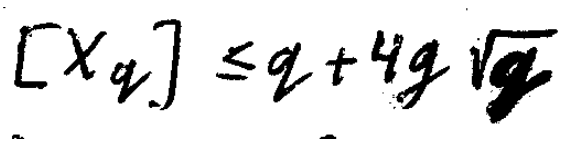


Original source details: GELFAND-P000532

**Manuscript excerpt.** Inside the absolute-value signs the manuscript reads $[X_q] - q$.

($g$ — genus of the curve, $[X_q]$ number of points of $X$ over $F_q$).
This is equiv. to the Weil conjecture.
Stepanov's proof (in Bombieri's presentation at the Bourbaki sem.). He proved

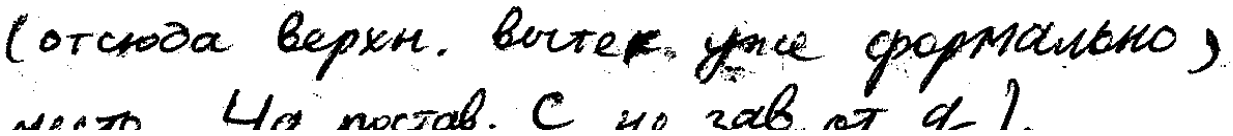
$$[X_q] \leq q + 4g\sqrt{q}$$



**Manuscript excerpt.** The heavily retraced letter under the radical reads $q$.

Original source details: GELFAND-P000532

(from this the upper bound follows formally).



Original source details: GELFAND-P000532

**Uncertain reading.** The final stroke in the Russian abbreviation for "follows" is retraced. The reading is provisional.

(It even suffices, instead of $4g$ to put $C$, independent of $q$).
$X$ — a curve, on it a divisor $D$. Number of funcs. divisible by the divisor $D$:

$$L(D) = \deg D - g + 1 \qquad \text{when } \deg D > 2g$$

(if there are many more poles than zeros).



Original source details: GELFAND-P000532

**Manuscript excerpt.** The left side is $L(D)$. The condition has a strict $>$ before $2g$.

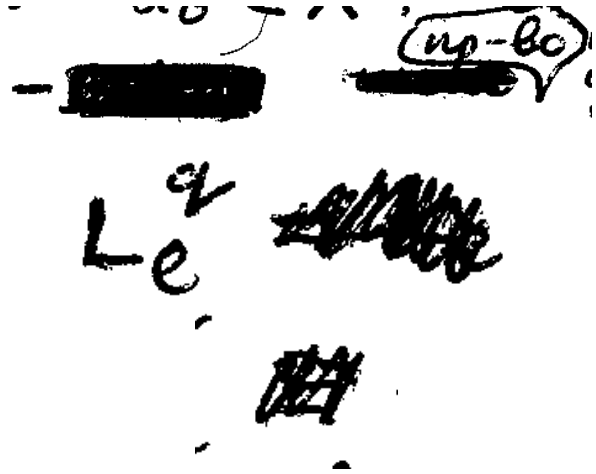
GELFAND-P000533

[source number: 38]

Let $x_0 \in X$, fixed.

$L_e$ — [inserted above:] space of funcs. with a pole of mult. $\leq e$ at pt. $x_0$.

$L_e^q$ has the same number of elements as $L_e$.

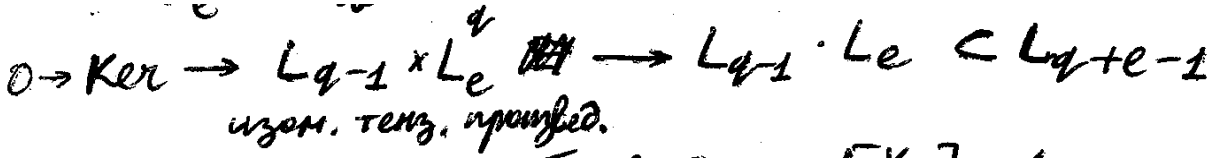


**Undeciphered passage.** Two heavily crossed-out entries follow the dash on the $L_e$ line. One follows $L_e^q$, and one precedes the product arrow. "Space" is circled above the second cancelled entry. The cancelled entries are illegible.

Original source details: GELFAND-P000533

$$0 \to \text{Ker} \to L_{q-1} \times L_e^q \longrightarrow L_{q-1} \cdot L_e \subset L_{q+e-1}$$

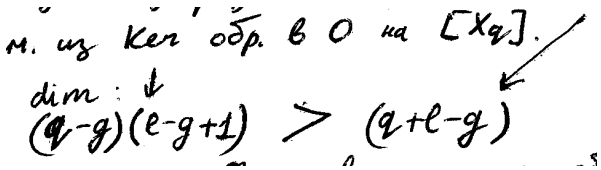
[under the middle term:] isom. tensor prod..



Original source details: GELFAND-P000533

**Manuscript excerpt.** The middle term uses $\times$. Below it is the note "isom. tensor prod." The index in $L_e$ is provisionally set as $e$.

Elems. of Ker are $0$ on $[X_q]$.



Original source details: GELFAND-P000533

**Diagram and typed description.**

dim : $\downarrow$ $\searrow$

$(q-g)(e-g+1) > (q+e-g)$

A short downward arrow points to the first product, and a longer diagonal arrow points to the right side.

And there is a kernel. Funcs. of low deg. are $0$ on $[X_q]$.



**Diagram and typed description.**

Original source details: GELFAND-P000533

GELFAND-P000534

19/XI-73. Gelfand's sem.. (last meeting this semester).

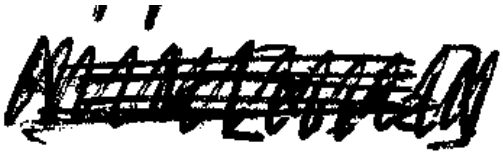


**Manuscript excerpt.** The Roman XI has a rule above and below.

Original source details: GELFAND-P000534

Serre's problem on polyn. in 3 variables has been solved.

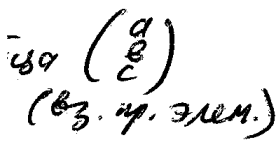
Vaserstein.



**Undeciphered passage.** An elongated entry to the right of the surname is crossed out with horizontal and slanting strokes. Its reading remains unresolved.

Original source details: GELFAND-P000534

Def. The stab. rank of the ring $A \leq 2$, if $\forall$ unimod. column $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ (rel. prime elems.)

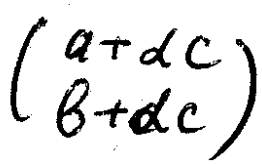


**Manuscript excerpt.** The parenthesis "(rel. prime elems.)" is below the three-entry column.

Original source details: GELFAND-P000534

$\exists \alpha, \beta \in A$, such that

$$\begin{pmatrix} a+\alpha c \\ b+\alpha c \end{pmatrix} \quad \text{is unimodular.}$$

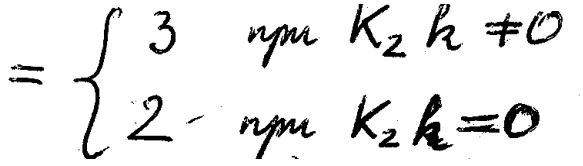


**Manuscript excerpt.** Both entries of the short column contain $\alpha$. The condition above gives $\alpha, \beta$.

Original source details: GELFAND-P000534

Thm.

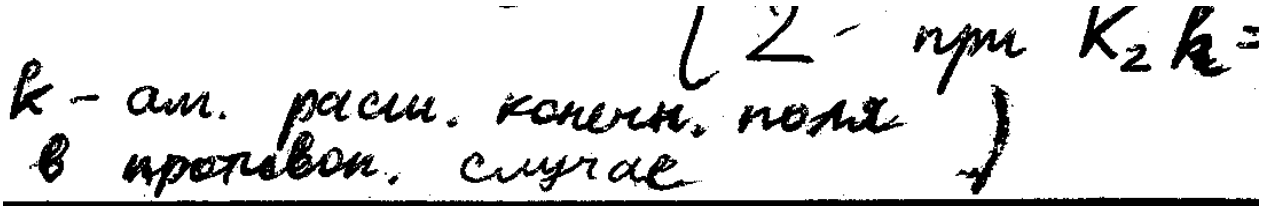
$$\text{st. rank } k[x_1, x_2] = \begin{cases} 3 & \text{when } K_2 k \neq 0 \\ 2 — & \text{when } K_2 k = 0 \end{cases}$$



**Manuscript excerpt.** There is a short stroke after 2 before "when".

Original source details: GELFAND-P000534

$$\left(K_2 k = \begin{cases} 0 & k \text{ — alg. ext. of a finite field} \\ \neq 0 & \text{otherwise} \end{cases}\right)$$



Original source details: GELFAND-P000534

**Uncertain reading.** The Russian abbreviation before “ext.” is provisionally read as “ал.” and translated as “alg.”.

GELFAND-P000535

[source number: 45]

Example over $\mathbb{R}$.



**Undeciphered passage.** A long entry after $\mathbb{R}$ is crossed out with slanting and horizontal strokes.

Original source details: GELFAND-P000535

$$\begin{pmatrix} x_1 \\ x_2 \\ 1 - x_1^2 - x_2^2 \end{pmatrix}$$

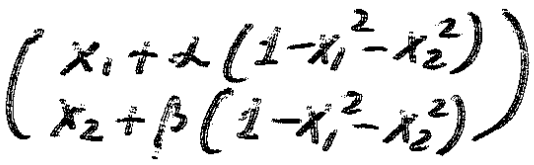
there do not exist $\alpha, \beta \in \mathbb{R}$, such that

$$\begin{pmatrix} x_1 + \alpha(1 - x_1^2 - x_2^2) \\ x_2 + \beta(1 - x_1^2 - x_2^2) \end{pmatrix}$$



**Undeciphered passage.** There is a dense cancellation between “not” and “exist”. The underlying writing is illegible.

Original source details: GELFAND-P000535



**Manuscript excerpt.** The second entry of the short column here contains $\beta$.

Original source details: GELFAND-P000535

(by degree).



**Manuscript excerpt.** Below the line with “do not exist” is the note “(by degree)”.

Original source details: GELFAND-P000535

There is a curve $C(x_1, x_2) \equiv C = 0$

Winding index with respect to the set $a(x_1, x_2) = 0$, $b(x_1, x_2) = 0$

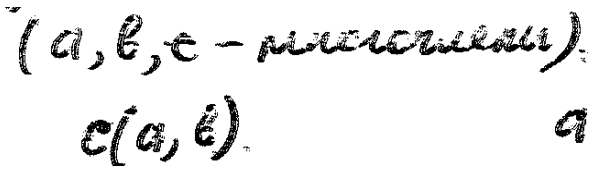


Original source details: GELFAND-P000535

**Manuscript excerpt.** The abbreviation before the equations reads “set”.

($a, b, c$ — polynomials).

$c(a, b)$. $a = x_1$, $b = x_2$ — number of turns about the origin.



Original source details: GELFAND-P000535

**Uncertain reading.** The sign before $(a, b)$ is provisionally set as $c$. Its loop also permits $e$.

$B$ — a ring

$K_1 \mathrm{Sp}\, B =$ Quotient of symplectic matrices of order $2n$ by the elementary ones.

Question: $z_1^2 + z_2^2 = 1$ index $0$ or non $0$?



Original source details: GELFAND-P000535

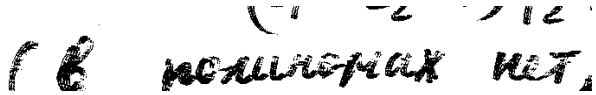
**Manuscript excerpt.** Both signs in the question are read as $0$. A faint closing stroke is visible at the right of each oval.

$\Longleftrightarrow \exists$ are there 2 funcs. $f_1(z_1, z_2)$ and $f_2(z_1, z_2)$, such that

(entire, entire of exp. growth, etc.)

$$\begin{matrix}(z_1^2 + z_2^2 - 1) f_1 + z_1 \\ (z_1^2 + z_2^2 - 1) f_2 + z_2\end{matrix}$$ have no common div.

(no for polynomials, yes for cont. funcs.).



Original source details: GELFAND-P000535

**Manuscript excerpt.** The first phrase reads “for polynomials”.

Kazhdan. Derived category.

$A$ — a ring, ~~$\forall$ — modules over $A$.~~



Original source details: GELFAND-P000535

**Uncertain reading.** The crossed-out initial sign is provisionally set as $\forall$. “Modul...” and “over $A$” are visible under the cancellation.

~~$\mathrm{Hom}(M, N)$ — functor~~



Original source details: GELFAND-P000535

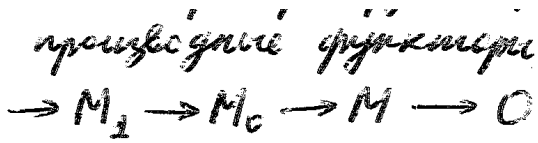
**Uncertain reading.** The word after the crossed-out $\mathrm{Hom}(M, N)$ is provisionally read as “functor”.

Consider the category of modules over $A$.

(Everything will be illustrated by the functor $\mathrm{Hom}(M, N)$).

Define the derived functors $\mathrm{Ext}^i(M, N)$:

$$\ldots \to M_1 \to M_0 \to M \to 0 \qquad (M_i \text{ — free})$$



Original source details: GELFAND-P000535

**Uncertain reading.** The subscript before $M$ is open at right. It is provisionally read as $0$.

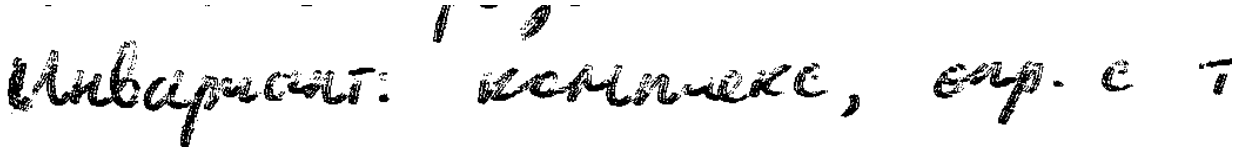
Consider the complex $C^i = \mathrm{Hom}(M_i, N)$

$$0 \to C^0 \to \ldots \to C^{i-1} \to C^i \to \ldots$$

Its cohomology grs. $\mathrm{Ext}^i(M, N)$. $\mathrm{Ext}^0(M, N) = \mathrm{Hom}(M, N)$.

Any 2 resol. are homotopic.

Invariant: complex, def. up to homotopy.



Original source details: GELFAND-P000535

**Manuscript excerpt.** The first word on the line reads “Invariant”.

There are complexes with isom. homologies that are not homotopic. (see topology).

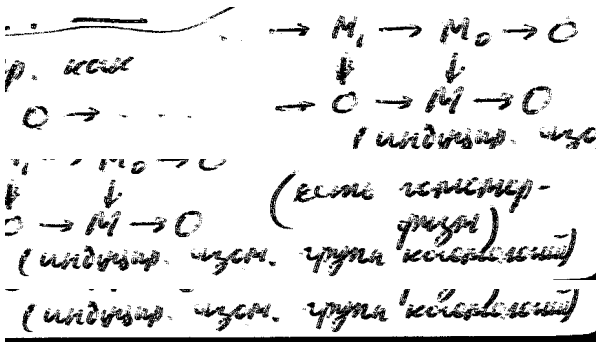
The natural object is a complex.

Let $\dot{M}$ — a complex $\quad 0 \to M_1 \to \ldots \to M_i \to 0$

$\dot{N}$ — — " — $\quad 0 \to N_1 \to \ldots \to N_j \to 0$

$\mathrm{Hom}(\dot{M}, \dot{N})$

(!) $M$ is interpr. as



Original source details: GELFAND-P000535

**Diagram and typed description.**

$$\begin{array}{ccccccc} \cdots & \longrightarrow & M_1 & \longrightarrow & M_0 & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow \cdots \longrightarrow & 0 & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

(if a homomorphism)

(induc. an isom. of cohomology groups)

The lower letter beneath $M_0$ is provisionally set as $M$. The small stroke cannot securely be separated from the arrow. The caption at right is provisional.

GELFAND-P000536

$M$ is replaced by a compl. of free modules, for which $\to 0 \to M \to 0 \to \dots$ is mapped with an induced isom. of cohomology grs..

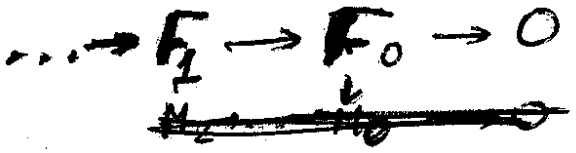
(quasi-isomorphic complexes)

Resolution — replacement of $M$ by a quasi-isom. compl. of free modules.

---

$\mathrm{Hom}(\dot{M}, \dot{N})$ in the derived category.

Replace $\dot{M}$ by a quasi-isom. complex of free modules



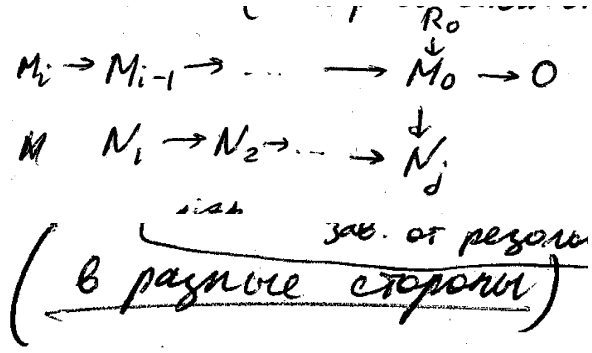
Original source details: GELFAND-P000536

**Diagram and typed description.**

$$\cdots \longrightarrow F_1 \longrightarrow F_0 \longrightarrow 0$$
$$\downarrow$$
$$\cancel{M_2 \longrightarrow M_0 \longrightarrow 0}$$

The first crossed-out subscript is provisionally read as 2. A downward arrow is visible below $F_0$. Below $F_1$ a separate arrow cannot be distinguished from the subscript stroke.

Construct the complex $C^i = \mathrm{Hom}^i(R, N)$ — derived functor. (maps with shift by $i$)



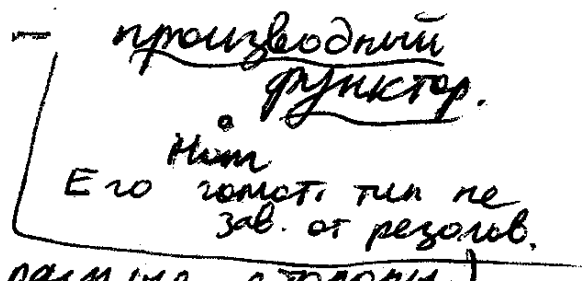

Original source details: GELFAND-P000536

**Diagram and typed description.**

$$\begin{array}{ccccccc} & & & & R_0 & & \\ & & & & \downarrow & & \\ M_i & \longrightarrow M_{i-1} & \longrightarrow \cdots & \longrightarrow & M_0 & \longrightarrow & 0 \\ & & & & \downarrow & & \\ M & N_i \longrightarrow N_2 & \longrightarrow \cdots & \longrightarrow & N_j & & \end{array}$$

(in opposite directions)

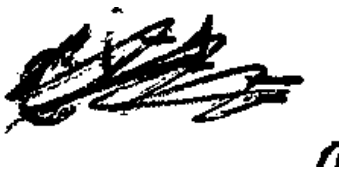
The first subscript below is provisionally set as $i$. An isolated $M$ stands to the left of the lower chain. A faint leftward arrow is visible under the note at right.




Original source details: GELFAND-P000536

**Diagram and typed description.**

Its [inserted above: $\overset{\circ}{\mathrm{Hom}}$] homot. type is independent of the resol.



Original source details: GELFAND-P000536

**Undeciphered passage.** A short, crossed-out fragment lies below the chains. The underlying writing is illegible.

2 complexes of free modules are quasi-isom. $\Longleftrightarrow$ they are homot. equiv.



Original source details: GELFAND-P000536

**Manuscript excerpt.** Two heads are visible on the retraced arrow, rendered as $\Longleftrightarrow$.

$K(A)$ — the categ. of complexes. Morphisms — homot. classes of morphisms.
The category $D(A)$. $\mathrm{Ob}\,\mathcal{D}(A) = \mathrm{Ob}(K(A))$, and $\mathrm{Mor}(\mathcal{D}(A))$ — a seq.

$$\dot{M} \to \dot{M}_1 \to \ldots \to \dot{M}_k \to \dot{N} = \mathrm{Mor}(\dot{M}, \dot{N})$$

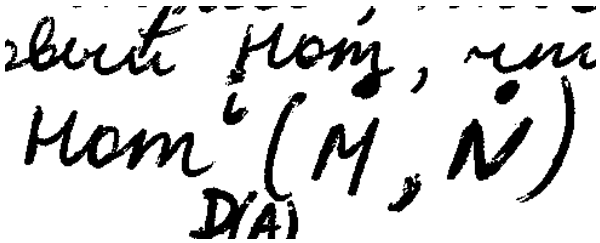
where $\to$ — a morph. to the right or a quasi-isom. to the left.
We want: to introduce a new Hom, so that it becomes a complex.
~~$\mathrm{Hom}_{D(A)}$~~



**Uncertain reading.** The crossed-out fragment shows Hom. The subscript resembles $D(A)$. The rest of the entry remains undeciphered.

Original source details: GELFAND-P000536

$$\mathrm{Hom}^i_{D(A)}(\dot{M}, \dot{N})$$



**Uncertain reading.** There is a dot over $N$. The mark over $M$ touches a letter on the line above and is provisionally rendered as a dot.

Original source details: GELFAND-P000536

. . . . . .

$F$ — a contravar. functor; $F : \underset{\text{modules over } A}{M(A)} \to \underset{\text{modules over } B}{M(B)}$

$\dot{M} \in D(A)$

It is bad to take $F(\dot{M})$, since quasi-isoms. will not go to quasi-isoms.

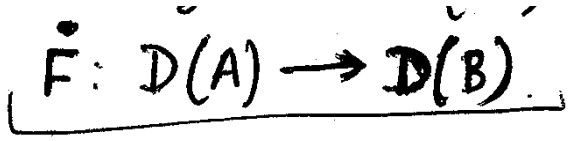


**Diagram and typed description.**
$\dot{F} : D(A) \longrightarrow D(B)$.

Original source details: GELFAND-P000536

(Its cohom. are the ordinary derived functors)

$\dot{M} \in D(A)$. Choose a representative of free modules $\dot{M}$ and apply to it $F$ — we obtain a new complex. Regard it as an element of $D(B)$. Well-definedness follows from the stated theorem on homotopy.

More gen. def. $F : A \to B$, $\dot{F} : D(A) \to D(B)$

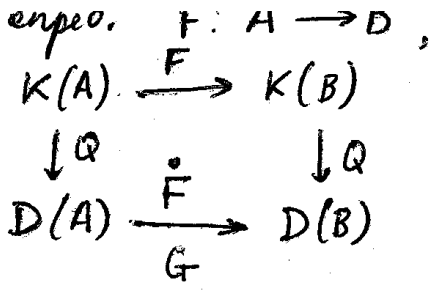
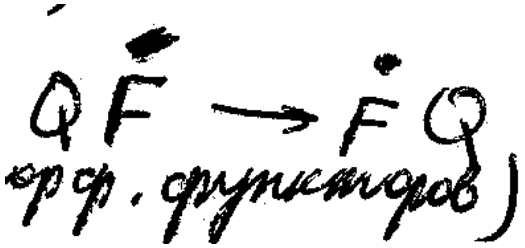



**Diagram and typed description.**

$$\begin{array}{ccc} K(A) & \xrightarrow{F} & K(B) \\ \downarrow Q & & \downarrow Q \\ D(A) & \underset{G}{\xrightarrow{\dot{F}}} & D(B) \end{array}$$

Original source details: GELFAND-P000536

Req.: $\exists$ a map $Q\overline{F} \to \dot{F}Q$ (morph. of functors)



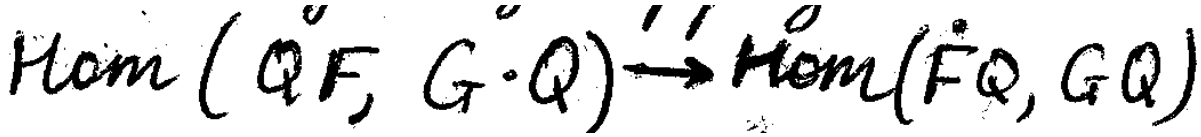

**Uncertain reading.** The first superscript mark is a heavy slanted stroke, provisionally set as a bar. The second $F$ has a dot.

Original source details: GELFAND-P000536

$\forall G$ we obtain an isomorphism

$$\mathrm{Hom}(QF, G \cdot Q) \longrightarrow \mathrm{Hom}(\dot{F}Q, GQ)$$

would be an isomorphism for $\forall G$.




Original source details: GELFAND-P000536

**Manuscript excerpt.** The left side reads $QF$ without a superscript mark, and the right side $\dot{F}Q$. The arrow points right.

GELFAND-P000537

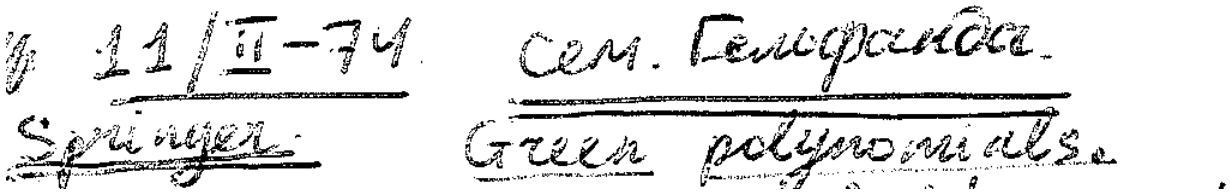
11/II-74. Gelfand's sem..




**Uncertain reading.** A crossed-out entry lies to the left of 11/II-74. Its first stroke resembles 1. The full entry remains unreadable.

Original source details: GELFAND-P000537

Springer. Green polynomials.



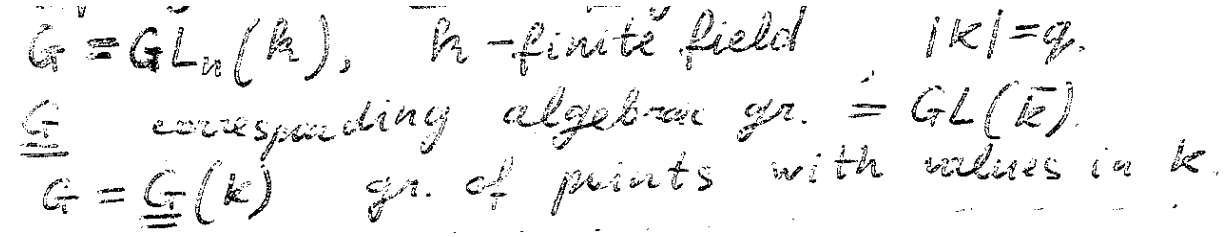

Original source details: GELFAND-P000537

**Manuscript excerpt.** The title and the lines about the group are written in English in the manuscript.

$$G = \mathrm{GL}_n(k), \qquad k \text{ — finite field}, \qquad |k| = q.$$

$G$ corresponding algeb-raic gr. $= \mathrm{GL}(\bar{k})$.

$G = G(k)$ gr. of points with values in $k$.




Original source details: GELFAND-P000537

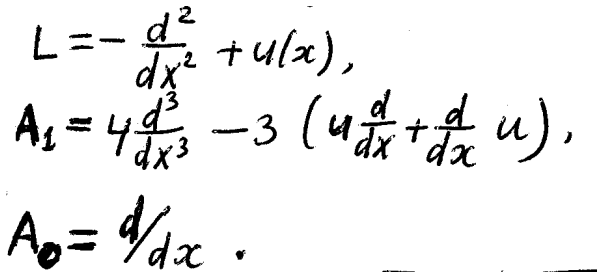
**Manuscript excerpt.** Two rules below $G$ distinguish the algebraic group from the plain $G$. After "gr." the manuscript gives $\mathrm{GL}(\bar{k})$ without the subscript $n$.

GELFAND-P000538

25/II-74. Gelfand's sem.

S. P. Novikov. (continued).

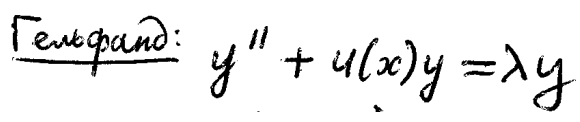
$$L = -\frac{d^2}{dx^2} + u(x),$$
$$A_1 = 4\frac{d^3}{dx^3} - 3\left(u\frac{d}{dx} + \frac{d}{dx}u\right),$$
$$A_0 = d/dx.$$



**Manuscript excerpt.** The definition of $L$ has a minus sign before the second derivative.

Original source details: GELFAND-P000538

Gelfand: $y'' + u(x)y = \lambda y$



**Manuscript excerpt.** There is no separate minus sign before $y''$ on this line.

Original source details: GELFAND-P000538

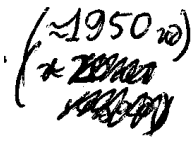


Original source details: GELFAND-P000538

**Undeciphered passage.** A small entry to the right of $\lambda_y$ lies under a heavy cancellation.

$$u(x+T) = u(x)$$

structure of the spectrum — shortest proof in Gelfand DAN ($\approx 1950$ ).



Original source details: GELFAND-P000538

**Undeciphered passage.** Two crossed-out entries lie below the year. A sign resembling $\approx$ precedes the upper one.

$u(x)$ — real and its closure (from $C_0^\infty$) is self-adj.

Another case:

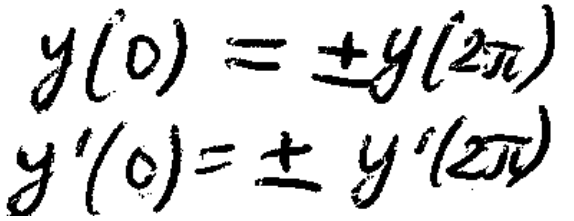


Original source details: GELFAND-P000538

**Diagram and typed description.**

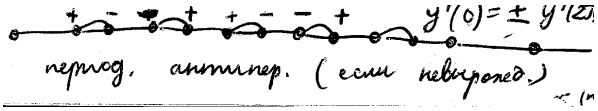



$$y(0) = \pm y(2\pi), \qquad y'(0) = \pm y'(2\pi)$$



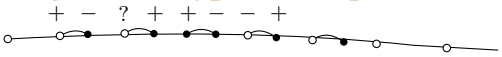
Original source details: GELFAND-P000538

**Manuscript excerpt.** Both boundary conditions carry a $\pm$ sign.




Original source details: GELFAND-P000538

**Diagram and typed description.**



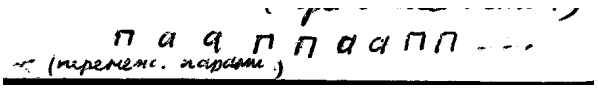

periodic, antiper. (if nondegenerate)

The question mark stands for the overwritten label above the second arc. The fill of partially retraced circles follows the visible center.

(alternation!)

п а а п п а а п п ...

$\sim$ (altern. in pairs)




Original source details: GELFAND-P000538

**Uncertain reading.** A short wavy stroke below the letters is rendered as $\sim$. The final Russian word is provisionally read as “in pairs”.

GELFAND-P000539

[source number: 59]

On the line, consid. the following bound. conditions:

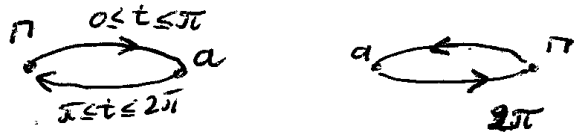
$$y(x+T) = e^{it}y(x)$$
$$y'(x+T) = e^{it}y'(x)$$



Original source details: GELFAND-P000539

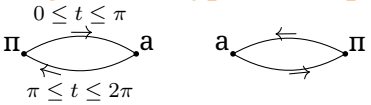
**Uncertain reading.** The sign after $i$ in the first exponent is heavily overwritten. The reading $t$ is provisional. The second exponent $it$ is independently legible.

$\forall t$ self-adj. prob. with discr. spectrum.



Original source details: GELFAND-P000539

**Diagram and typed description.**

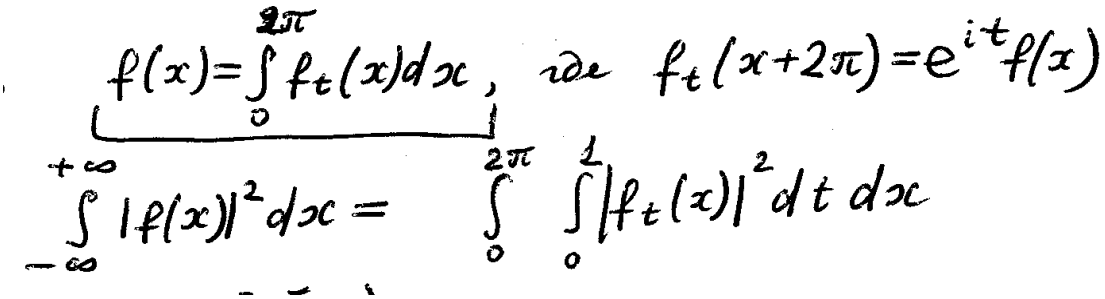



Lemma. $\forall f(x)$,

$$f(x) = \int_0^{2\pi} f_t(x)\,dx, \qquad \text{where } f_t(x+2\pi) = e^{it}f(x)$$

At the same time

$$\int_{-\infty}^{+\infty} |f(x)|^2\,dx = \int_0^{2\pi}\int_0^1 |f_t(x)|^2\,dt\,dx$$



Original source details: GELFAND-P000539

**Manuscript excerpt.** The first integral has $dx$, and the right side of the condition has $f(x)$ without a subscript. In the second equality the upper limits are $2\pi$ and $1$, with differentials $dt\,dx$.

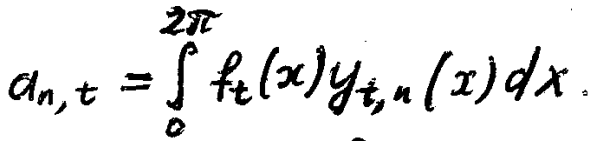
(reduc. to a comp. group — Weyl)

$0 \le x \le 2\pi \Rightarrow$ expand in a Fourier series in the eigenfunctions of the problem.

Obtain an integral over the whole line now $\| \; f_t(x) = \sum_n a_{n,t} y_{t,n}(x)$

$y_{t,n}$ — Bloch funcs.

$$a_{n,t} = \int_0^{2\pi} f_t(x) y_{t,n}(x)\, dx.$$

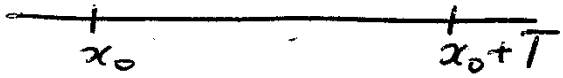


Original source details: GELFAND-P000539

**Manuscript excerpt.** There is no bar over $y_{t,n}(x)$ in the integral.

$\rightleftarrows$ is calculated $\frac{d\lambda}{dt}$.

S.P.N.



Original source details: GELFAND-P000539

**Diagram and typed description.**




Consid. $y(x_0, x)$

$Ly = \lambda y, \quad \lambda = k^2$

$y|_{x=x_0} = 1$
$y'|_{x=x_0} = ik$ — one eigenfunc. (closer to $e^{ikx}$).



Original source details: GELFAND-P000539

**Undeciphered passage.** A heavily crossed-out entry before “and”.

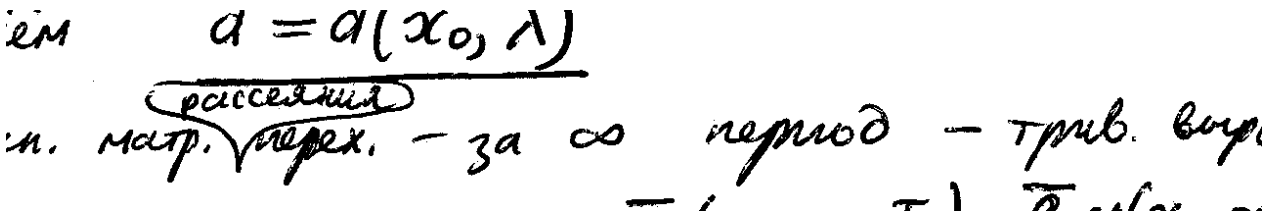
and $\bar{y}(x_0, x)$ — another eigenfunc.

Monodromy matr.

$$y(x_0, x+T) = a y(x_0, x) + b \bar{y}(x_0, x),$$

moreover $a = a(x_0, \lambda)$.

(ord. transfer matr. [above the line: scattering] — over an $\infty$ period — triv. degener. case).



Original source details: GELFAND-P000539

**Manuscript excerpt.** “Scattering” is circled and linked by a stroke to the line at “transfer”.

$$\bar{y}(x_0, x+T) = \bar{b}y(x_0, x) + \bar{a}\bar{y}(x_0, x).$$

$$T = \begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix} \text{ — monodromy matr.} \qquad T = T(x_0, \lambda)$$

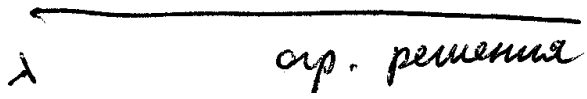
$\det = 1$ (the Wronskian is constant)

$$|a|^2 - |b|^2 = 1;$$
$$\operatorname{Sp} T = 2a_R = a + \bar{a};$$

Eigenvalues

$$\alpha_\pm(\lambda) = a_R \pm \sqrt{a_R^2 - 1}$$

consid. the graph in $\lambda$




Original source details: GELFAND-P000539

**Diagram and typed description.**

$\lambda$

There is a small hook at the left of the line.

bounded solutions $\iff |a_R| \le 1 \implies$ obtain $e^{\pm it}$
unbounded solutions $\iff |a_R| > 1 \implies$ obtain $e^{\pm t}$

GELFAND-P000540

$(y, y') = (1, ik)$

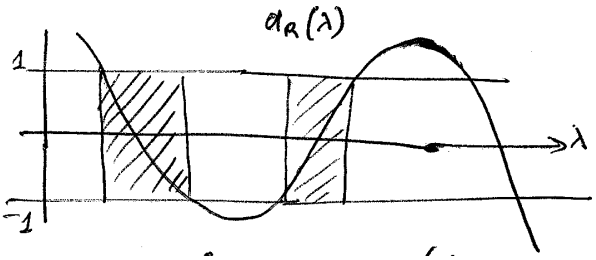
Relation to the spectrum.



Original source details: GELFAND-P000540

**Uncertain reading.** Before “For” is a dense mark with two lower ends. The sign is undeciphered.

For $|a_R| \le 1$ there is Bloch spectrum.




Original source details: GELFAND-P000540

**Diagram and typed description.**

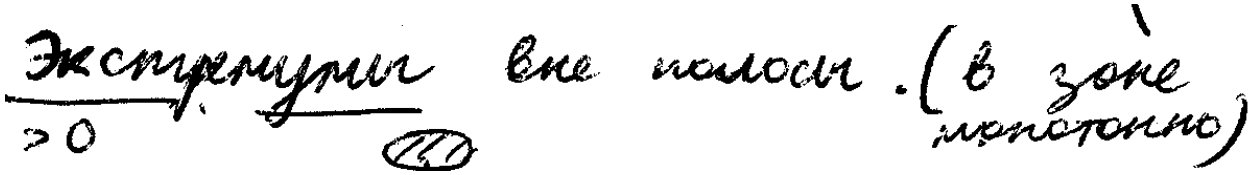
$a_R(\lambda)$

1

λ

−1

Extrema outside the strip. (monotonically in the band)



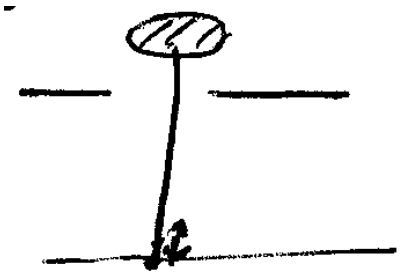

Original source details: GELFAND-P000540

**Manuscript excerpt.** “Extrema” is underlined.

G. $u(x) = \text{const}$ — spectrum $> 0$

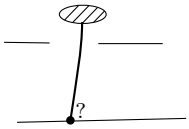
$u(x) < 0$ is possible.

stability bands extend into the neg. region



Original source details: GELFAND-P000540

**Diagram and typed description.**




The question mark at the lower point stands for the undeciphered mark.

$$-y'' + (a + b\cos 2x)y = 0$$

bands of stability and instab.

1st stab. band when $a + b\cos 2x < 0$

(does not occur for constant ones)

N. monot. in the band — proved for small potentials

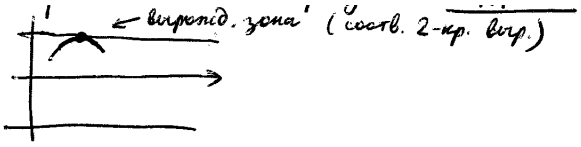


Original source details: GELFAND-P000540

**Manuscript excerpt.** The Russian abbreviations read “монот.” and “доказ.”.

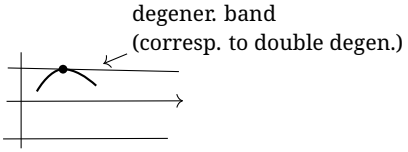
G. Contest for a proof (for next time)

N. Periodic and antiperiodic $|a_R| = 1$ — band edges.



Original source details: GELFAND-P000540

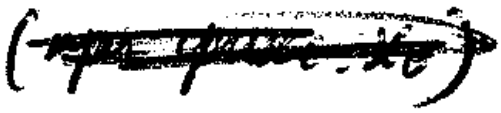
**Diagram and typed description.**




For a const. or zero one it is always thus:

$$\begin{pmatrix} e^{iT} & 0 \\ 0 & e^{-iT} \end{pmatrix} \qquad a_R = \cos T$$

On inverse problems: $a_R(k) = a_R(x_0, k)$ does not determine the potential



Original source details: GELFAND-P000540

**Undeciphered passage.** After “potential” there is a crossed-out entry in parentheses. Its reading is unresolved.

(≈ inform. about the spectrum).

1) nondegener. $\lambda_n : b(x_0, \lambda) \neq 0$

2) doubly degener. $\lambda_n : b(x_0, \lambda_n) \equiv 0$.



**Uncertain reading.** A separate heavily overwritten sign precedes “doubly degener.”.

Original source details: GELFAND-P000540

What scattering data determine $u(x)$ (apart from the spectrum). Dubrovin and Faddeev's students are working on it. _____ · _____

If from some index onward $\forall$ eigenvalues are doubly degener., then the forbidden bands contract to a point. _____ · _____

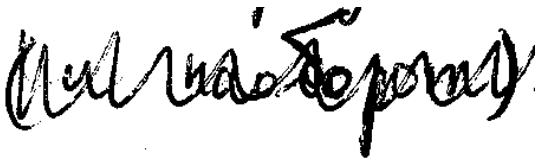
Theorem.

1) Consid. $\sum_{i=0}^{n} Q_i(u, \dots, u^{(k_i)}) \frac{d^i}{dx^i}$ and let

$$[A, L] = P(u, \dots, u^{(\ell)}) \qquad \text{(multiplication oper.)}$$

and consid. the diff. eq. $P(u, \dots, u^{(\ell)}) = 0$ and suppose it is periodic. Then it is a finite-gap potential.

~~or conversely~~ (in fact, the converse is true too).



**Manuscript excerpt.** “Or conversely” is crossed out with a wavy stroke.

Original source details: GELFAND-P000540

GELFAND-P000541

[source number: 60]

one of them at each odd order.

(and any lin. combination)

2) They are all Hamilton. syst. with $n$ degrees of freedom. _____ · _____

N.'s question: can any potential be approx. by finite-gap ones? _____ · _____

$$P_0 = u', \qquad P_1 = 6uu' - u''' + cu', \dots$$

2) $\exists$ a matrix of order 2

$$\Lambda(u, \dots, u^{(k)}, \delta) = \begin{pmatrix} \lambda & \mu \\ \bar{\mu} & \bar{\lambda} \end{pmatrix}$$

polynomial of degree $2k+1$ in $\delta$

$$\lambda + \bar{\lambda} = 0.$$

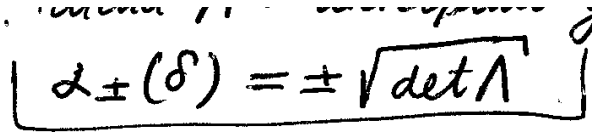
and another matrix

$$Q(u, \delta) = i\delta \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \frac{iu}{2\delta} \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix},$$

such that $P(u, \dots, u^{(2k+1)}) = 0$ is equiv. to the eq. $\Lambda' = [\Lambda, Q]$

(in partic., the eigenvalues of $\Lambda$ — integrals of the eq.)

$\alpha_\pm(\delta) = \pm\sqrt{\det \Lambda}$ — a polynomial in $\delta$.



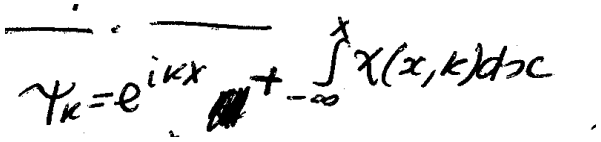
**Manuscript excerpt.** The symbol reads $\alpha_\pm(\delta)$. The radicand is written $\det \Lambda$.

Original source details: GELFAND-P000541

Commuting integrals — max set. _____ · _____

Consid. an eigenfunc.:

$$\psi_k = e^{ikx + \int_{-\infty}^{x} \chi(x,k)\,dx}, \qquad L\psi = k^2\psi.$$



**Undeciphered passage.** Below the exponent line after $ikx$ there is a heavy cancellation. The plus and the integral are at exponent height.

Original source details: GELFAND-P000541



**Undeciphered passage.** A separate crossed-out mark precedes the parenthesis. There is an extra heavy stroke at the end of the word for "correction".

Original source details: GELFAND-P000541

(correction to the phase $\int_{-\infty}^{x} \chi(x,k)\,dx$)

$I = \int \chi(x,k)\,dx$ conserved

$$\chi(x,k) = \sum_{n\geq 1} \frac{\chi_n(x)}{k^n}, \qquad \chi_n(x) \text{ — a polynomial}$$

$$I_n = \int_{-\infty}^{\infty} \chi_n(x)\,dx$$

as a func. of $u$ — int. of the KdV eq.

Let $(*)$ $\underline{\dot{u} = [A, L]}$ — mult. op.;



**Undeciphered passage.** A crossed-out word precedes $L$. Its reading is unresolved.

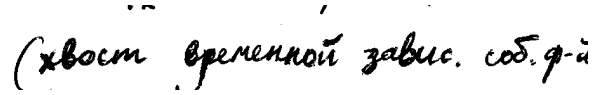
Original source details: GELFAND-P000541

$L$ — Sturm–Liouville oper..

$\psi_k$ — its eigenfunc. (evolution of $\psi_k$ — according to the potential $u$).

$$\dot{\psi}_k = A\psi_k + \lambda\psi_k + \mu\bar{\psi}_k, \qquad \Lambda = \begin{pmatrix} \lambda & \mu \\ \bar{\mu} & \bar{\lambda} \end{pmatrix}.$$

(tail of the time dep. of the eigenfuncs.)



**Uncertain reading.** The initial letter is heavily overwritten. “Tail” is a provisional reading.

Original source details: GELFAND-P000541

Apply $d/dx$ — obtain an eq.

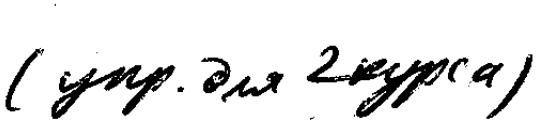
$\psi_k(x_0, x)$

At $x = x_0$ $(1, ik)$

$\underline{\dot{T} = [\Lambda, T]}$ — evolution of the monodromy matr. $T$

|| *Novikov’s thm.*

$$\frac{dT}{dx_0} = [Q, T], \qquad Q \text{ — stand. matrix (exercise for the second year)}$$



**Manuscript excerpt.** The parenthesis reads “exercise for the second year”.

Original source details: GELFAND-P000541

$\Lambda' = [\Lambda, Q]$. Band edges are the roots of a polynomial. (for stat. solns.)

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Yakubovich, Starzhinskii: "Diff. eqs. with periodic coefficients".



Original source details: GELFAND-P000541

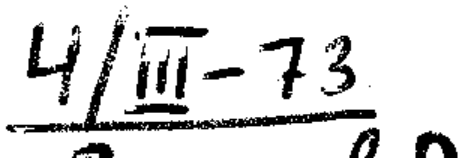
**Manuscript excerpt.** The upper rule touches the surname "Yakubovich".

GELFAND-P000542

4/III-73 Gelfand's sem..

Zakharov V.

**Manuscript excerpt.** The year is written "73".

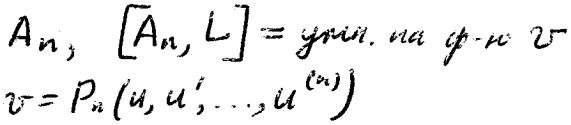


Original source details: GELFAND-P000542

G. $L = -\dfrac{d^2}{dx^2} + u(x)$. Then $\exists$ oper. $A_n$, $[A_n, L] =$ mult. by the func. $v$

$$v = P_n(u, u', \ldots, u^{(n)})$$

**Manuscript excerpt.** In both lines the letter reads $v$.



Original source details: GELFAND-P000542

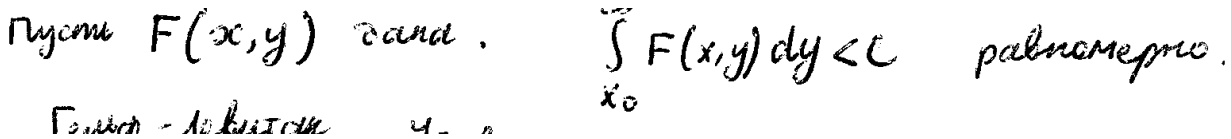
$$A_n = \frac{d^{2n+1}}{dx^{2n+1}} + \ldots \qquad \text{lower terms}$$

How to find $A_n$? How are $A_n$ and $P_n$ related?

Relation to the trace formula? Other eqs?

3. Results of A. B. Shabat and Zakharov.

Let $F(x, y)$ be given. $\int_{x_0}^{\infty} F(x, y)\, dy < C$ uniformly.



Original source details: GELFAND-P000542

**Manuscript excerpt.** The word after $F(x, y)$ reads "given".

Gelf.—Levitan. Eq.

$$K(x,y) + F(x,y) + \int_x^\infty K(x,s)F(s,y)\,ds = 0 \tag{1}$$

$x$ — parameter.

We shall consid. them as oper. kernels.

$$1 + \hat{F} = (1 + \hat{K}_+)^{-1}(1 + \hat{K}_-), \qquad \hat{K}_\pm \text{ — Volterra.}$$

$$\begin{aligned} K_+(x,y) &= 0 \quad \text{when } y < x \\ K_-(x,y) &= 0 \quad \text{when } y > x \end{aligned}$$

This is the same thing,

$$K + F + K * F = 0 \qquad \text{form (1)}$$

Differentiate in $t$ (parameter)

$$D_tK + D_tF + D_tK * F + K * D_tF = 0 \tag{2}$$

Let $D$ and $\tilde{D}$ be other operators: $D$ acts on $F$, $\tilde{D}$ acts on $K$. We seek to have

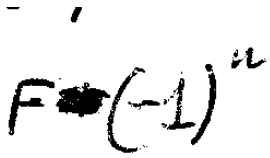
$$\tilde{D}K + DF + \tilde{D}K * F + K * DF = 0 \tag{3}$$

Let

$$DF = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}, \qquad \tilde{D}K = \frac{\partial K}{\partial x} + \frac{\partial K}{\partial y}.$$

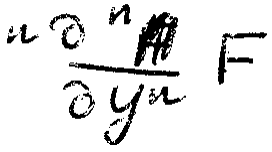
Then all is in order (direct calculation) shows that identity (3) holds.

$$DF = \frac{\partial^n}{\partial x^n}F \,?\, (-1)^n \frac{\partial^n}{\partial y^n}F \tag{4}$$



Original source details: GELFAND-P000542

**Uncertain reading.** The sign between the first $F$ and $(-1)^n$ is heavily obliterated.



Original source details: GELFAND-P000542

**Undeciphered passage.** In the numerator of the second differential operator, to the right of $\partial^n$, there is a crossed-out sign.

For it one can construct $\tilde{D}$.

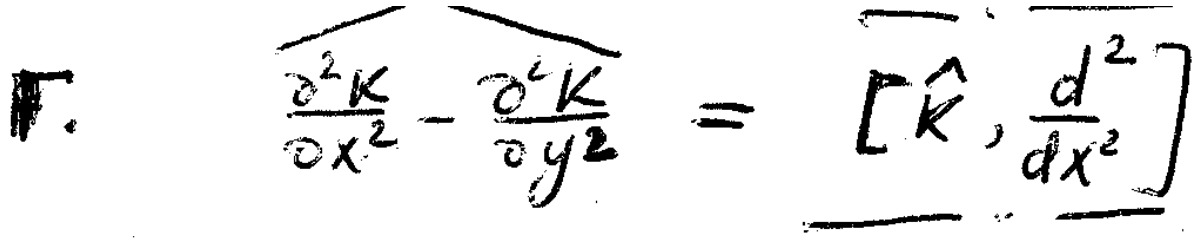
$DF = [D_n\, F]$, more precisely $DF =$ the kernel of $[\hat{D}_n, \hat{F}]$, where $\hat{D}_n = \dfrac{\partial^n}{\partial x^n}$.

Namely

$$\tilde{D}K = L_n K + (-1)^n \frac{\partial^n}{\partial y^n} K, \tag{5}$$

G.

$$\widehat{\frac{\partial^2 K}{\partial x^2} - \frac{\partial^2 K}{\partial y^2}} = \left[\hat{K}, \frac{d^2}{dx^2}\right]$$



Original source details: GELFAND-P000542

**Manuscript excerpt.** An arch spans the difference of derivatives. On the right $\hat{K}$ is first in the brackets.

where $L_n = \dfrac{\partial^n}{\partial x^n} + \ldots$ (lower terms)

$K, F$ can also be matrices.

GELFAND-P000543

[source number: 61]

$$L_n = \frac{\partial^n}{\partial x^n} + n \underbrace{\left(\frac{d}{dx} K(x,x)\right)}_{u_0(x)} \frac{\partial^{n-2}}{\partial x^{n-2}} + \ldots \tag{6}$$

There are recurr. formulas for the lower terms.

$L_n$ — nonlinear operator in $K$, but its coeffs. (all of them) are made from the quantities

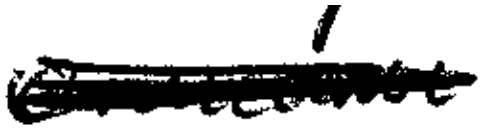
$$\xi_n(x) = \left(\frac{\partial}{\partial x} - \frac{\partial}{\partial y}\right)^n K\Big|_{y=x}$$

$$u_0 = \frac{d\xi_0}{dx}$$

The coeffs. are already constant (of these).

These are not all the operators that differ. this convolution.

These $D$ — linear opers. with const. coeffs. Variable ones are possible too. This is a Lie algebra



**Undeciphered passage.** A crossed-out entry follows the opening parenthesis. Its reading is unresolved.

Original source details: GELFAND-P000543

(all $D$, that can be extended (A. Shabat) — they are described).

Suppose that (1) is uniquely solvable for $K$ ($F$ given). And let $F$ be such that $\exists D\colon DF = 0$. Hence

$$\underline{\tilde{D}K = 0} \tag{7}$$

(from (2) and unique solvability of the eq.)

Hence

$$\underline{DF = 0 \longleftrightarrow \tilde{D}K = 0.} \tag{8}$$

Obtain some linearizable eqs. But they are uninteresting.

Let

$$D_1F = 0 \quad \text{and} \quad D_2F = 0 \tag{9}$$



**Uncertain reading.** A separate loop-like stroke stands to the left of “Let”. Its meaning is unresolved.

Original source details: GELFAND-P000543



**Uncertain reading.** The crossed-out word provisionally reads “where”.

Original source details: GELFAND-P000543

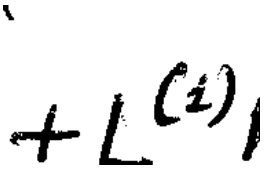
2 parameters are given $t, z$; $\quad F = F(x, y; t, z)$ and

$$D_1F = \alpha\frac{\partial F}{\partial t} + ([\hat{L}_0^{(1)}, \hat{F}])_{\text{kernel}} \tag{10}$$

$$D_2F = \beta\frac{\partial F}{\partial z} + ([\hat{L}_0^{(2)}, \hat{F}])_{\text{kernel}} \tag{11}$$

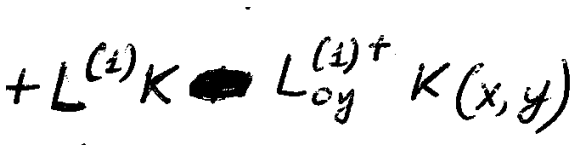
$$\hat{L}_0^{(1,2)} = \sum C^{(n)}\frac{\partial^n}{\partial x^n}$$

$$\tilde{D}_1K = \alpha\frac{\partial K}{\partial t} + L^{(i)}K\ ?\ L_{0y}^{(1)+}K(x,y) \tag{12}$$



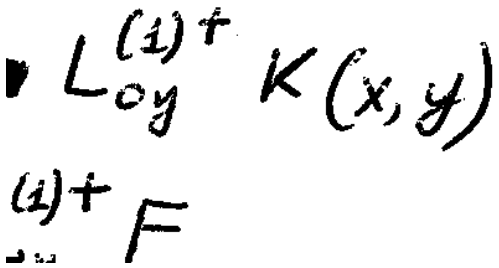
**Manuscript excerpt.** The index above the first $L$ has a distinct dot and reads $i$.

Original source details: GELFAND-P000543



**Uncertain reading.** The sign between $L^{(i)}K$ and the next term is heavily obliterated.

Original source details: GELFAND-P000543



Original source details: GELFAND-P000543

**Uncertain reading.** In (12) and on the next line the last operator carries a superscript provisionally read as $+$. The superscript is absent in (13).

$$[\hat{L}^{(1)}, \hat{F}] = L_x^{(1)} F - L_y^{(1)+} F$$

$$\tilde{D}_2 K = \beta \frac{\partial K}{\partial z} + L^{(2)} K - L_{0y}^{(2)} K \tag{13}$$

$$\boxed{D_1 F = 0,\ D_2 F = 0 \Longrightarrow \tilde{D}_1 K = 0,\ \tilde{D}_2 K = 0.} \tag{14}$$

Theorem.

$$\boxed{\beta \frac{\partial L^{(1)}}{\partial z} - \alpha \frac{\partial L^{(2)}}{\partial t} = [L^{(1)}, L^{(2)}]} \tag{15}$$

$L^{(1)}, L^{(2)}$ — linear operators (for given $K$).

In $L^{(1)}$ and $L^{(2)}$ the coeffs. are expressed in terms of

$$\xi_n(x) = \left( \frac{\partial}{\partial x} - \frac{\partial}{\partial y} \right)^n K(x,y) \Big|_{y=x}$$

GELFAND-P000544

This is a syst. of nonlin. diff. eqs. for $\xi_n(x, y, t, z)$.



Original source details: GELFAND-P000544

**Uncertain reading.** The crossed-out argument after $x$ provisionally reads $y$.



Original source details: GELFAND-P000544

**Undeciphered passage.** A short crossed-out entry follows the first line. Its reading is unresolved.

The syst. for $K$ reduces to the syst. for $F$. using the G.–L. eq.



Original source details: GELFAND-P000544

**Manuscript excerpt.** ~~from $F$ construct $K \Rightarrow \xi_n$~~



Original source details: GELFAND-P000544

**Undeciphered passage.** There is a heavily crossed-out fragment after the arrow at the end of the line.

Give a pair of lin. diff. ops. with const. coeffs. $L_0^{(1)}$ and $L_0^{(2)}$
from the pair construct the syst. of eqs. (15). Let $D_1F = D_2F = 0$, where
$F = F(x, y; t, z)$

From $F$ construct $K$, and from $K$ construct a soln. of the system.
Obtain a solution of the system.
1st nontriv. — eq. Kadomtsev–Petriashvili (?).

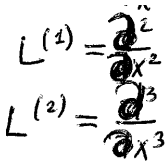



Original source details: GELFAND-P000544

**Manuscript excerpt.** The first name provisionally reads “Kadomtsev”. The second appears as “Petriashvili”. The parenthesized question mark is in the manuscript.

$$L_0^{(1)} = \frac{\partial^2}{\partial x^2}, \qquad L_0^{(2)} = \frac{\partial^3}{\partial x^3} + \lambda\frac{\partial}{\partial x}.$$

$$L^{(1)} = \frac{\partial^2}{\partial x^2} + u(x), \qquad \text{where } u(x) = 2\frac{d}{dx}K(x,x),$$
$$L^{(2)} = \frac{\partial^3}{\partial x^3} + \frac{3}{4}\left(u\frac{\partial}{\partial x} + \frac{\partial}{\partial x}u\right) + \lambda\frac{\partial}{\partial x} + w,$$



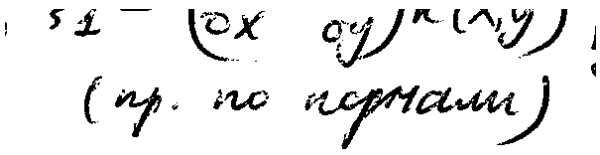
Original source details: GELFAND-P000544

**Uncertain reading.** The derivative signs in the leading terms of both operators are retraced. They are rendered as $\partial$.

where

$$w = \frac{3}{2}\frac{d}{dx}(\xi_1 + \xi_0^2)$$
$$\xi_0 = K(x,x), \qquad \xi_1 = \left(\frac{\partial}{\partial x} - \frac{\partial}{\partial y}\right)K(x,y)\Big|_{y=x}$$

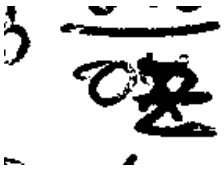
(deriv. in the normal direction)




Original source details: GELFAND-P000544

**Manuscript excerpt.** The parenthesis reads “deriv. in the normal direction”.

$$\begin{cases} \frac{3}{4}\beta\frac{\partial u}{\partial z} = -\frac{\partial w}{\partial x}; \\ \beta\frac{\partial w}{\partial y} = \frac{1}{4}\left(u_{xxx} + 3uu_x\right) + \lambda u_x + \alpha\frac{\partial}{\partial t}u \end{cases}$$



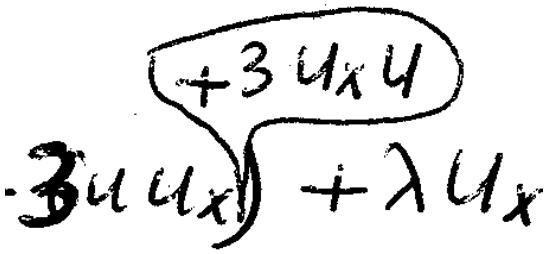
Original source details: GELFAND-P000544

**Uncertain reading.** In the first equation the variable below the derivative sign is overwritten. The final outline reads $z$.



Original source details: GELFAND-P000544

**Manuscript excerpt.** In the second equation the variable below the derivative sign clearly reads $y$.



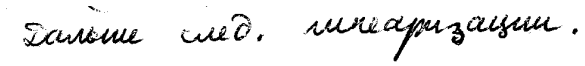
Original source details: GELFAND-P000544

**Uncertain reading.** The coefficient of $uu_x$ is overwritten and reads 3. At the closing parenthesis is a circled insertion $\boxed{+3u_xu}$

$u$ can be matrix-valued

$$\begin{pmatrix} u_0 & u_1 \\ 0 & u_0 \end{pmatrix}$$ for $u_1$ obtain a linearized eq. for $u_0$.

Next come the following linearizations.

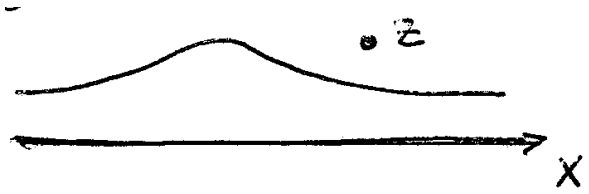


Original source details: GELFAND-P000544

**Manuscript excerpt.** The abbreviation before "linearizations" reads "след.".

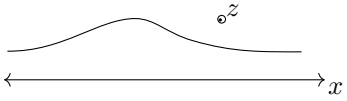
$\beta$ real or imaginary, $\lambda$...

All have a phys. meaning.

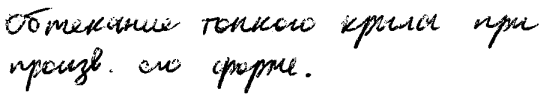


Original source details: GELFAND-P000544

**Diagram and typed description.**




Flow around a thin wing of arbitrary shape.



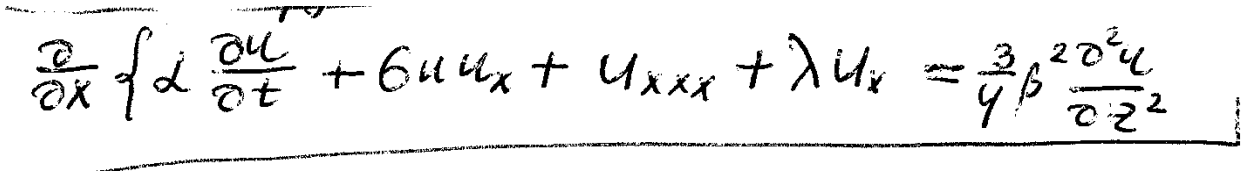
Original source details: GELFAND-P000544

**Manuscript excerpt.** The manuscript describes a wing of arbitrary shape.

KdV if independent of $z$.

Nonlin. string.

$$\frac{\partial}{\partial x}\left\{\alpha\frac{\partial u}{\partial t} + 6uu_x + u_{xxx} + \lambda u_x = \frac{3}{4}\beta^2\frac{\partial^2 u}{\partial z^2}.\right.$$



Original source details: GELFAND-P000544

**Manuscript excerpt.** The sum starts with an opening brace. The manuscript has no closing brace before the equals sign.

A medium with sound dispersion and shallow water.

Karpman's book — see the derivations there. (book on nonlin. waves)



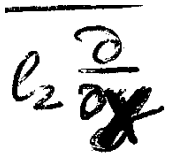
**Uncertain reading.** The middle letter of the surname is overwritten. The reading "Karpman" is provisional.

Original source details: GELFAND-P000544

$$L_0^{(1)} = \ell_1 \frac{\partial}{\partial x} + \ell_2 \frac{\partial}{\partial y}, \qquad [\ell_1, \ell_2] = 0, \qquad \ell_i \text{ — matrices (diagonal)},$$

$$L^{(1)} = \ell_1 \frac{\partial}{\partial x} + [\ell_1, \xi_0], \qquad \xi_0(x) = K(x,x), \qquad K \text{ — a matrix},$$

$$L^{(2)} = \ell_2 \frac{\partial}{\partial x} + [\ell_2, \xi_0].$$



**Uncertain reading.** The variable in the denominator of the second term of $L_0^{(1)}$ is overwritten. The reading $y$ is provisional.

Original source details: GELFAND-P000544

GELFAND-P000545

[source number: 62]

$$\ell_1 = \begin{pmatrix} a_1 & & 0 \\ & \ddots & \\ 0 & & a_n \end{pmatrix}; \qquad \ell_2 = \begin{pmatrix} b_1 & & 0 \\ & \ddots & \\ 0 & & b_n \end{pmatrix}.$$

The 1st nontriv. one arises in a matrix of size $3 \times 3$

$$\frac{\partial \xi}{\partial t} + \vec{V} \nabla \xi \qquad u_1 = \xi_{13}, \quad u_2 = \xi_{12}, \quad u_3 = \xi_{23}. \qquad \xi^+ = \xi$$



**Manuscript excerpt.** The superscript on $\xi$ is clearly a $+$.

Original source details: GELFAND-P000545

$$\begin{cases} \dfrac{\partial u_1}{\partial t} + \vec{V}_1 \nabla u_1 = i\lambda u_2 u_3, \\ \dfrac{\partial u_2}{\partial t} + \vec{V}_2 \nabla u_2 = i\lambda u_1 u_3^*; \\ \dfrac{\partial u_3}{\partial t} + \vec{V}_3 \nabla u_3 = i\lambda u_1 u_2^* \end{cases}$$



**Manuscript excerpt.** The rewritten subscript of the last factor reads 2.

Original source details: GELFAND-P000545

$u_i(t,x,z)$, $\nabla$ — 2-dimensional, $\vec{V}_i$ — arb. vectors

Describes the interaction of waves of 3 types in a nonlin. medium.

Dissipation cannot be introduced (ever). _____ · _____

Conjecture: they are all Hamiltonian. From one $L$ one constructs the symplect. form, and from the other, the Hamiltonian. _____ · _____

$u_{tt} - u_{xx} = \sin u$ also comes from the matrices

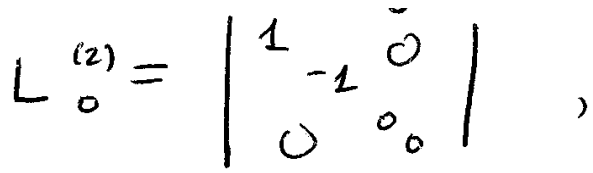


**Diagram and typed description.**

$$L_0^{(2)} = \begin{vmatrix} 1 & & 0 \\ & -1 & \\ 0 & & 0 \end{vmatrix}$$

Original source details: GELFAND-P000545

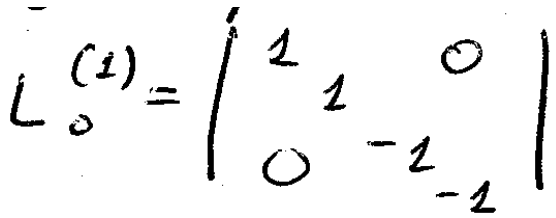
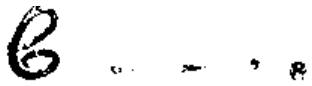


**Diagram and typed description.**

$$L_0^{(1)} = \begin{vmatrix} 1 & & 0 \\ & 1 & \\ 0 & & -1 \end{vmatrix}$$

Original source details: GELFAND-P000545

and an eq. for propag. of a laser pulse in . . .



**Manuscript excerpt.** The manuscript has a series of dots after “in”.

Original source details: GELFAND-P000545

_____ · _____

Bloch eq. for density matrices (quant. mech. in a 2-level system). _____ · _____

Matrix ops. of high orders.

$$L_0^{(1)} = \ell_1 \frac{\partial}{\partial x}, \qquad L_0^{(2)} = \frac{\partial^2}{\partial x^2}.$$

One obtains

$$i\frac{\partial \xi}{\partial t} + \xi_{xx} \pm |\xi|^2 \xi = 0 \qquad \text{(nonlin. Schrödinger eq.)}$$

there is a scheme for solving the Cauchy prob. _____ · _____

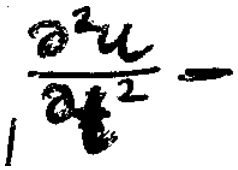
11/III-74. Gelfand's sem..




**Diagram and typed description.**
Povzner. Witham.

Original source details: GELFAND-P000546

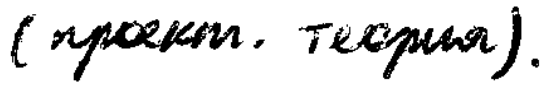
$$\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} + T'(u) = 0$$



**Uncertain reading.** The variable in the first denominator is overwritten and rendered as $t$.

Original source details: GELFAND-P000546

1) a wave travels $x \sim t$ (pract. theory).



**Manuscript excerpt.** The parenthesis reads "pract. theory".

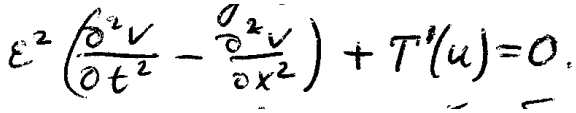
Original source details: GELFAND-P000546

and the asympt. of what happens far away

$$x \longrightarrow \frac{x}{\epsilon}, \qquad t \longrightarrow \frac{t}{\epsilon}$$

the asympt. is now sought in a finite region and as $\epsilon \to 0$.
$f(x - vt)$, $v$ depends little on $t$ and $x$ (varies slowly in the sense of $\epsilon$).
The method consists of the following:

$$\epsilon^2 \left( \frac{\partial^2 v}{\partial t^2} - \frac{\partial^2 v}{\partial x^2} \right) + T'(u) = 0.$$



**Manuscript excerpt.** Both numerators carry $v$, and the last term is $T'(u)$.

Original source details: GELFAND-P000546

study the asympt. in a finite region.

Add. assumption (justif. by scientific experience):

$$u(x,t) = f\left(\frac{\Theta_0(x,t)}{\epsilon}; x, t\right)$$

$$f(x_{\text{old}} - ct_{\text{old}}) = f\left(\frac{x-ct}{\epsilon}\right)$$

— exact trav. waves are a special case with $\Theta_0(x,t) = x - ct$



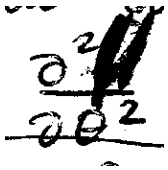
**Manuscript excerpt.** Both coordinate subscripts read “cr”, translated as “old”.

Original source details: GELFAND-P000546

$$\frac{\Theta_0(x,t)}{\epsilon} = \Theta \qquad f(\Theta; x, t).$$

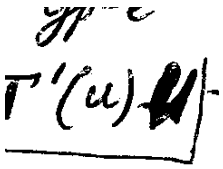
hence one obtains a diff. eq.

$$\epsilon^2(\omega_2^2 - \omega_1^2)\frac{\partial^2 u}{\partial \Theta^2} + T'(u)u + [\epsilon(u_x, u_t)\ldots] = 0$$



**Uncertain reading.** The letter in the derivative numerator is heavily overwritten. The reading $u$ is provisional.

Original source details: GELFAND-P000546



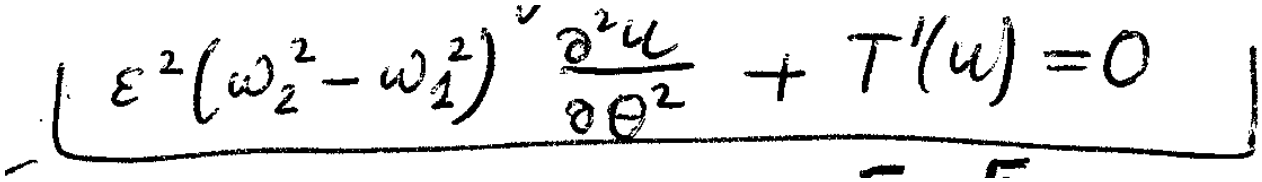
**Uncertain reading.** Immediately after $T'(u)$ is a heavily overwritten mark. The reading $u$ is provisional.

Original source details: GELFAND-P000546

where $\omega_2 = \Theta_{0t}, \quad \omega_1 = \Theta_{0x}$.

As $\epsilon \to 0$ one obtains

$$\epsilon^2(\omega_2^2 - \omega_1^2)\frac{\partial^2 u}{\partial \Theta^2} + T'(u) = 0$$



Original source details: GELFAND-P000546

**Manuscript excerpt.** The factor $\epsilon^2$ remains in the displayed equation after $\epsilon \to 0$.

ordinary eq. of 2nd order, $E_1, E_2$ — arb. consts.

$$u = u(E_1, E_2; x, t, \Theta)$$

Equation for $\Theta$.

Now substitute $\Theta = \dfrac{\Theta_0(x,t)}{\epsilon}$.

$$u = u\left(E_1, E_2; \frac{\Theta_0(x,t)}{\sqrt{\omega_2^2 - \omega_1^2} \cdot \varphi(E) \cdot \epsilon}\right)$$

(the solution is obtained in this form)

Add. eq.:

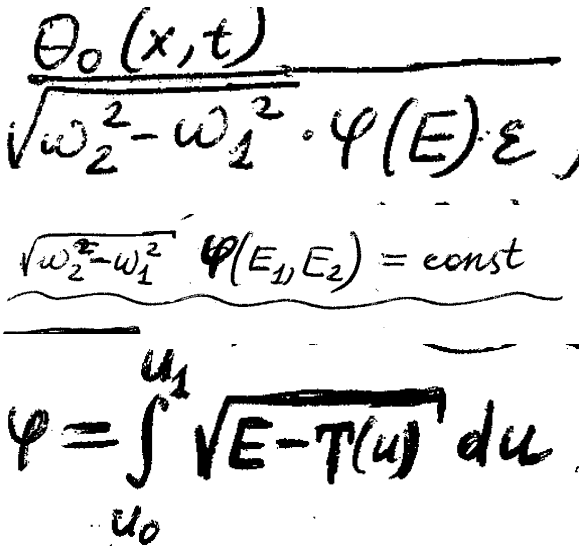


**Uncertain reading.** A crossed-out symbol resembling $\Theta_0$ precedes the square root.

Original source details: GELFAND-P000546

$$\sqrt{\omega_2^2 - \omega_1^2}\,\varphi(E_1, E_2) = \text{const}$$

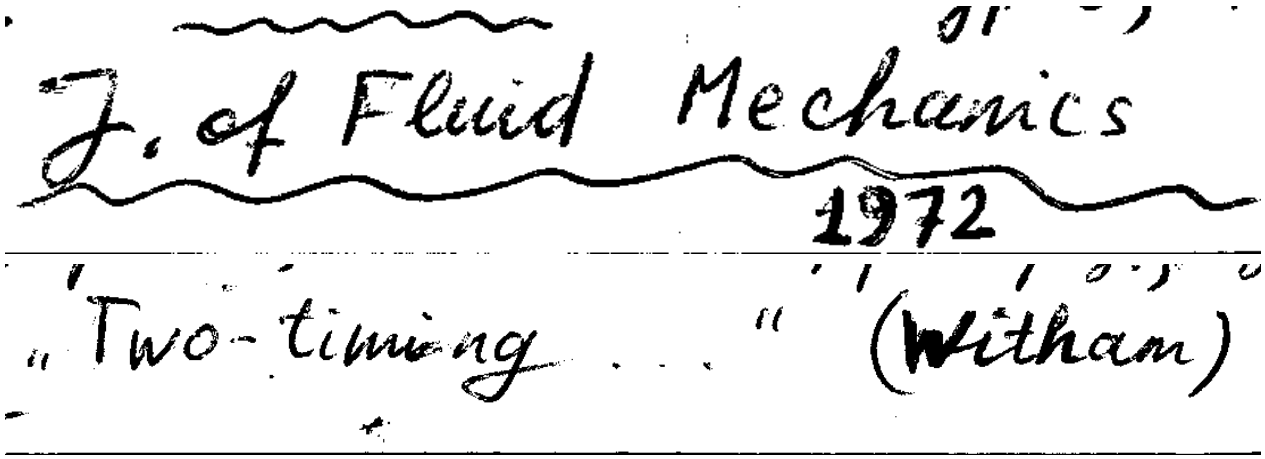
$$\varphi = \int_{u_0}^{u_1} \sqrt{E - T(u)}\, du$$



**Manuscript excerpt.** The letter $\varphi$ has a closed right loop in all three places. The middle occurrence is retraced.

Original source details: GELFAND-P000546

Ref.: Witham. This eq., Korteweg, heavy fluid, radiophys., lasers.
J. of Fluid Mechanics 1972 "Two-timing ..." (Witham)




Original source details: GELFAND-P000546

**Manuscript excerpt.** The parenthesized name and the year follow the manuscript. "1972" is written below the journal title.

GELFAND-P000547

[source number: 66]

V. N. Tulovsky. On Hörmander's work.
Wave fronts.

$$u \in \mathcal{D}'(\Omega), \qquad \Omega \subset \mathbb{R}^n.$$

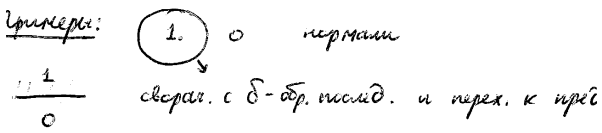
$(x_0, \xi_0) \notin WF(u)$, if $\exists \varphi \in C_0^\infty(\Omega), \quad \varphi(x_0) \neq 0$

$$\left|\langle u, \varphi e^{-i\langle x,\xi\rangle}\rangle\right| \leq C_N(1+|\xi|)^{-N}$$

in a small conic neighborhood of the point $\xi_0$:

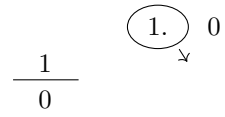
$$\left|\frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|}\right| < \epsilon.$$

Examples:




Original source details: GELFAND-P000547

**Diagram and typed description.**



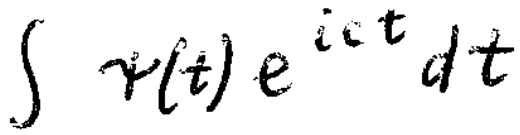

normals
convol. with a $\delta$-like seq. and pass. to the limit.

G.

$$\int \varphi(x_1, x_2) e^{i(x_1\xi_1 + x_2\xi_2)}\, dx_1\, dx_2$$

stat. int. over the plane.

$$\int \psi(t) e^{ict}\, dt$$



**Uncertain reading.** A faint sign appears between $i$ and $t$, provisionally read as $c$.

Original source details: GELFAND-P000547

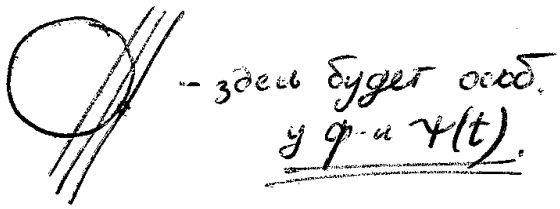
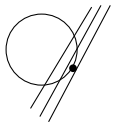



**Diagram and typed description.**



— there will be a sing. here of the func. $\psi(t)$.

Original source details: GELFAND-P000547

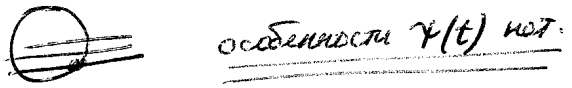



**Diagram and typed description.**



no singularity of $\psi(t)$.

Original source details: GELFAND-P000547

Tul. $\pi WF(u) = \operatorname{sing\,supp} u.$

$$\pi(x, \xi) = x.$$

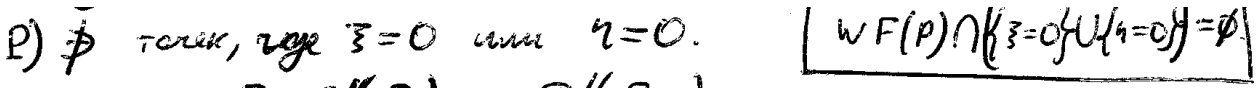
---

$$P : C_0^\infty(\Omega) \longrightarrow C^\infty(\Omega_1), \qquad \exists K_P \in \mathcal{D}'(\Omega \times \Omega_1),$$
$$\langle K_P, u(x)v(y)\rangle = \langle Pu, v\rangle$$
$$WF(K_P) \underset{\text{def}}{=} WF(P)$$

In the space $(x, y, \xi, \eta)$.

Suppose $WF(P) \not\ni$ points where $\xi = 0$ or $\eta = 0$.

$$\boxed{WF(P) \cap (\{\xi = 0\} \cup \{\eta = 0\}) = \varnothing}$$




Original source details: GELFAND-P000547

**Manuscript excerpt.** A slashed ∋ precedes “points”.

Then one can extend $P : \mathcal{E}'(\Omega) \to \mathcal{D}'(\Omega_1)$.

Def.

$$WF'(P) = \{(x, y, \xi, \eta) : (x, y, \xi, -\eta) \in WF(P)\}.$$

Def.

$$WF'(P) \circ WF(u) = \{(y, \eta) : \exists (x, y, \xi, \eta) \in WF'(P),\ (x, \xi) \in WF(u)\}.$$

Theorem.

$$WF(Pu) \subset WF'(P) \circ WF(u).$$

Suppose there are 2 operators:

$$Q : \underset{y}{\mathcal{E}'(\Omega_1)} \longrightarrow \underset{z}{\mathcal{D}'(\Omega_2)},$$
$$P : \underset{x}{\mathcal{E}'(\Omega)} \longrightarrow \underset{y}{\mathcal{E}'(\Omega_1)}$$

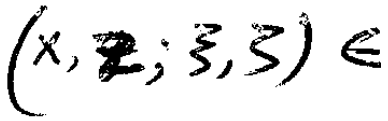
$$WF'(Q \circ P) \subseteq WF'(P) \circ WF'(Q).$$



Original source details: GELFAND-P000547

**Manuscript excerpt.** The left side has $Q \circ P$ in parentheses. On the right $WF'(P)$ comes first.

$$(x, z; \xi, \zeta) \in WF'(P) \circ WF'(Q) \underset{\text{def}}{\Longleftrightarrow} \exists\, (x, y, \xi, \eta) \in WF'(P),$$
$$(y, z, \eta, \zeta) \in WF'(Q).$$



**Uncertain reading.** The overwritten second coordinate on the left provisionally reads $z$.

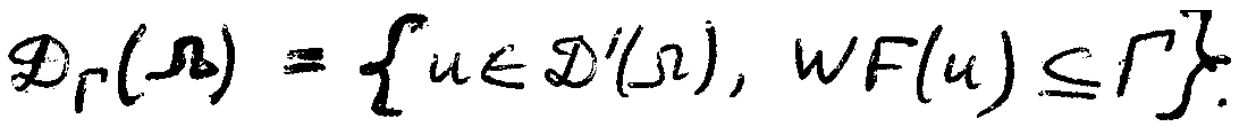
Original source details: GELFAND-P000547

GELFAND-P000548

Product of generalized funcs..

$\Gamma \subseteq T^*(\Omega) \setminus 0$, $\Gamma$ — a conic closed set.

$$\mathcal{D}_\Gamma(\Omega) = \{u \in \mathcal{D}'(\Omega),\ WF(u) \subseteq \Gamma\}.$$



Original source details: GELFAND-P000548

**Manuscript excerpt.** The first $\mathcal{D}_\Gamma$ is written without a prime. The inner $\mathcal{D}'$ carries one.

Theorem: let $\Gamma_1, \Gamma_2$ — 2 closed conic sets in $T^*(\Omega) \setminus 0$ and suppose

$$\Gamma_1 + \Gamma_2 \subseteq \Omega \times (\mathbb{R}^n \setminus 0)$$

$$(x,\xi) \qquad (x,\eta) \qquad \Gamma_1 + \Gamma_2 = \{(x, \xi+\eta)\}, \qquad \xi + \eta \neq 0$$

Then there exists a cont. mapping

$$\mathcal{D}_{\Gamma_1}(\Omega) \times \mathcal{D}_{\Gamma_2}(\Omega) \longrightarrow \mathcal{D}_{\Gamma_1 \cup \Gamma_2 \cup (\Gamma_1+\Gamma_2)}(\Omega),$$

extending the product of ordinary funcs.
(and the product of a generalized func. by a smooth one)
(It is unique owing to the topology consid.)

Def. of Fourier integral ops.

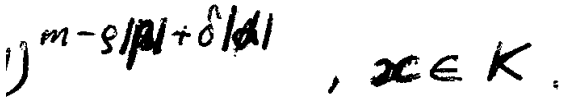
$\varphi(x,\Theta), \quad (x,\Theta) \in \Omega \times (\mathbb{R}^N \setminus 0)$ — a phase func., if it is real, homogeneous of order 1 [in $\Theta$] and $\varphi'_{x,\Theta} \neq 0$.



Original source details: GELFAND-P000548

**Manuscript excerpt.** The circled insertion "in $\Theta$" is attached to "homogeneous".

$$S^m_{\rho,\delta}(X \times \mathbb{R}^N) \ni a(x,\Theta),$$
$$\left|\partial_x^\alpha \partial_\Theta^\beta a(x,\Theta)\right| \leq C_{\alpha,\beta}(1+|\Theta|)^{m-\rho|\beta|+\delta|\alpha|}, \qquad x \in K.$$



Original source details: GELFAND-P000548

**Uncertain reading.** Two indices in the exponent and the variable before $\in K$ are retraced. The reading is $m - \rho|\beta| + \delta|\alpha|$ and $x \in K$.

$$\langle A, u\rangle = \int e^{i\varphi(x,\Theta)} a(x,\Theta) u(x)\, dx\, d\Theta$$
$$= \lim_{\epsilon \to 0} \int e^{i\varphi(x,\Theta)} a(x,\Theta) \chi(\epsilon\Theta) u(x)\, dx\, d\Theta$$

where $\chi \in C_0^\infty(\mathbb{R}^N)$, $\chi = 1$ when $|\Theta| \leq 1$.

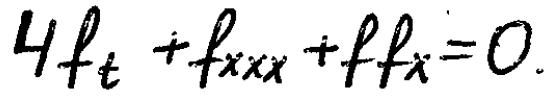


Original source details: GELFAND-P000548

**Manuscript excerpt.** The last condition clearly reads $|\Theta| \leq 1$.

18/III-74. Gelfand's sem.. A. B. Shabat.

$$4f_t + f_{xxx} + ff_x = 0.$$



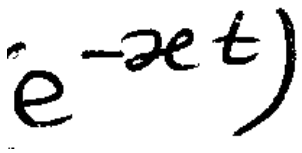
**Manuscript excerpt.** The plus sign is followed immediately by $ff_x$.

Original source details: GELFAND-P000549

$$f(x,t) = g(x-ct) = \frac{2\lambda^2}{\mathrm{ch}^2[\lambda(x-\lambda^2 t - h)]};$$

2-solit. solution

$$f(t,x) = \dots + \dots + O(e^{-\varkappa t}) \quad \text{as } t \to +\infty.$$



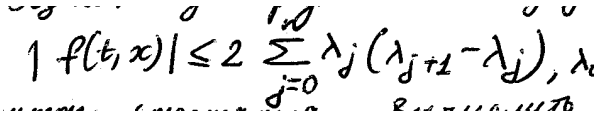
**Uncertain reading.** The letter in the exponent has two loops. The reading $\varkappa$ is provisional.

Original source details: GELFAND-P000549

one can prove that it is unique.

1) Estimate the total amplitude at the instant of collision.

$$|f(t,x)| \le 2\sum_{j=0}^{n} \lambda_j(\lambda_{j+1} - \lambda_j), \qquad \lambda_0 = 0$$



**Uncertain reading.** A small sign above the sum touches a descender from the adjacent line. The reading $n$ is provisional.

Original source details: GELFAND-P000549

(the amplit. do not add as in the lin. case)

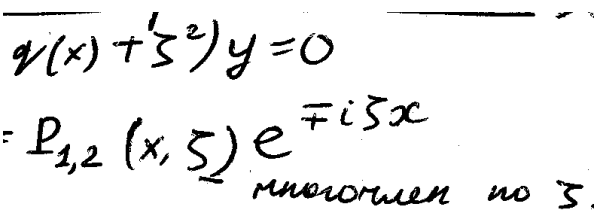
2) The centers will shift. Calculate the shift. The answer is stand. — scattering.

Bargmann potential $q(x)$, if $\exists 2$ solns. of the eq.

$$y'' + (q(x) + \zeta^2)y = 0$$

of the form

$$y_{1,2} = P_{1,2}(x,\zeta)e^{\mp i\zeta x} \qquad \text{polynomial in } \zeta.$$



**Uncertain reading.** The spectral letter provisionally reads $\zeta$.

Original source details: GELFAND-P000549



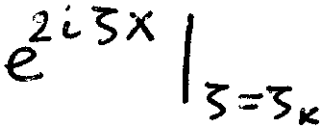
**Undeciphered passage.** A neighboring entry is cut off at the left edge. The closing parenthesis and equals sign are visible $\ldots) = \ldots$

Original source details: GELFAND-P000549

GELFAND-P000550

Let $P, Q$ be such that:

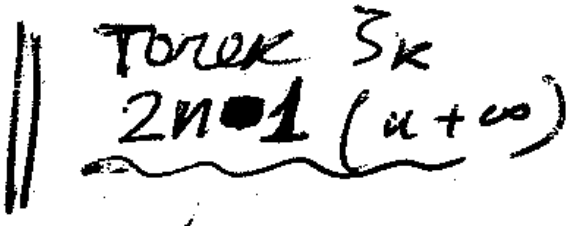
$$c_k e^{2i\zeta x}\Big|_{\zeta=\zeta_k} = \frac{P(x,\zeta)}{Q(x,\zeta)}\Big|_{\zeta=\zeta_k}, \qquad \frac{P}{Q}\Big|_{\zeta=\infty} = 1$$



**Uncertain reading.** The spectral letter provisionally reads $\zeta$.

Original source details: GELFAND-P000550

|| Points $\zeta_k$ $2n\,?\,1$ (and $+\infty$)



**Uncertain reading.** Ink obscures the sign between $2n$ and $1$.

Original source details: GELFAND-P000550

For any choice of parameters $c_k, \zeta_k$ one obtains a fund. syst. of solutions of the S.–L. oper. with Bargmann pot.

$$y_1 = P(x,\zeta)e^{i\zeta x}$$
$$y_2 = Q(x,\zeta)e^{-i\zeta x}$$

1.

$$P'_x Q - Q'_x P - 2i\zeta PQ = \prod(\zeta - \zeta_k).$$

Diff. with respect to $x$.

$$\frac{d}{dx}\frac{P}{Q}\Big|_{\zeta=\zeta_k} = \frac{P'_x Q - Q'_x P}{Q^2}\Big|_{\zeta=\zeta_k} = 2i\zeta\,\frac{P}{Q}\Big|_{\zeta=\zeta_k}$$

is a rat. func. in $\zeta$ with zeros at the [crossed out] giv. points



**Uncertain reading.** The short crossed-out entry before “giv.” is illegible.

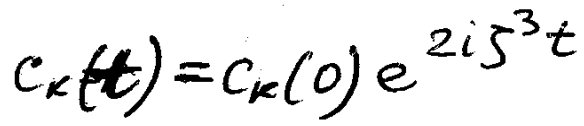
Original source details: GELFAND-P000550

$$\left.\frac{P_xQ - Q_xP - 2i\zeta PQ}{Q^2}\right|_{\zeta=\zeta_k} = 0.$$

and one cond. at $\infty$.

Now let $c_k$ depend on $t$ (the parameter):

$$c_k(t) = c_k(0)e^{2i\zeta^3 t}$$



**Uncertain reading.** The argument of $c_k(t)$ reads $t$ over earlier strokes. The underlying entry is illegible.

Original source details: GELFAND-P000550

$$\left.\frac{P}{Q}\right|_{\zeta=\zeta_k} = \left. c_k(t)e^{2i\zeta x}\right|_{\zeta=\zeta_k}$$

Then also

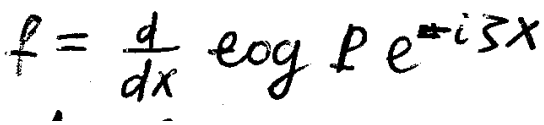
$$\left(4\frac{\partial}{\partial t} + \frac{\partial^3}{\partial x^3} - 3q\frac{\partial}{\partial x}\right)P = (\qquad)Q = 0$$



**Undeciphered passage.** The manuscript leaves the parentheses before $Q$ empty.

Original source details: GELFAND-P000550

Then $f = \frac{d}{dx}\log Pe^{?i\zeta x}$ satisf.



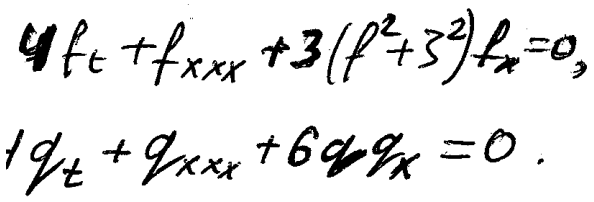
**Uncertain reading.** The sign before $i\zeta x$ is overwritten and its reading remains unresolved.

Original source details: GELFAND-P000550

$$4f_t + f_{xxx} + 3(f^2 + \zeta^2)f_x = 0,$$

while $f' + f^2 = q$ satisf. the eq.

$$4q_t + q_{xxx} + 6qq_x = 0.$$



**Manuscript excerpt.** In the first equation the $4$, the $+$ sign and the $3$ are retraced. In the second, $qq_x$ is retraced.

Original source details: GELFAND-P000550

GELFAND-P000551

25/III-74. Gelfand's sem..

Kirillov. On inf. groups. Descr. of repres. of the fol. groups

$$G_1 = \text{Diff}\, M, \qquad G_2 = H^M = C^\infty(M, H) \qquad (H \text{ is a group}).$$



**Uncertain reading.** A separate two-stroke glyph stands to the left of the diagram. Its reading remains unresolved.

Original source details: GELFAND-P000551

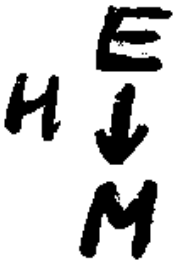


**Diagram and typed description.**

$$\begin{array}{ll} & E \\ H & \downarrow \\ & M \end{array}$$

Original source details: GELFAND-P000551

principal bundle and its automorphism group

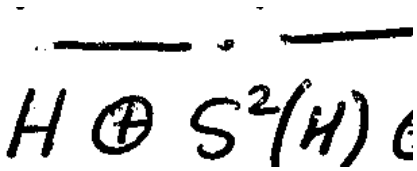
It projects onto $G_1$ with fiber $G_2$



**Undeciphered passage.** The rectangular entry below the diagram is densely crossed out and illegible.

Original source details: GELFAND-P000551

$$e^H = H_0 \oplus H \oplus S^2(H) \oplus S^3(H) \oplus \dots$$

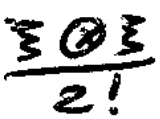


**Uncertain reading.** Two horizontal strokes and a small loop between them stand above $H, \oplus, S^2(H)$. Their meaning and attachment remain unresolved.

Original source details: GELFAND-P000551

($H_0$ is one-dimensional; $S^k(H)$ is a symm. power

$$1 \qquad \xi \qquad \frac{\xi \otimes \xi}{2!} \qquad\qquad |\xi|^2 = \sum \frac{|\xi_n|^2}{n!}$$



**Uncertain reading.** The circled sign in the numerator provisionally reads $\otimes$.

Original source details: GELFAND-P000551

Gurevich. On the K.–d.–V. eq. by Whitham's method.



**Uncertain reading.** The ending of the abbreviated word for "equation" is overwritten in the Russian. Its exact reading remains unresolved.

Original source details: GELFAND-P000551

GELFAND-P000552

1/*IV*-74. Gelfand's sem..

Yura Koblents. On the asymptotics of the eigenvalues of the Laplace oper..

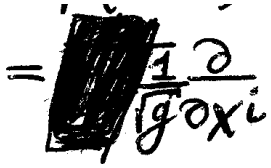
(from Leningrad)

$M$ — closed Riem. manif., $\Delta$ — Beltrami–Laplace oper..

$$\lambda_0 = 0, \qquad 0 \le \lambda_1 \le \lambda_2 \le \dots \qquad \text{— spectrum.}$$

$$Z(s) = \operatorname{Tr}\exp(-s\Delta) = \sum_i e^{-\lambda_i s}$$

$$\Delta f = \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^i}\left(\sqrt{g}\, g^{ij} \frac{\partial f}{\partial x^j}\right), \qquad \text{where } g = \det \|g_{ij}\|,$$



**Undeciphered passage.** There is a densely crossed-out entry between the equals sign and $1/\sqrt{g}$. A stroke on the right cannot confidently be separated from the cancellation.

Original source details: GELFAND-P000552

$$E(m, m', t) = \sum e^{-\lambda_i t} \varphi_i(m) \varphi_i(m').$$



**Uncertain reading.** The letter in both factors provisionally reads $\varphi$.

Original source details: GELFAND-P000552

$$Z(s) = \int E(m, m, s)\, dm$$

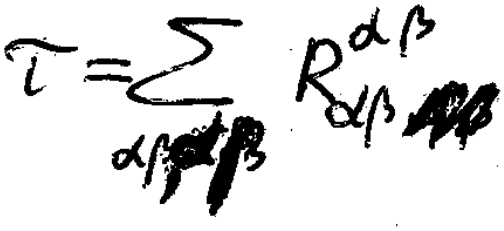
Minakshisundaram, 1949

$$Z(t) = (4\pi t)^{-n/2}(a_0 + a_1 t + a_2 t^2 + \ldots), \qquad t \to 0.$$

$$n = \dim M, \qquad a_0 = \operatorname{vol} M, \qquad \operatorname{Re} t > 0.$$

$$a_1 = \int_M \frac{\tau}{6}\, dV, \qquad \text{where } \tau \text{ — scal. curvature}$$

$$\tau = \sum_{\alpha\beta[?]\beta} R^{\alpha\beta}_{\alpha\beta[\ldots]}$$



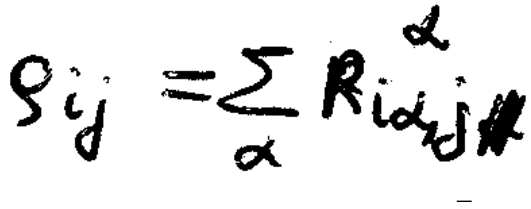
**Uncertain reading.** The sum index shows an initial $\alpha\beta$ and a final $\beta$, with overwritten strokes between them. Two signs after the lower $\alpha\beta$ of $R$ are crossed out.

Original source details: GELFAND-P000552

$$a_2 = \frac{1}{360} \int \left(2|R|^2 - 2|\rho|^2 + 5\tau^2\right)$$

$R$ — curvature tensor; $\rho$ — Ricci tensor,

$$\rho_{ij} = \sum_{\alpha} R^{\alpha}_{i\alpha j}$$



**Uncertain reading.** There is a heavily overwritten mark after the lower $j$ of $R$. The original stroke is illegible.

Original source details: GELFAND-P000552

$$|R|^2 = \sum R^2_{\alpha\beta\gamma\delta} \qquad \text{in norm. coord.} \qquad |R|^2 = R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$$

$$|\rho|^2 = \rho_{\alpha\beta}\rho^{\alpha\beta}$$

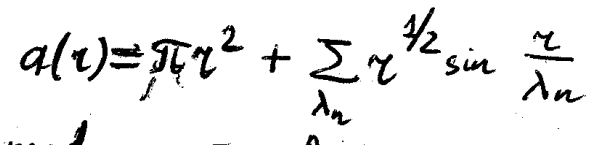
$$E(t, m, m') = (4\pi t)^{-n/2}\left(u_0(m, m') + u_1(m, m')t + u_2(m, m')t^2 + \ldots\right)$$

in a neighb. of the diagonal.

$a_3$ — Sakai and the speaker (independently).

G. Number of int. points in a disk:

$$a(r) = \pi r^2 + \sum_{\lambda_n} \tau^{1/2} \sin \frac{\tau}{\lambda_n}$$

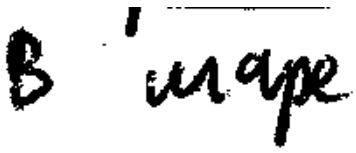


**Uncertain reading.** The letter at the start of the formula provisionally reads $r$. Both occurrences in the summand have a broad wavy top, provisionally read as $\tau$.

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($\lambda_n$ — zeros of the Riemann $\zeta$-func.) mod the Riemann hypothesis Schnirelmann.

In the ball $\lambda_n$ — lengths of closed billiard paths.



**Manuscript excerpt.** The first letter of “шаре” has three strokes.

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$$\sum \delta(\lambda - \lambda_k) * \frac{\varepsilon}{\lambda^2 + \varepsilon^2} = f(\lambda)$$

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[source number: 83]

$$f(\lambda) = f_0(\lambda) + \sum_k c_k \lambda^{\alpha_k} \sin(\ell_k \lambda + \omega_k)$$

There is exper. confirmation (Balian and Bloch)

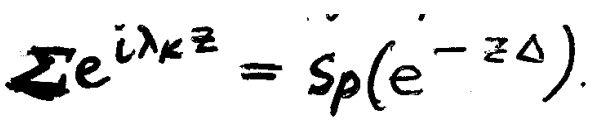


**Uncertain reading.** The end of the surname Balian is retraced. The reading of the final n is provisional.

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(and even a precise statement).

$$\sum e^{i\lambda_k z} = \mathrm{Sp}(e^{-z\Delta}).$$



**Manuscript excerpt.** The exponent on the left is $i\lambda_k z$, and the one on the right is $-z\Delta$.

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contin. downward (from the upper half-plane)

Twin probl.:

$$\lambda_{k+1} - \lambda_k > c\lambda_k^{-\nu} \qquad \text{for general operators}$$



**Uncertain reading.** The letter after the minus has two converging strokes and provisionally reads $\nu$.

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but on the circle they are double. (stably asympt. double spectrum).

$T^2$ $\quad g_{ij} \sim \delta_{ij}$ $\quad$ the theorem holds

Theorem.

$$\min(\lambda_{k+1} - \lambda_k, \lambda_k - \lambda_{k-1}) < C_N \cdot k^{-N}, \qquad \forall N$$

(Lazutkin, Schnirelmann)



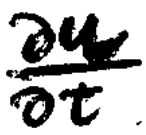
**Uncertain reading.** The strokes leading into the last letters of “Schnirelmann” are overwritten. The underlying strokes are illegible.

Original source details: GELFAND-P000553

(a small but finite perturbation) — stab.

For $T^3$ this is no longer so. (apparently).

$$\frac{\partial u}{\partial t} = i\sqrt{-\Delta}\, u; \qquad u|_{t=0} = \delta_x$$

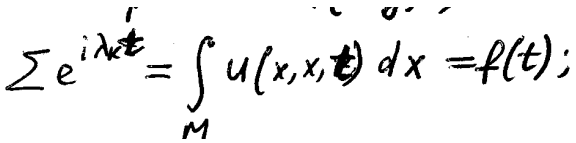


**Uncertain reading.** The overwritten letter after $\partial$ provisionally reads $u$.

Original source details: GELFAND-P000553

Its soln. $\quad u(x, y, t)$

$$\sum e^{i\lambda_k t} = \int_M u(x, x, t)\, dx = f(t);$$



**Uncertain reading.** The sign in the exponent of the sum and the final argument of the integrand are retraced. Both provisionally read $t$.

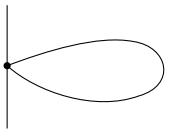
Original source details: GELFAND-P000553

singul. of $u(x, y, t)$ where $|x - y| = t$



Original source details: GELFAND-P000553

**Diagram and typed description.**



for some $t$ bifurcations occ.



Original source details: GELFAND-P000553

**Uncertain reading.** The Russian ending of “bifurcations” is overwritten. The reading is provisional.

singul. move at speed 1